

Fontaine–Mazur modularity at the prime 2

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Abstract

We prove that every continuous, irreducible, odd two-dimensional 2-adic representation of $G_{\mathbb{Q}}$, unramified outside finitely many primes and de Rham at 2 with distinct Hodge–Tate weights, is modular up to Tate twist. This resolves the odd, regular two-dimensional Fontaine–Mazur conjecture over $\mathbb{Q}$ at 2, including all residual representations.

1 Introduction

The Fontaine–Mazur conjecture predicts that irreducible p -adic Galois representations satisfying suitable finiteness and local conditions arise from algebraic geometry [24]. Modular forms supply a central class of examples through Deligne’s cohomological construction [19, Sections 3.9–3.12 and 5.2–5.5]. Over $\mathbb{Q}$, its odd two-dimensional form has a more precise modularity formulation. A continuous representation

$$r : G_{\mathbb{Q}} \longrightarrow \mathrm{GL}_2(\overline{\mathbb{Q}}_p)$$

is *odd* if $\det r(c) = -1$ for a complex conjugation c . It is *regular de Rham* at p if its restriction to $G_{\mathbb{Q}_p}$ is de Rham and has two distinct Hodge–Tate weights. The conjecture asserts that an irreducible such representation, unramified outside finitely many primes, comes from a classical cuspidal eigenform, allowing a Tate twist.

Theorem 1.1. *Let*

$$r : G_{\mathbb{Q}} \longrightarrow \mathrm{GL}_2(\overline{\mathbb{Q}}_2)$$

be continuous and irreducible. Suppose that r is odd, is unramified outside finitely many primes, and is de Rham at 2 with distinct Hodge–Tate weights. Then r is isomorphic to a Tate twist of the 2-adic Galois representation associated with a classical cuspidal eigenform.

Thus Theorem 1.1 resolves the odd, regular two-dimensional Fontaine–Mazur conjecture over $\mathbb{Q}$ at 2. Its scope includes every semisimple residual representation, including scalar and reducible ones. Together with the odd-prime results discussed below, it gives the stated modularity conclusion at every prime. The geometric realization of modular representations also gives the geometric-origin conclusion in this odd, regular setting.

Theorem 1.1, in particular its absence of a residual-image hypothesis, supplies the weight-two modularity input in the height argument of *Hilbert’s tenth problem over the rational numbers* [44, Section 4], whose undecidability proof also uses substantial additional arithmetic and logical constructions.

1.1 Background and the dyadic problem

The deformation-theoretic approach to modularity lifting developed by Wiles and Taylor–Wiles compares Galois deformation rings with Hecke algebras [63, 57]. Serre’s conjecture concerns the residual starting point [51]; its proof by Khare–Wintenberger and Kisin supplies modularity for odd absolutely irreducible residual representations, including characteristic two [31, 32, 35]. The existence of a residual modular form still leaves the problem of proving modularity for a specified characteristic-zero lift. For a reducible residual representation, even the modular seed must be obtained by a different construction.

Skinner and Wiles developed ordinary families and pro-modularity arguments for residually reducible representations and then for nearly ordinary deformations of irreducible residual representations [53, 54]. Allen adapted this family method to nearly ordinary dyadic representations with solvable absolutely irreducible residual image [1]. Kisin brought local deformation geometry and the p -adic Langlands correspondence into the regular Fontaine–Mazur problem [34]. Emerton’s local–global compatibility work separated occurrence in completed cohomology from the production of classical vectors [22]. These developments explain why both stages of the proof below are needed, and why ordinary auxiliary families can be useful without ordinarity being imposed on the final target.

At odd primes, Pan’s work treats the residually reducible case and, together with preceding modularity lifting theorems, establishes the regular conjecture apart from one local residual configuration at 3 [45, Theorems 1.0.2 and 1.0.4]. Zhang removes that configuration, obtaining the stated odd, regular theorem for every odd prime [64, Theorem 1.0.2]. At 2, Tung removes the remaining local restriction in a modularity lifting theorem whose global residual image is nonsolvable [60, Theorem A]. Thus local scalar residual behavior and a globally scalar residual representation must be distinguished: a local theorem covering the former does not by itself settle the latter.

The proof has a useful precedent in the passage from unrestricted pro-modularity to classicality developed by Pan, and in Zhang’s potential pro-modularity argument [45, 64]. The present dyadic problem requires integral control of the local actions and of deformation families even when their residual representation is scalar. We first construct potential occurrence in completed definite forms, then extract classical vectors, and finally descend. The distinction between occurrence and classicality also appears in Pan’s theorem for representations in the completed cohomology of modular curves: that theorem assumes occurrence as a hypothesis [46, Theorem 7.1.2].

Thorne’s theorem supplies an essential set of classical points: ordinary, potentially crystalline representations of weights $\{0, 1\}$ are modular at every prime, without a residual irreducibility hypothesis [59, Theorem D]. After a global normalization and passage to the dual, our family construction makes this theorem applicable at a dense set of points, and the subsequent specialization argument recovers the original target’s arbitrary weight gap. Prescribing global lifts with chosen local components, as in Emerton–Gee–Pan–Zhu [23, Theorem 1.1.1], is a different conclusion; their dyadic theorem also retains a nonsolvable residual-image hypothesis.

The local input is the p -adic Langlands correspondence of Colmez [17], in the all-prime form of Colmez, Dospinescu and Paškūnas [18]. Paškūnas and Tung’s block finiteness theorem [48] supplies an integral local pseudodeformation action even in scalar residual blocks. This resolves a local representation-theoretic obstacle, but leaves a global one: a trace does not record a residual extension class, and globally reducible families can be too large for a direct connectedness argument. We work with actual representations and temporarily record local invariant lines to obtain the necessary dimension loss. These auxiliary marks are used in connectedness; the propagation theorem imposes no ordinary condition at the dyadic places.

1.2 From occurrence to classicality and descent

A *pro-modular locus* is a closed trace locus in a Galois pseudodeformation spectrum defined by a completed Hecke algebra of definite quaternionic forms. No local condition is imposed at 2. Its characteristic-zero points give Hecke eigensystems in completed forms. The central problem is to pass from this occurrence statement to a classical cuspidal form with the prescribed Galois representation. We do so in three stages, summarized in Figure 1.

Potential occurrence for a whole family. The first stage places the target and the deformation families needed later in one pro-modular trace locus over a common solvable totally real extension F , completely split at 2. Two constructions work together. Localized propagation starts at a characteristic-two curve whose generic representation remains absolutely irreducible under finite restriction. If the curve is pro-modular and has finite local images at the allowed non-dyadic primes, patching gives potential pro-modularity of every irreducible deformation locus containing it (Theorem 5.1). The local deformation problem at 2 is unrestricted.

To produce the curves to which this result applies, we construct abelian 2-extensions $\mathbb{Q} \subset A \subset F$, keeping the number of allowed ramified primes under control while their degrees grow. In the scalar and reducible cases we mark a small proportion of the dyadic places. Recording an invariant line at those places retains information that traces alone would lose. We select the open part of the residual fiber on which the marked lines are transverse to every globally invariant line. A Selmer calculation makes the globally reducible locus too small to disconnect these charts. A modular seed and a curve cut from a component containing the target enter the same connected residual fiber. Applying propagation along it yields Theorem 7.1. The conclusion concerns the whole closed trace image of the prescribed family at one common tame level; it does not require that this family itself have the dimension of the larger component used to find the curve.

Classical vectors at the target. When the local representation at 2 is irreducible, the local Langlands correspondence supplies locally algebraic vectors. For a reducible local target, we choose an invariant line with the larger cyclotomic exponent and retain its character as a parameter. The resulting family records trace and character jointly and is finite over a one-dimensional weight space. Its continuous members are not all de Rham; the useful points are a dense set of specially chosen weights. After a global twist the corresponding representations have cyclotomic exponents $\{1, 0\}$, or Hodge–Tate weights $\{-1, 0\}$. Their *duals* have Hodge–Tate weights $\{0, 1\}$ and satisfy the ordinary potentially crystalline hypotheses of Thorne’s theorem.

Those classical points determine the order of the two characters in the ordinary induction. A finite closed joint support transports this order to the target. A quotient, lattice and Tate-module construction then supplies a genuine eigenvector in the target eigenspace. Finally, a direct polynomial-orbit argument turns its character and exponent inequality into a nonzero locally algebraic vector. This last step avoids assuming that algebraic vectors commute with specialization.

Descent of a classical packet. The vector gives a regular cuspidal automorphic representation over F . We descend along prime-degree cyclic steps in a solvable tower. At each step, a finite-character discrepancy between the descended Galois representation and the target is removed by a Hecke-character twist. Descent is applied only after classicality has been established; no descent statement for pro-modular loci is needed.

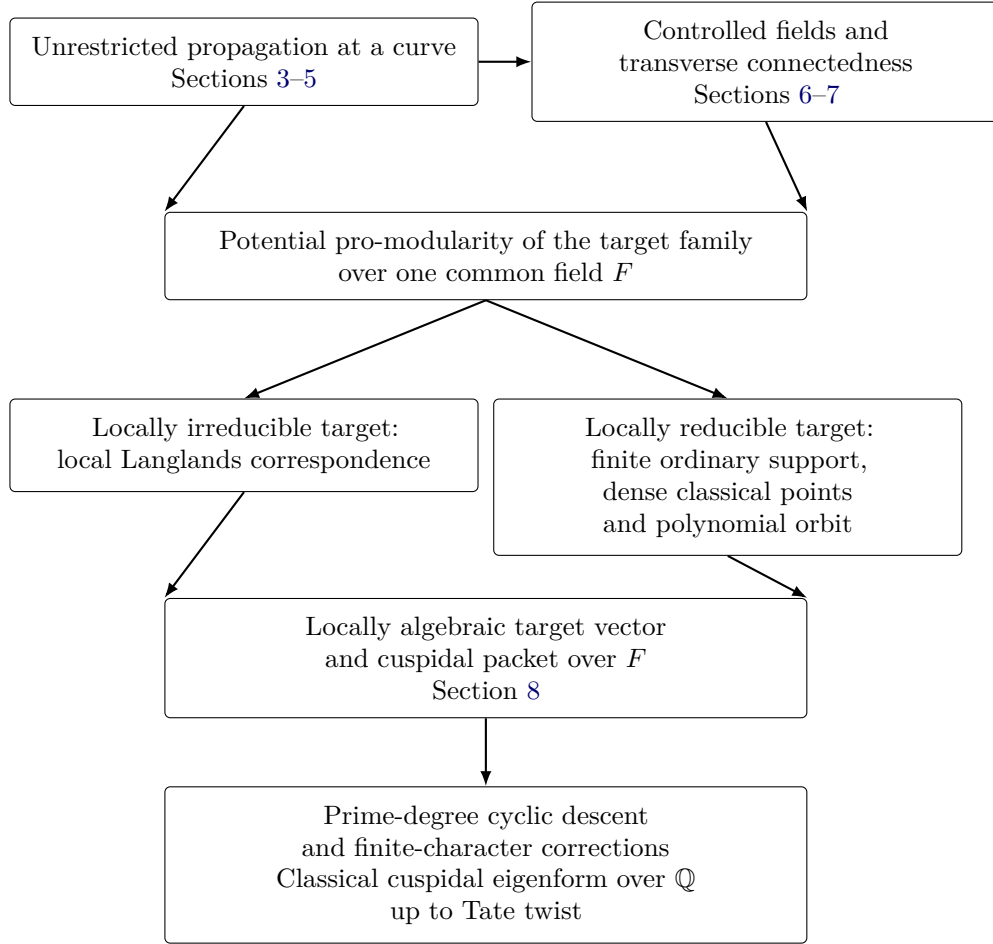


Figure 1: The proof route. The connectedness construction uses the propagation theorem; the two upper boxes are cooperating inputs to occurrence. The target enters through that construction. The common field F is solvable, totally real, of even degree and completely split at 2. The ordinary family is used in the reducible local branch. Both branches produce a classical packet before descent.

1.3 The constructions used in the proof

Several established methods must be kept compatible at the characteristic-two curve. Böckle–Iyengar–Paškūnas develop local deformation theory with Laurent-series residue fields and coefficient orders, including the auxiliary-variable comparison of completions [11, Section 3.5]. We use this framework with an extra coefficient variable and global representation and incidence spaces. Lemma 3.4 proves the resulting identification of continuous deformation and obstruction problems, including the topology on the residue field and the dimension change.

The propagation construction follows the auxiliary-level method of Taylor–Wiles, Kisin’s formulation relative to local deformation rings, and the characteristic-two determinant-twist argument of Khare–Wintenberger [57, 36, 32]. Pan’s localization at one-dimensional primes is the direct global precedent [45]; Zhang’s potential formulation is used in the same overall progression [64]. At the Laurent-series residue field we patch finite jets, meaning quotients by fixed powers of the local maximal ideal. The exactness argument works with filtrations whose successive quotients are fixed old-level modules. A uniform Artin–Rees bound preserves their actual old deformation actions. Determinant twists and explicit quadratic sign intertwiners then give support on every local component. All of these action comparisons are retained integrally.

The marked-line construction serves a different purpose. Its local factors are integral even over the characteristic-two real-place base, and transversality gives a strict Selmer loss at every mark. The number of marks grows more slowly than the total degree, so this loss is compatible with the global dimension estimates. The argument uses the unrestricted propagation theorem after entering a transverse chart; it does not replace that theorem by ordinary patching.

Organization. Section 2 fixes conventions and removes the virtually solvable case. Section 3 constructs the representation spaces, their curve completions and the local models. Section 4 compares the actual Galois and local block actions on completed forms and proves the Hecke-dimension bound. Section 5 proves localized propagation. Section 6 supplies the field and Selmer estimates, and Section 7 combines them with a modular seed to obtain potential pro-modularity. Section 8 constructs classical vectors and carries out solvable descent.

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2 Conventions and initial reductions

We fix the coefficient and automorphic conventions before introducing deformation spaces. We then remove the virtually solvable case and specify the trace loci used throughout the proof. Table 1 collects the normalizations that link deformation theory, completed forms and classicality.

Object	Convention
Frobenius	Geometric by default; local reciprocity sends a uniformizer to geometric Frobenius. The tame relation with exponent q_v uses arithmetic Frobenius explicitly.
Weights	$\varepsilon = x x $, cyclotomic exponent 1, $\text{HT}(\varepsilon) = \{-1\}$, and $ 2 = 2^{-1}$.
Determinants and blocks	$\chi = \det r$, completed-form central character $\psi = \chi\varepsilon$, local block parameter $r \otimes \varepsilon$ with determinant $\chi\varepsilon^2$.
Ordinary target	At a locally reducible de Rham target choose the line character η_1 with exponent $a > b$, where $\eta_2 = \chi/\eta_1$ has exponent b . The favorable upper-Borel, unnormalized inducing pair is $(\eta_2\varepsilon, \eta_1)$.
Algebraic factor	$\text{Sym}^{a-b-1} \det^{b+1}$. The continuous family retaining η_1 need not be de Rham.
Curve coefficients	$K = k((t))$ has its Laurent topology; $\mathcal{L} = \widehat{\mathcal{O}[[z]]_{(\varpi)}}$, with $z \mapsto t$. The auxiliary ultrapower in Section 5 is specified separately.
Dual actions	Group duals use the indicated contragredient action. On scalar eigenspaces the commuting Hecke, torus and U operators are transposed without inversion, as in Section 8.
Support	Images in pseudodeformation spectra are closed images; pro-modular support containments are set-theoretic.

Table 1: Conventions used at the interfaces of the proof. The character orientation in the table concerns the chosen target and the normalized dense points, not every member of the family.

2.1 Coefficients and cyclotomic exponents

Write $\mathcal{O}$ for the ring of integers in a sufficiently large finite extension $E/\mathbb{Q}_2$, ϖ for a uniformizer, and k for its residue field. Finite coefficient extensions are allowed. A lattice means a finite free $\mathcal{O}$ -module of full rank. The cyclotomic character is denoted by ε . We count weights as *cyclotomic exponents*, so that ε has exponent 1. In the Hodge–Tate convention $\text{HT}(\varepsilon) = \{-1\}$, exponents are the negatives of Hodge–Tate weights.

We use geometric Frobenius and local reciprocity sending a uniformizer to geometric Frobenius. Thus, as a character of $\mathbb{Q}_2^\times$,

$$\varepsilon = x|x|, \quad x(a) = a.$$

The determinant of a Galois representation is denoted by χ . For a representation in Theorem 1.1, de Rham class field theory gives

$$\chi = \chi_{\text{fin}} \varepsilon^w$$

for a finite-order character χ_{fin} and an integer w . Indeed, Tate’s local algebraicity theorem [52, Chapter III, Section 1.2] gives the integral cyclotomic exponent at 2. Removing this power leaves a character locally constant at every finite place. Finite ramification and global reciprocity make it a character of a finite ray class group, hence of finite order. In the convention for quaternionic forms used below, the central character is

$$\psi = \chi\varepsilon.$$

The normalized local block parameter attached to a Galois representation ρ is that of $\rho \otimes \varepsilon$. These conventions agree with [45, Sections 3.1–3.5].

For the statement of Theorem 1.1, one may use the arithmetic normalization V_f of the representation of a primitive eigenform of weight k and nebentype ψ_f : at a good prime q ,

$$\mathrm{tr} V_f(\mathrm{Frob}_q^{\mathrm{ar}}) = a_q(f), \quad \det V_f = \psi_f \varepsilon^{k-1}.$$

These trace and determinant formulas use the normalization of [20, Theorem 6.1]; arithmetic Frobenius is fixed there in Section 3.1. The cyclotomic exponents of V_f are $\{0, k-1\}$. Evaluating the same representation at geometric Frobenius inverts its eigenvalues; it does not change the representation into its dual. If a construction instead produces $V_f^\vee$, the wedge pairing gives

$$V_f^\vee \simeq V_f \otimes \psi_f^{-1} \varepsilon^{1-k} \simeq V_g \otimes \varepsilon^{1-k},$$

where g is the primitive form associated with the finite-character twist $f \otimes \psi_f^{-1}$. Thus an expression $r \simeq V_f^\vee \varepsilon^n$ becomes $r \simeq V_g \varepsilon^{n+1-k}$. The up-to-Tate-twist conclusion of the main theorem is unchanged by this conversion.

Lemma 2.1. *A continuous finite-dimensional representation of a profinite group on $\overline{\mathbb{Q}}_2^n$ is defined over a finite extension of $\mathbb{Q}_2$ and admits a stable lattice there.*

Proof. There are countably many finite subextensions of $\overline{\mathbb{Q}}_2/\mathbb{Q}_2$. For each one E , the set of group elements whose representation matrices lie in $\mathrm{GL}_n(E)$ is a closed subgroup. These subgroups cover the profinite group. By Baire's theorem, one has nonempty interior, hence is open. Enlarge E to contain the finitely many matrix entries of coset representatives. Then every matrix lies in that finite extension.

The image is compact. Its entries, and the entries of its inverses, have bounded valuations. The $\mathcal{O}$ -submodule generated by the orbit of $\mathcal{O}^n$ therefore lies between two scalar multiples of $\mathcal{O}^n$. It is a stable lattice. $\square$

We apply Lemma 2.1 whenever a characteristic-zero representation is first introduced. Over a curve field, the analogous compact-image lattice argument gives an integral realization over $k'[[t]]$.

2.2 The virtually solvable case

For a representation over any field, after extending scalars to an algebraic closure, call it *virtually solvable* if the identity component of the reduced Zariski closure of its image is solvable. The closure here is taken over the algebraic closure, so that the resulting algebraic group is smooth. This property is unchanged by restriction to a finite-index subgroup.

Lemma 2.2. *Theorem 1.1 holds for virtually solvable representations. A representation satisfying its hypotheses and not virtually solvable remains absolutely irreducible and not virtually solvable over every finite extension of $\mathbb{Q}$.*

Proof. Let H be the algebraic monodromy group of an irreducible two-dimensional representation over an algebraic closure. Lie–Kolchin and the structure of connected solvable groups [43, Theorems 17.4 and 16.33] give the following alternatives. If H° is solvable and has a nontrivial unipotent radical, the latter has a unique fixed line; normality makes that line H -stable, contradicting irreducibility. The remaining possibilities are that H° is scalar, or that its two distinct character lines are permuted by H .

In the scalar case the projective image is finite. After a finite extension of the base field, the representation is a scalar character on a two-dimensional space. Its two Hodge–Tate weights are equal, which is excluded. In the second case the action on the two lines gives a quadratic extension $L/\mathbb{Q}$ and

$$r \simeq \text{Ind}_{G_L}^{G_{\mathbb{Q}}} \theta$$

for a continuous character θ . The character is de Rham and finitely ramified, hence is the 2-adic avatar of an algebraic Hecke character, by the following rank-one consequence of Tate’s theorem and the construction in [52, Chapter III, Sections 1.2 and 2.3]. The integral algebraic exponents on open local units obey the global unit relation. Principal ideals in a suitable ray subgroup therefore receive the prescribed algebraic monomials. A positive power of each of finitely many ray-class representatives is principal in that subgroup, so its character value is algebraic. Adjoining these values places all ideal-character values in one number field and constructs the algebraic Hecke character. This argument does not assume that the given character was initially rational over a number field. If L is real quadratic, its algebraic exponents at the two real embeddings are equal by the unit theorem: a unit of infinite order forces their difference to vanish in the algebraic character relation. This again contradicts regularity. For imaginary quadratic L , the theta-series construction gives the required cuspidal eigenform, with its algebraic Tate normalization; equivalently, use quadratic automorphic induction [28, Proposition 12.1].

Finally, restriction to finite index preserves H° . If a restriction of a non-virtually-solvable representation were reducible, its identity component would lie in a Borel subgroup and be solvable. This is impossible. $\square$

Henceforth the target r is not virtually solvable. In particular, its restriction to every field used below is absolutely irreducible.

2.3 Fields, deformation spaces, and trace loci

All auxiliary base fields F are totally real, solvable over $\mathbb{Q}$, of even degree, and split completely at 2. We may replace them by Galois closures. Let S contain the real places, all places above 2, and a finite set of other places. Set

$$d = [F : \mathbb{Q}], \quad u = \#\{v \in S : v \nmid 2, v \text{ finite}\}, \quad s = 2d + u.$$

The group $G_{F,S}$ is the Galois group of the maximal extension of F unramified outside S .

A two-dimensional pseudorepresentation means a determinant in the sense of Chenevier, with its determinant character retained. In particular, no division by 2 is used to recover a determinant from traces. Fixing χ and a semisimple residual representation gives a complete local pseudodeformation ring, denoted by $R_F^{\text{ps},\chi}$. Framed representation rings and schemes, including those with specified local invariant lines, will be constructed in Section 3. Whenever a component is mapped to a pseudodeformation spectrum, its *image* means the closure of its image unless stated otherwise.

There is a unique quaternion algebra D_F/F , up to isomorphism, ramified at all real places and at no finite place. The evenness of d makes this possible, by the classification by ramification sets [61, Main Theorem 14.6.1]. A completed Hecke algebra of definite $D_F^\times$ -forms, with central character $\psi = \chi\varepsilon$, defines a closed trace locus in $\text{Spec } R_F^{\text{ps},\chi}$. We call such a locus *pro-modular*. The tame level is open, is hyperspecial away from S , and is unrestricted at the places above 2. A closed trace locus is *potentially pro-modular* if its restriction becomes pro-modular after a further solvable totally real extension split completely at 2. All support containments in this terminology are set-theoretic.

A *characteristic-two curve* is a one-dimensional domain quotient of a complete local k -algebra. After finite extension, its normalization has the form $k'[[t]]$; its generic point has residue field

$k'((t))$. We also use the word curve for this generic point and its associated representation when there is no ambiguity. Whenever such a representation is used, we choose a compact-image lattice. Agreement of traces and determinants modulo the maximal ideal determines the semisimple residual representation, but need not determine a residual extension class. The latter distinction will be important in the marked-line connectedness argument.

The reductions now leave a fixed determinant and a target that remains absolutely irreducible after every finite restriction. We next construct the spaces in which its trace, and when needed its local line character, can vary. The distinction between these data is essential: a trace space is sufficient for completed-form occurrence, whereas the final reducible classicality argument also needs the chosen character.

3 Deformations and local geometry

We establish the deformation-theoretic tools used in both propagation and connectedness. The two points requiring particular care are deformation theory at a characteristic-two curve and the geometry of several local conditions imposed simultaneously. We retain determinant characters throughout. In particular, in characteristic two a pseudorepresentation means a two-dimensional determinant, not merely its trace.

There are three outputs. First, finite trace-and-character spaces let us compare dimensions before and after forgetting frames and lines. Second, curve completion gives legitimate continuous local and global deformation problems with a common coefficient field. Third, two distinct local geometries serve different later uses: unrestricted dyadic factors enter propagation, whereas marked scalar factors enter connectedness.

3.1 Representation spaces and formal dimensions

The groups occurring below, namely local Galois groups and the groups $G_{F,S}$ with S finite and containing the dyadic and infinite places, satisfy Mazur's finiteness condition Φ_2 : for every open subgroup H , the set $\text{Hom}_{\text{cts}}(H, \mathbb{Z}/2\mathbb{Z})$ is finite. We use continuous deformation functors. A frame is a specified basis; a local line, when recorded, is part of the deformation data. This is the framed form of the deformation framework of [39].

Proposition 3.1 (Representation spaces). *Let G satisfy Φ_2 , let $\overline{D}$ be a continuous two-dimensional determinant over the finite field k , and fix a continuous lift χ of its determinant character. The following statements hold.*

1. *The determinant deformation functor is represented by a complete Noetherian local $\mathcal{O}$ -algebra R^{ps} . There is a finite R^{ps} -algebra $\mathcal{E}$ carrying its universal Cayley–Hamilton representation. Matrix representations of $\mathcal{E}$ compatible with the universal determinant form an affine scheme of finite type over R^{ps} . Its completion at a closed residual matrix representation is that representation's framed deformation space.*
2. *Invariant lines at finitely many closed subgroups can be imposed by closed incidence conditions in products of projective lines. On the absolutely irreducible global locus, existence of such a line with a specified character is closed in the trace and character parameters.*
3. *Suppose the subgroups' abelianizations have finitely generated maximal pro-2 quotients. Adjoining the characters on the specified lines gives a finite extension of the completed trace-and-determinant image.*

4. *At an absolutely irreducible representation over a coefficient field, determinants classify unframed deformations. After a splitting extension, the framed fiber is a conjugacy orbit of dimension three.*

These assertions do not require the residual determinant to be multiplicity-free.

Proof. For representability under Φ_2 , we use [16, Propositions 3.3 and 3.7, Remark 3.5], with the numbering of the cited arXiv version. Finiteness of the universal Cayley–Hamilton algebra and the algebraization of its compatible matrix representations are [62, Propositions 3.2 and 3.6, Theorems 3.7 and 3.8]. Compatibility here includes the characteristic-polynomial equations; no generalized matrix algebra at a scalar residual point is being assumed. Fixing χ is a base change of these statements.

For clarity, finite type has a direct algebraic explanation. Choose finitely many algebra generators for the finite algebra $\mathcal{E}$, assign a 2×2 matrix to each, and impose the multiplication and characteristic-polynomial identities. These are polynomial equations over the Noetherian base. Completing at a residual matrix point recovers continuity and the chosen residual representation, including its extension class.

Line invariance and the prescribed action on a line are closed equations in the relative projective line. Over the absolutely irreducible locus the Cayley–Hamilton algebra is an Azumaya algebra of rank four; one may split it faithfully flat locally and then descend the proper incidence image. This proves the closedness assertion.

If $g_1, \dots, g_a$ topologically generate the relevant abelian character quotient, every recorded character value $\lambda(g_j)$ satisfies

$$X^2 - T(g_j)X + \chi(g_j) = 0.$$

Modulo the maximal ideal of the trace image, the corresponding parameters are nilpotent after the chosen residual roots have been fixed. The closed fiber of the completed character algebra is therefore finite. Complete Nakayama proves finiteness of that algebra over the trace image. Finite torsion generators cause no change.

Finally, the absolutely irreducible determinant theorem identifies the Cayley–Hamilton algebra with a matrix algebra after a splitting extension [16, Theorem 2.22 and Corollary 2.23]. Two compatible representations are conjugate and their stabilizer consists of scalars. This gives the final assertion. $\square$

In dimension two the fixed-determinant ring is topologically generated by traces. There is no division by two in this statement: for group elements g_j the determinant of a formal sum is

$$D\left(\sum_j t_j g_j\right) = \sum_j t_j^2 \chi(g_j) + \sum_{j < l} t_j t_l (T(g_j)T(g_l) - T(g_j g_l)).$$

This identity also explains why the determinant character must remain part of the data in characteristic two.

For a representation chart with recorded local lines, its completed trace-and-character image is the closed algebra generated by the global traces and the character values on those lines. Passing to this image forgets the frame and the positions of the lines, but retains their characters. Under the abelianization hypothesis of Proposition 3.1(3), forgetting the characters is then a finite map and does not change the dimension of the image. The next lemma bounds the dimension lost in the first passage.

Lemma 3.2 (Formal dimension comparison). *Let X be one of the finite-type representation spaces of Proposition 3.1, including any recorded local lines, over an excellent complete local trace-and-character base R . Let x be a closed point above the closed point of R , and let P be a prime of $\hat{\mathcal{O}}_{X,x}$. Suppose its generic global representation is absolutely irreducible. If $p = P \cap R$, then*

$$\dim(\hat{\mathcal{O}}_{X,x}/P) \leq \dim(R/p) + 3 + \ell,$$

where ℓ is the number of recorded lines not known to have distinct characters generically. A line with distinct characters contributes zero to this bound. The analogous statement holds with any established bound for the relevant generic representation fiber.

Proof. Contract P to a prime q of $\mathcal{O}_{X,x}$. There is a surjection from the completion of $\mathcal{O}_{X,x}/q$ to $\hat{\mathcal{O}}_{X,x}/P$, and completion preserves dimension [58, Tags 02IJ and 07NV]. The finite-type dimension formula over the excellent local base therefore bounds the latter dimension by $\dim(R/p)$ plus the generic-fiber dimension of the contraction. Over an algebraic closure of $\text{Frac}(R/p)$ the representation is unique up to a three-dimensional conjugacy orbit. Each unspecified invariant line has dimension at most one in $\mathbf{P}^1$; with distinct characters its choices are finite. This gives the inequality. Notice that the proof contracts the formal prime before applying the finite-type formula. It does not assert that an arbitrary formal prime itself algebraizes. $\square$

Thus passing from a representation chart to its trace-and-character image loses at most the three frame dimensions and the dimensions of the lines whose character has not separated. The next lemma supplies curves and coefficient points on which these comparisons can be tested.

Lemma 3.3 (Curves and coefficient points). *Let A be a complete Noetherian local domain with finite residue field.*

1. *If A has characteristic two and positive dimension, then its dimension-one domain quotients have generic points dense in $\text{Spec } A$. Their normalizations are $k'[[t]]$ for finite extensions k'/k .*
2. *If A is a flat $\mathcal{O}$ -algebra, continuous maps from A to rings of integers in finite extensions of E are dense on $\text{Spec } A$.*

In either assertion the points can be chosen outside any prescribed proper closed subset.

Proof. Use formal Noether normalization, over k in the first case and over $\mathcal{O}$ in the second; the coefficient and finite normalization facts used here are [58, Tags 032A, 032D, 07QW and 07QV]. Thus A is finite over a power-series subring S . Given $0 \neq f \in A$, integrality supplies a nonzero element of $fA \cap S$: use an integral equation for f with nonzero constant term in the fraction field extension, and clear denominators.

In the first case choose a one-parameter specialization of S on which that element is nonzero. Such a specialization exists by choosing positive weights for the variables and, if necessary, coefficients in a finite residue extension so that the lowest weighted part does not vanish. Its image is finite over $k[[t^a]]$ for some $a > 0$. Lying-over gives a one-dimensional prime of A avoiding f . Excellence makes its normalization finite; a complete equicharacteristic discrete valuation ring with finite residue field has the asserted form.

In the second case choose sufficiently small integral values for the variables on which the selected nonzero series does not vanish. Existence follows successively from the one-variable preparation theorem. A prime above this specialization has a finite extension of E as fraction field. Integrality puts the image of A in its ring of integers; locality and finiteness make the map continuous. Again f does not vanish. Applying this to an element of the ideal of the excluded closed subset proves density. $\square$

3.2 Continuous deformations at a curve

Deformation theory over Laurent-series residue fields, with integral residual bases and finite coefficient orders, is developed in [11, Section 3.5, Proposition 3.34 and Lemmas 3.35–3.37]. We give the completion argument needed here with its global frames and incidence data. Keeping the Laurent topology makes continuous cocycles and obstructions the correct ones.

Let $\rho_* : G \rightarrow \mathrm{GL}_2(K)$ be a continuous curve representation, where $K = k((t))$. After a finite extension of constants and of the curve, choose a stable lattice and write its matrices in $\mathrm{GL}_2(k[[t]])$. Let $\bar{\rho}$ be their reduction at $t = 0$. A restriction of the curve to a local group need not generate all of K from its matrix entries. Introduce a new variable z , send it to t , and carry this same variable through every local and global deformation ring. Their enhanced curve maps then have the common residue field K . The enhanced coefficient ring is

$$\mathcal{L} = \widehat{\mathcal{O}[[z]]}_{(\varpi)}.$$

It is a complete discrete valuation ring with residue field K , and

$$\mathcal{L}/\varpi^a = (\mathcal{O}/\varpi^a)((z)).$$

Finite-length $\mathcal{L}$ -modules carry their Laurent-series topology. In particular K is not given the discrete topology.

Lemma 3.4 (The curve completion). *Let R be the framed deformation ring of $\bar{\rho}$, with the specified determinant and any of the incidence data above. Let $\mathfrak{p}$ be the kernel of the enhanced curve map $R[[z]] \rightarrow k[[t]]$. Then*

$$R_* := (\widehat{R[[z]]})_{\mathfrak{p}}$$

classifies continuous deformations of ρ_ to finite-length local Artinian $\mathcal{L}$ -algebras with residue K . This identification commutes with restriction to local groups, with frames and invariant lines, and with fixing the determinant. The analogous assertion holds for determinants; on the absolutely irreducible locus it agrees with unframed matrix deformation theory.*

If a component of the original complete local ring has dimension e and contains the curve, its enhanced completed localization has dimension e . At a characteristic-zero coefficient point the corresponding assertion uses the usual completion over its finite 2-adic coefficient field, without adjoining z ; a horizontal integral component of dimension e then gives a completed local component of dimension $e - 1$.

Proof. We first prove the integral-order assertion underlying the functorial claim. Let C be one of the Artinian algebras, with maximal ideal $\mathfrak{n}$, $\mathfrak{n}^N = 0$, and $\varpi^a C = 0$. Write

$$A_0 = \mathrm{im}((\mathcal{O}/\varpi^a)[[z]] \rightarrow C).$$

The inverse image of $k[[t]]$ under $C \rightarrow K$ is $A_0 + \mathfrak{n}$. A bounded subset of this inverse image lies in a finite A_0 -order: subtract coefficientwise lifts of its residual entries, choose a finite A_0 -lattice $U \subset \mathfrak{n}$ containing the resulting bounded differences, and put

$$B = A_0 + U + U^2 + \cdots + U^{N-1}. \tag{1}$$

Powers here mean finite module spans of products. Nilpotence makes B an algebra finite over A_0 . Its quotient by $B \cap \mathfrak{n}$ is $k[[t]]$, with nilpotent kernel. Hence B is local with residue field k . The finite local ring B/zB is Artinian, so a power of $\mathfrak{m}_B$ is contained in zB . Its maximal-ideal topology is therefore its finite-module topology; B is complete and compact.

Apply this to the compact set of matrix entries of a continuous lift of ρ_* . These entries have integral residues because the residual basis was chosen above. Their determinants have residual values in $k[[t]]^\times$, and so are units of B . We obtain a continuous representation in $\mathrm{GL}_2(B)$ reducing to precisely $\bar{\rho}$. Negative powers of z in nilpotent entries belong to $B \cap \mathfrak{n}$ and disappear at that closed residual point. There is no uniform lower valuation bound as C varies, nor is one needed. The same construction simultaneously contains any finite collection of local data, and two such orders admit a common enlargement.

Universality gives a map $R \rightarrow B$. Adjoin z using its coefficient value. Elements outside $\mathfrak{p}$ have nonzero residue in K , so are units in C , while $\mathfrak{p}^N$ maps to zero after increasing N . The map factors uniquely through R_* . Conversely, a local map $R_* \rightarrow C$ factors through a finite jet. Write the original formal coordinates as integral lifts of their curve values plus nilpotent increments. Expansion in the increments has only finitely many terms at this jet depth; the residual coordinates, after their constants are removed, lie in (t) . Substitution in each original formal series consequently converges in the Laurent-series topology. It is continuous, and so gives a continuous lift of the universal representation. The two constructions are inverse. They plainly preserve determinant equations, line incidences, and restriction maps. For determinants use their characteristic coefficients, followed on the absolutely irreducible locus by the matrix-algebra identification in Proposition 3.1.

For dimensions, adjoining z increases a component dimension from e to $e + 1$, while the enhanced curve quotient has dimension one: the map on $k[[z]]$ is already onto $k[[t]]$. The dimension formula for complete equidimensional excellent local rings gives height e at the enhanced prime. Localization and completion have dimension e ; see also the dimension and catenarity facts in [58, Tags 00KW, 032C, 0ECF and 07NV]. At a horizontal characteristic-zero coefficient point the integral quotient again has dimension one, but no new variable has been adjoined. Its height and completed dimension are therefore $e - 1$. The deformation functor uses the usual 2-adic topology. $\square$

The restriction to the inverse image of $k[[t]]$ in (1) is essential. For example, the bounded singleton $\{z^{-1}\}$ cannot lie in a finite order having integral residual image.

For a representation V over its coefficient field L , write $h^i(G, V) = \dim_L H_{\mathrm{cts}}^i(G, V)$, omitting G when understood. For global V , write $h_v^i(V) = \dim_L H_{\mathrm{cts}}^i(G_{F_v}, V)$. These are ordinary cohomology dimensions; at real places the duality pairings use Tate cohomology, as specified below.

Lemma 3.5 (Curve-field cohomology). *Let V be a finite-dimensional continuous K -representation of a local Galois group or of $G_{F,S}$. Continuous cohomology is finite dimensional and satisfies local duality, local and global Euler characteristics, and Poitou–Tate duality with arbitrary linear local conditions. The twist in characteristic two is the residual Tate twist. At real places local duality uses Tate cohomology; the h_v^0 appearing in the Euler and Selmer formulas below still denotes the dimension of ordinary invariants. In particular, if F is totally real of degree d and $\dim_K V = r$, then*

$$h_v^0(V) - h_v^1(V) + h_v^2(V) = -r[F_v : \mathbb{Q}_2] \quad (v \mid 2), \quad (2)$$

$$h^0(V) - h^1(V) + h^2(V) = -rd + \sum_{v \mid \infty} h_v^0(V). \quad (3)$$

At a real place the local contribution to a Selmer dimension difference is $\dim L_v - h_v^0(V)$, where $L_v \subset H^1(F_v, V)$. The same formulas, with the usual real-place interpretation, hold for finite 2-adic coefficient fields.

Proof. Choose a stable $k[[t]]$ -lattice $T \subset V$. Continuous cochains with values in T are the inverse limit of the cochain complexes for $T/t^n T$. Finite-level cohomology is finite, so the derived inverse

limits vanish and cohomology commutes with this inverse limit. The resulting compact cohomology modules are finite over $k[[t]]$: their quotients by t inject into the appropriate finite-level cohomology group, and compact Nakayama applies. Every continuous cochain into V has bounded image, since its domain is compact. Thus its cochain complex is obtained by inverting t , and the same is true of cohomology.

Apply finite-coefficient local duality and Poitou–Tate to $T/t^n T$ and the dual torsion lattice, then pass to the inverse limit and invert t . This gives the perfect field-valued pairings, with Tate cohomology at real places, and the exact global sequence. Taking lengths in the finite-level Euler formulas and dividing by n gives the displayed dimensions; bounded torsion contributes zero in the limit. The real-place terms pass through the same calculation and must not be discarded at 2. Intersections with lattices and saturation allow any linear local condition and its annihilator. The finite-coefficient inputs are [40, I, Corollary 2.3, Theorems 2.8, 2.13, 4.10 and 5.1]. The identical lattice argument with an $\mathcal{O}$ -lattice gives the finite-2-adic-field version. $\square$

3.3 Relative presentations

Curve completion has now identified the deformation functors, and the preceding cohomology calculation supplies their tangent and obstruction spaces. We turn these data into a bound on actual relations cutting out the global ring over its local base. At real places the displayed invariant dimensions are ordinary H^0 , while the duality argument uses Tate-modified groups.

Put $D = \text{ad}^0(\rho)$ for a fixed-determinant two-dimensional representation. Its dimension is three, also in characteristic two. Its dual there is $\text{ad}(\rho)/\text{scalars}$, not in general D itself.

Proposition 3.6 (Relative relation bounds). *Let F be totally real of degree d , let S contain the real and dyadic places, and let ρ be a representation over a finite residue field, a curve field as above, or a finite 2-adic field. Use one global frame and a fixed determinant. Let $T \subset S$ consist of all real places and m dyadic places, each isomorphic to $\mathbb{Q}_2$. Over the completed product of the unrestricted local framed deformation rings at T , the global framed ring has a power-series presentation with g relative variables and r relations satisfying*

$$g - r \geq 3 - 6m - h^0(D^\vee(1)). \quad (4)$$

Local quotient conditions and local line charts may then be imposed by base change without changing this bound. This includes replacing the local base by its reduction.

For the variable-determinant problem with independent frames at the places of S , the relative tangent dimension over the local framed base is

$$h_{\text{Sel}}^1(\text{ad } \rho) + \sum_{v \in S} h_v^0(\text{ad } \rho) - h^0(\text{ad } \rho), \quad (5)$$

where the Selmer condition is zero at S and any other prescribed conditions are included in the Selmer group. On the absolutely irreducible locus, forgetting these independent frames has smooth fiber dimension $4|S| - 1$.

Proof. The relative tangent space in the one-frame problem is the kernel of restriction on $Z^1(D)$. We spell out the obstruction bound since it is also used at enhanced curves. Across a small extension, choose continuous matrix lifts with the prescribed determinant. Multiplication errors form a continuous 2-cocycle. Given the specified local lifts, its class belongs to the kernel of restriction on $H^2(D)$. If that class vanishes, correct the matrices to a global representation. Its differences from the chosen local lifts are 1-cocycles, and the second obstruction belongs to the cokernel of restriction

on $Z^1(D)$. Continuous set sections of finite-length coefficient modules suffice: coefficientwise digit sections supply them. The determinant map on invertible matrices is smooth, so determinant correction needs no division by two.

Consequently the relation number is at most the sum of these two obstruction dimensions. One can verify that this bounds actual minimal relations as follows. For a minimal relative presentation $H \twoheadrightarrow R$ with kernel I , test the small extension

$$H/(\mathfrak{m}_H I + \mathfrak{m}_H^j) \longrightarrow H/(I + \mathfrak{m}_H^j)$$

for j large enough that its kernel is $I/\mathfrak{m}_H I$. The two obstruction maps must jointly detect its dual relation directions. Indeed, quotient the kernel onto an undetected one-dimensional direction. Vanishing of both obstructions would give a lift compatible with the local base. Its relative coordinates differ from the tautological ones by elements of the small kernel, which is killed by $\mathfrak{m}_H$. Changing these coordinates therefore alters a relation only through its residual linear terms in the relative variables. Those terms vanish by minimality, so the nonzero surviving relation cannot become zero. This contradicts the existence of the lift and proves the relation bound for the complete rings.

Poitou–Tate bounds the first obstruction dimension by

$$h^2(D) - \sum_{v \in T} h_v^2(D) + h^0(D^\vee(1)).$$

Subtracting it and the Z^1 cokernel from the tangent dimension gives

$$\begin{aligned} g - r &\geq 3(1 - |T|) - (h^0 - h^1 + h^2)(D) \\ &\quad + \sum_{v \in T} (h_v^0 - h_v^1 + h_v^2)(D) - h^0(D^\vee(1)) \\ &= 3(1 - d - m) + 3d - 3m - h^0(D^\vee(1)). \end{aligned}$$

For the last equality, the real terms in (3) cancel the real local terms; the dyadic Euler characteristic is -3 . This proves (4) in both coefficient characteristics. Base change carries the same presentation to the extra local conditions.

For independent frames, first choose a global cohomology class vanishing at the prescribed local places. Local choices of trivialization contribute the local invariant spaces, and a common global infinitesimal automorphism is quotiented out. This gives (5). If the global representation is absolutely irreducible, the common automorphisms are exactly the scalars; the quotient of the $|S|$ frame spaces therefore has dimension $4|S| - 1$. $\square$

Corollary 3.7 (A characteristic-zero line chart). *For a two-dimensional representation of $G_{\mathbb{Q}_2}$ over a finite 2-adic field with a chosen invariant line and fixed determinant, every component of its framed line-deformation chart has dimension at least five. The odd fixed-determinant real chart has dimension two. In particular, over $\mathbb{Q}$, the relative bound for these two local charts is -3 whenever $h^0(D^\vee(1)) = 0$.*

Proof. Hold the line fixed in an adapted basis. The coefficient Lie algebra is the trace-zero upper triangular algebra of dimension two. The local Euler formula gives $\dim Z^1 - \dim H^2 = 4$, so the relation bound gives local dimension at least four. Moving the line adds one formal variable. At a real place the odd involution has two distinct eigenvalues in characteristic zero and its conjugacy orbit has dimension two. The final assertion is (4) with $d = m = 1$. $\square$

3.4 Small pseudodeformation loci and control of inertia

Lemma 3.8 (Virtually solvable loci). *Let F be totally real of degree d , let S contain its dyadic and infinite places, and fix the determinant. In the characteristic-two pseudodeformation space, the irreducible dihedral curve points lie in a closed subset of dimension at most d . Finite-projective-image curve points give only the residual pseudorepresentation. If $F/\mathbb{Q}$ is abelian, the reducible locus is closed and has dimension at most one.*

Proof. There are only finitely many quadratic extensions of F unramified outside S . This follows from Hermite–Minkowski, since fixed local degree bounds the finitely many possible local discriminants. For each such extension F'/F , use the character deformation space of $G_{F',S}$ with the product of the character and its conjugate fixed. Class field theory identifies its free pro-2 parameters with a quotient of the dyadic local units; non-dyadic unit groups and class groups contribute only finite groups. The dyadic unit rank over F' is $2d$, and its minus rank for the quadratic involution is d . Fixing the product fixes the plus part up to finite groups: multiplication by two is an isogeny of the relevant free $\mathbb{Z}_2$ -modules even though two is not a unit. Thus the reduced special character space has dimension at most d .

Its map to pseudorepresentations has finite image algebra. Indeed the finitely many generating character values satisfy the characteristic polynomials of the induced traces, and complete Nakayama applies as in Proposition 3.1. Its image is consequently closed and has dimension at most d . Taking the finite union gives the dihedral bound. The argument applies to curve points after finite coefficient and curve extensions, and Lemma 3.3 gives the asserted closed-locus formulation.

If the projective image is finite, each scalar in the representation has square fixed by its determinant. The determinant has finite image in the special fiber, so the scalar image, and hence the whole image, is finite. Its traces are algebraic over a finite field. An element of $k'[[t]]$ algebraic over a finite field is constant; its constant reduction is already prescribed. Thus these traces give exactly the residual pseudorepresentation.

Finally, a reducible determinant is a pair of characters with fixed product, so one character determines the other. For an abelian totally real F , Brumer’s theorem on Leopoldt’s conjecture [15], in the form proved in [25, Theorem 1 and Proposition 8, pp. 9-08–9-09 and 9-14], gives free pro-2 character rank one. The same finite character-image argument therefore gives a closed locus of dimension at most one. $\square$

Lemma 3.9 (Inertia and solvable base change). *Consider finitely many complete deformation or pseudodeformation families over finite-residue coefficient rings, and a finite set of places not above two. After a fixed finite local extension at each place, all their field-valued representations have unipotent inertia. For a characteristic-two curve in such a family whose determinant has finite image, imposing the residual value of a Frobenius trace makes the local image finite.*

Any further finite local extensions needed to kill these finite images, to kill the finitely many inertial types, to make the fixed determinant unramified, or to arrange residue cardinalities congruent to 1 modulo 4, can be realized inside a totally real solvable base change which splits completely at the dyadic places. One may take its Galois closure over $\mathbb{Q}$. New ramification of the base field away from the original allowed set creates no new ramification of the restricted representations.

Proof. Use the finite universal Cayley–Hamilton algebra $\mathcal{E}$ for each family. Its reduction modulo the base maximal ideal is finite. After killing that finite residual group image locally, the group acts through the pro-2 group $1 + \mathfrak{m}\mathcal{E}$. At a place of residue characteristic $\ell \neq 2$, wild inertia and the non-2 part of tame inertia therefore act trivially. Write τ for the remaining tame generator and ϕ for arithmetic Frobenius, so that

$$\phi\tau\phi^{-1} = \tau^q.$$

At any field-valued point the multiset of eigenvalues of τ is stable under the q th-power map. Each eigenvalue consequently has order dividing $q^2 - 1$. A further fixed tame extension kills this semisimple inertia at every point, leaving unipotent inertia. The same finite extension works for the finite list of source families.

In characteristic two a unipotent 2×2 matrix has square one. After these extensions the inertia image is therefore finite: it is generated by the image of the single pro-2 tame generator. The original inertia image is finite as well, since the inertia subgroup upstairs has finite index in the original one. Now use the Frobenius chosen in the original local group, whose trace is prescribed by the curve constraint. Its characteristic polynomial has coefficients in a finite field, since that trace is the residual constant and the determinant has finite image. Its eigenvalues have finite order, and its possible Jordan part has order at most two. This original Frobenius consequently has finite order. The full local image is an extension of the finite inertia image by the finite quotient generated by Frobenius, and is therefore finite.

Every finite Galois group over a nonarchimedean local field is solvable: wild inertia is an ℓ -group, and the successive tame inertia and residue quotients are cyclic. A composition tower therefore realizes each required local extension through cyclic prime-degree steps. At each step the prime-degree case of the Grunwald–Wang theorem [42, Chapter VIII, Theorem 2.4] prescribes the finitely many local extensions while imposing splitting at dyadic and real places. For degree two this is simply weak approximation for squareclasses; no higher-power Grunwald–Wang assertion is needed. Composita and Galois closure preserve solvability, total reality, and complete dyadic splitting. An unramified local extension of even degree makes an odd residue cardinality 1 modulo 4. The determinant’s finite non-dyadic inertia can be killed at the same time.

Finally, inertia in the enlarged field maps into inertia downstairs. A representation unramified at a place outside the original allowed set remains unramified after restriction, even if the field extension itself ramifies there. This proves the last assertion. $\square$

The unipotent assertion may be carried to closures of restricted pseudorepresentation images. On the globally absolutely irreducible locus this can be checked by characteristic polynomials and by traces against a matrix-algebra basis of group elements. For example, the equations

$$\mathrm{tr}((\rho(\tau) - 1)^2 \rho(g)) = 0$$

for such a basis force $(\rho(\tau) - 1)^2 = 0$ by the perfect trace pairing on M_2 . Thus these are closed identities on the locus where the corresponding trace Gram matrix is invertible. This justifies the set-theoretic use of the uniform inertia assertion on the source-image closures below.

3.5 The local product used in propagation

We now define the local model independently of any patched global ring. Fix d real factors and d dyadic factors, the latter all over $\mathbb{Q}_2$. At each use the prescribed fixed determinant, odd at the real places. Start with integral residual bases over a common finite field, take the completed product of the framed local rings, and form its enhanced completion at any chosen curve as in Lemma 3.4. Denote the resulting $\mathcal{L}$ -algebra by A^{rd} .

We enlarge $\mathcal{O}$ once so that it contains i with $i^2 = -1$. Choose u odd residue cardinalities $q_j \equiv 1 \pmod{4}$, and $g \geq 0$. Put

$$A = A^{\mathrm{rd}}[[X_1, \dots, X_g]].$$

For $\eta \in \{1, i\}$ define B_η by adjoining at each of the u places two formal invertible matrices τ_j, ϕ_j , both reducing to the identity at the curve, and imposing

$$\mathrm{tr} \tau_j = \eta + \eta^{-1}, \quad \det \tau_j = 1, \quad \phi_j \tau_j \phi_j^{-1} = \tau_j^{q_j}. \quad (6)$$

There is no condition on $\det \phi_j$. This is the raw completed matrix quotient: we do not silently replace it by its reduction. At characteristic-zero field points the i -condition gives the two distinct inertia eigenvalues $i, -i$, whereas the 1-condition makes inertia unipotent. Both i and $-i$ reduce to one in characteristic two. The two conditions thus have identical special-fiber equations but different characteristic-zero component geometry, described below.

Proposition 3.10 (Local component geometry). *The ring A^{rd} is a flat complete-intersection normal domain over $\mathcal{L}$, of dimension $1 + 8d$, and its special fiber is a normal domain of dimension $8d$. Set*

$$e = 1 + 8d + 4u + g.$$

Then the following statements hold for the model (6).

1. *$\text{Spec } B_i$ is irreducible of dimension e , and its unique irreducible component is horizontal.*
2. *The two rings have a canonical common special fiber, compatible with their universal matrices and the variables $X_1, \dots, X_g$. Its irreducible components have dimension $e - 1$ and are indexed by a choice, at each tame place, of the unramified component or the closure of the nontrivial-unipotent component.*
3. *The irreducible components of $\text{Spec } B_1$ are horizontal and have dimension e . They are indexed by the same choices. The specialization of each has precisely its corresponding special-fiber component as underlying support. In particular the generic point of a special-fiber component lies on exactly one component of $\text{Spec } B_1$.*

The assertions remain valid after finite extension of the constants and curve field and the resulting enhanced completion. No condition on the dyadic curve representations is required.

Proof. *The unrestricted real and dyadic base.* We first establish the assertion about A^{rd} . An odd real framed ring is given by trace zero and determinant -1 . These equations are equivalent to the involution relation with that determinant, by Cayley–Hamilton and invertibility. At the scalar residual point it has the form

$$\mathcal{O}[[a, b, c]]/(a^2 + 2a + bc).$$

Its special fiber $a^2 + bc = 0$ is a geometrically normal surface: it is an integral hypersurface whose singular locus has codimension two [58, Tag 031S]. At a non-scalar residual involution one obtains a smooth chart of the same relative dimension two.

For a dyadic factor, the fixed-determinant framed deformation theorem [11, Theorem 1.5] gives a flat complete-intersection normal domain of dimension seven and a normal special-fiber domain of dimension six. The theorem applies at 2, including scalar and reducible residual representations; the prime restriction on a different density theorem is not being used. The same statement holds after each finite coefficient extension.

The completed tensor product of these special-fiber rings over the finite, hence perfect, field is a normal domain. Here we use the completed tensor-product normality theorem [55, Theorem 3.3 and Proposition 3.1]; the residue tensor product is the field itself, so the product is local and connected. Its dimension is $8d$. The mixed-characteristic completed product is a flat complete intersection: concatenate presentations, first checking the regular sequences modulo ϖ , and then lift them together with the nonzerodivisor ϖ . Its special fiber is the normal domain just obtained, and the regular-element normality criterion makes the product normal. It is a domain as well: the lowest nonzero ϖ -adic terms of a putative product of two nonzero elements could not multiply to zero in the special fiber.

Adjoining z , localizing at the enhanced curve, and completing preserve normality by excellence [58, Tags 07QW and 0C23]. They preserve the asserted dimensions by Lemma 3.4, and the same applies to the special fiber. The element ϖ remains a nonzerodivisor, so the enhanced ring is flat over the discrete valuation ring $\mathcal{L}$. This proves the assertion about A^{rd} and also shows why no special dyadic hypothesis is needed. The construction can be repeated after any of the finite extensions in the statement.

The integral incidence for the i -type. We prove the tame assertions using proper incidences, keeping their images distinct from the incidences themselves. In the i -case, Cayley–Hamilton gives $\tau_j^2 = -1$, hence $\tau_j^{q_j} = \tau_j$. Adjoin a ϕ_j -stable i -eigenline for each τ_j . This gives a closed proper incidence Y_i in a product of projective lines over $\text{Spec } B_i$. In an adapted basis on a slope chart, its matrices are

$$\tau_j = \begin{pmatrix} i & n_j \\ 0 & -i \end{pmatrix}, \quad \phi_j = \begin{pmatrix} 1 + d_j + x_j & h_j \\ 0 & 1 + d_j \end{pmatrix},$$

and their sole remaining equation is

$$x_j n_j = 2ih_j. \tag{7}$$

The line slopes are free chart coordinates.

Here is a completion argument which also works for all places at once. On a product of affine slope charts put

$$C = A[\widehat{s_1, \dots, s_u}]^{m_A}.$$

This is completion along the entire closed fiber, not at a selected slope origin. Restricted coefficient series embed it into $A[[s_1, \dots, s_u]]$, so C is a domain. Exactness of completion identifies $C/\varpi C$ with the corresponding restricted-series ring over $A/\varpi A$, also a domain. Completion of the incidence chart along that fiber is

$$C[[x_j, n_j, h_j, d_j : 1 \leq j \leq u]] / (x_j n_j - 2ih_j)_{j=1}^u. \tag{8}$$

Reduce each monomial by replacing paired factors $x_j n_j$ by $2ih_j$ until they never occur together. Assign weights 1, 1, 2 to x_j, n_j, h_j . Reduction preserves weight, so it converges in the power-series topology and yields a normal form. The difference between a monomial and its reduction is a sum of the relations (7) times series whose weights are lower by two; these series converge as well. The substitution

$$h_j \mapsto x_j n_j / (2i)$$

embeds these normal forms into $\text{Frac}(C)[[x_j, n_j, d_j : 1 \leq j \leq u]]$. Indeed the two output exponents recover the exponent of h_j as their minimum, and each output coefficient receives only finitely many input terms. This proves uniqueness of the normal form and shows that (8) is a domain. This argument needs no uniform bound on denominators. It also shows that the successive relations are nonzerodivisors and gives dimension $\dim A + 4u = e$ on the incidence.

We explain why chart completions suffice. At every point of the closed fiber, localization of the chart maps faithfully flatly to a localization of its completed chart. The latter is a domain, so the incidence is locally integral there. Every irreducible component of a proper scheme over a local base meets the closed fiber; so does any nonempty closed intersection of components. Local integrality therefore makes the components disjoint. The closed fiber is $(\mathbf{P}_K^1)^u$, which is connected, hence there is only one component. A nonempty nilradical support would also meet the closed fiber, so the incidence is reduced. Thus Y_i is integral.

After inverting two the eigenline is uniquely given by the idempotent $(\tau_j + i)/(2i)$, and the incidence projection is an isomorphism. In characteristic two, commuting matrices have a common

eigenline after algebraic extension, so the proper projection covers every geometric point as well. Its image is therefore all of $\text{Spec } B_i$ set-theoretically. The integral incidence is horizontal and generically finite over that image. This proves assertion (1). It does not assert that the raw ring B_i is reduced.

The horizontal unipotent patterns. For $\eta = 1$, write $N_j = \tau_j - 1$. The equations give $N_j^2 = 0$ and $\phi_j N_j = q_j N_j \phi_j$. At a field point either $N_j = 0$, or its kernel and image are the same line. The first alternative is the unramified factor, with ϕ_j unrestricted. For the closure of the second alternative use the proper line incidence with adapted matrices

$$\tau_j = \begin{pmatrix} 1 & n_j \\ 0 & 1 \end{pmatrix}, \quad \phi_j = \begin{pmatrix} q_j(1 + d_j) & h_j \\ 0 & 1 + d_j \end{pmatrix}. \quad (9)$$

At a nonzero N_j the relation forces exactly this diagonal ratio. Its chart has three formal matrix coordinates and one slope.

Choose either alternative at each place. The resulting proper incidence has free completed slope charts over A , so the same closed-fiber argument proves that it is integral, horizontal, and of dimension e . Its projection is an isomorphism where every chosen ramified N_j is nonzero, because the line is then $\text{im } N_j = \ker N_j$. Each image is consequently an integral horizontal closed subset of dimension e . These finitely many images cover all geometric points of $\text{Spec } B_1$.

Specialization and uniqueness of the containing component. The dimension calculation alone is not enough for support transfer: we must know exactly which horizontal component contains each generic special-fiber point. It remains to identify their specializations and distinguish them. The two raw presentations reduce to the same one modulo ϖ , since $i \equiv 1$ and q_j is odd. At one place write

$$\tau = \begin{pmatrix} 1 + a & b \\ c & 1 + a \end{pmatrix}, \quad \phi = \begin{pmatrix} 1 + d & e' \\ f & 1 + g' \end{pmatrix}.$$

The common special matrix factor is

$$K[[a, b, c, d, e', f, g']]/(a^2 + bc, b(d + g'), c(d + g'), bf + ce'). \quad (10)$$

Its two reduced components are

$$U : a = b = c = 0, \quad R : g' = d, \quad a^2 + bc = bf + ce' = 0.$$

Both have dimension four and are geometrically integral. For R , prove this first in the homogeneous polynomial quotient. On $b \neq 0$ one eliminates $c = a^2/b$ and $f = a^2 e'/b^2$, obtaining a domain. The locus $b = 0$ has dimension three, whereas every component of the two-equation quotient has dimension at least four. There is thus only one component, generically reduced. The quotient is a complete intersection and has no embedded primes, so it is a domain. Completion is the product of its finite homogeneous pieces; lowest-degree nonzero terms show that it remains a domain. The same proof works over every field extension. This argument is made before completion; it does not substitute into arbitrary formal series after inverting b .

For several places over $A/\varpi A$, the polynomial products of the chosen U and R factors are flat over that base and have geometrically integral generic fiber. They are domains, and the same homogeneous completion argument applies, completing the base coefficients and then the positive-degree variables. Each pattern therefore gives an integral special-fiber component of dimension $e - 1$. Reduction of its proper incidence is integral and maps onto precisely this component. Proper surjectivity onto the image is preserved by base change, so this is the entire underlying specialization of the corresponding horizontal image.

The patterns are distinct. On the generic ramified special piece $\tau_j \neq 1$, whereas every unramified piece has $\tau_j = 1$. On the generic unramified piece $\text{tr } \phi_j \neq 0$, whereas every ramified special piece has $\text{tr } \phi_j = 0$. Thus no horizontal image contains the generic special point of a different pattern. The images already cover the raw spectrum, proving assertions (2) and (3), including uniqueness of the horizontal component through each special-fiber generic point. $\square$

Remark 3.11. The common raw special fiber need not be reduced. In (10), the element $a(d + g')$ can be nonzero with square zero. The proper incidence is also not a normalization: over $\tau = \phi = 1$ its fiber contains $\mathbf{P}^1$. Our conclusions concern precisely the spectra, components, and universal matrix actions required for support propagation, and do not make either of these stronger assertions.

3.6 Several marked scalar places

The marked local condition used in connectedness is different from the unrestricted dyadic condition. At a scalar residual dyadic representation, record an invariant line and retain no other local restriction. We need the reduced characteristic-two local base, jointly with the real factors, to be integral. Connectedness will use this entire completed product as a domain, so integrality of each individual dyadic factor does not suffice.

Lemma 3.12 (Simultaneous marked factors). *Over a finite characteristic-two field, take d fixed-determinant real framed factors and m dyadic framed factors whose residual matrices are scalar, with one invariant line recorded at each dyadic factor. The determinant is fixed, and after the residual scalar twist its dyadic reduction is one. The reduction of their completed local product is an integral complete-intersection base of dimension*

$$2d + 5m.$$

It remains integral after a finite extension of the residue field. Equivalently, one may first kill the nilpotent torsion parameters in the diagonal character rings; the resulting completed product is already reduced and integral and has the same spectrum as the original product.

Proof. Move a marked line in its one-dimensional formal chart and choose the corresponding adapted basis. After the scalar twist, the diagonal entries are characters λ, λ^{-1} congruent to one. Their reduced character ring is

$$C_0 = k[[x, y]].$$

Indeed $\mathbb{Q}_2^\times$ has maximal pro-2 quotient $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}_2^2$, and the first factor contributes only a nilpotent character parameter in characteristic two.

The maximal pro-2 Galois group of $\mathbb{Q}_2$ has three generators and one relation. The odd-degree dyadic case was established by Serre [50, Sections 4.4 and 8]; in Labute's classification, the $q = 2$, odd-local-degree case gives generators with relation $g_1^2 g_2^4 [g_2, g_3] = 1$ [37, §5, Theorem 8]. If $b(g)$ is the upper-right entry of the triangular matrix with diagonal entries $\lambda(g), \lambda(g)^{-1}$, put $c(g) = \lambda(g)b(g)$. Matrix multiplication gives the left-cocycle identity

$$c(gh) = c(g) + \lambda(g)^2 c(h).$$

Its values $u_j = c(g_j)$ satisfy one homogeneous linear equation

$$a_1(x, y)u_1 + a_2(x, y)u_2 + a_3(x, y)u_3 = 0. \tag{11}$$

All these deformations factor through the maximal pro-2 quotient: their reduction is the identity and their finite coefficient quotients have pro-2 congruence image. Thus this presentation does

describe the complete triangular deformation ring after the diagonal reduction, rather than just a subfunctor of it.

Write $B = 1 + x = \lambda(g_2)$ and $C = 1 + y = \lambda(g_3)$. The reduced diagonal relation gives $\lambda(g_1) = B^{-2}$. Using $[g_2, g_3] = g_2^{-1} g_3^{-1} g_2 g_3$ and this left-cocycle identity, direct substitution gives

$$(a_1, a_2, a_3) = \left(\frac{x^4}{B^4}, \frac{x^6}{B^8} + \frac{y^2}{B^2 C^2}, \frac{x^2}{B^2 C^2} \right).$$

Multiply (11) by the unit $B^2 C^2$ and set

$$v = u_3 + \frac{x^2 C^2}{B^2} u_1 + \frac{x^4 C^2}{B^6} u_2.$$

This triangular formal change of variables transforms the relation into $x^2 v + y^2 u_2 = 0$. Thus the coefficient ideal is exactly (x^2, y^2) and its radical is (x, y) ; the ideal itself is not radical.

Let A_∞ be the completed real special-fiber product. The real calculation in Proposition 3.10 makes it an excellent Cohen–Macaulay domain of dimension $2d$. Adjoin the independent pairs x_j, y_j and the line variables s_j , obtaining

$$A_0 = A_\infty[[x_j, y_j, s_j : 1 \leq j \leq m]].$$

Before completing the upper entries, form the graded algebra

$$P = A_0[u_{jr} : 1 \leq j \leq m, 1 \leq r \leq 3] / (f_1, \dots, f_m), \quad (12)$$

where each f_j is the corresponding relation (11). The coefficients of f_j involve only x_j, y_j and have common-zero locus (x_j, y_j) .

On a base stratum where exactly r coefficient rows vanish, the base loses $2r$ dimensions and the upper-entry fiber gains only r dimensions. Thus the corresponding locus in $\text{Spec } P$ has dimension at most

$$\dim A_0 + 2m - r.$$

Every minimal prime of the homogeneous relation ideal is homogeneous and is contained in $(\mathfrak{m}_{A_0}, u_{jr})$. In that excellent equidimensional local ambient ring, the height theorem for the m equations makes every component have dimension at least $\dim A_0 + 2m$. No component can therefore lie over a nonempty rank-drop stratum. Where all coefficient rows are nonzero, the quotient is a vector bundle of rank $2m$ over an integral base. There is consequently only one component, generically reduced.

The same argument for every initial subset of the equations shows that their heights are the expected ones. They form a regular sequence in the Cohen–Macaulay ambient ring. The quotient has no embedded associated primes, and generic reducedness now proves that P is a domain.

Finally, each homogeneous piece P_n is a finite, torsion-free A_0 -module. Over $F_0 = \text{Frac}(A_0)$ the m independent linear relations eliminate one upper-entry variable in each triple, so

$$\prod_{n \geq 0} P_n \hookrightarrow \prod_{n \geq 0} (P_n \otimes_{A_0} F_0) \simeq F_0[[v_1, \dots, v_{2m}]]$$

is an embedding of rings. Since A_0 and its finite modules are complete, the left side is precisely the completed local product in question. It is a domain, of dimension $\dim A_0 + 2m = 2d + 5m$. It is also a complete intersection by the regular sequence already exhibited.

Killing the original diagonal torsion parameters changes no prime ideal: each is nilpotent, and the ideal they generate is nilpotent. The obtained quotient is already reduced, proving the assertion about the original reduction. The real normality and the cocycle and dimension arguments hold after every finite constant extension, proving the final assertion. $\square$

The two local outputs will be used separately. Propagation uses the unrestricted real-dyadic base and the correspondence between the horizontal tame components and their common special fiber. Connectedness uses the integral marked base of dimension $2d + 5m$. The latter statement includes no normality assertion, and neither construction replaces the raw tame rings by their incidences.

4 Completed forms and their local parameters

In this section F, S, χ, ψ have the meaning fixed in Section 2. We construct completed forms and identify their integral local-block action with the restriction of the global Galois action. This comparison gives finiteness over the local pseudodeformation base and describes the Banach coefficient fibers needed for classicality. We then prove a lower bound on every Hecke component and exact auxiliary-level identities preserving the old deformation action; these are the inputs to patching. All local groups at 2 are $\mathrm{GL}_2(\mathbb{Q}_2)$, since 2 splits completely in F .

4.1 Integral forms and classical density

Fix an open tame level U^2 , hyperspecial outside S . We may also prescribe a finite-order transformation character at the tame level, compatible with the central character. Write $\tilde{S}_\psi(U^2, \mathcal{O})$ for the continuous $\mathcal{O}$ -valued functions on

$$D_F^\times \backslash (D_F \otimes_F \mathbb{A}_F^\infty)^\times$$

with the prescribed right tame transformation and central transformation $f(gz) = \psi(z)f(g)$. Its scalar extension to E is a unitary Banach representation of

$$G_2 = \prod_{v|2} \mathrm{GL}_2(F_v).$$

The torsion version has values in $E/\mathcal{O}$ and is smooth. The compact dual of the integral space is denoted by M ; equivalently, it is the Pontryagin dual of the torsion space. We also use this notation for a residual Hecke summand.

Proposition 4.1. *For the tame level U^2 , central character $\psi = \chi\varepsilon$, and full compact forms module M just defined, the following properties hold. In the second assertion weights and local parameters use the normalization of Section 2.*

1. *For a sufficiently small product K of uniform open subgroups of $\mathrm{SL}_2(F_v)$, the full compact module M is finite free over $\mathcal{O}[[K]]$. The integral and torsion function spaces are admissible and commute with reduction of coefficients.*
2. *Locally algebraic vectors are classical definite quaternionic forms of the corresponding algebraic weight. Their regular noncharacter eigenpackets transfer to Hilbert cuspidal eigenforms. They have Galois representations of determinant χ , with the local compatibility described below.*
3. *The regular classical vectors are dense in the Banach space. They remain dense if, at each $v \mid 2$, one fixes a smooth principal-series type with two distinct inertial characters and varies the algebraic factor with the prescribed central degree.*
4. *Every residual local factor $\mathbb{T}$ of the completed Hecke algebra away from S is a complete quotient of $R_F^{\mathrm{ps}, \chi}$. It is reduced and $\mathcal{O}$ -torsion-free, and its spectrum is the closure of its classical coefficient points.*

Proof. Finiteness of quaternion ideal classes gives finitely many double cosets at compact open level [61, Main Theorem 17.7.1]; passing to a smaller compact level has finite fibers. Stabilizers modulo the center are finite as well. Their reduced norms have even valuation at every finite place. The group

$$\{a \in F^\times : v(a) \in 2\mathbb{Z} \text{ for every finite } v\}/(F^\times)^2$$

is finite: the class of the half-valuation ideal maps it to $\text{Cl}(F)[2]$, with kernel given by unit squareclasses, and both groups are finite [41, Theorems 4.4 and 5.1]. The kernel of reduced norm modulo squares has, after scalar rescaling, norm-one representatives in a fixed compact subgroup; these are finite by total definiteness [61, Lemma 17.7.13]. Shrink K until it meets none of these finite projective stabilizers nontrivially. Choosing local sections for the central and tame transformations identifies the function space with continuous functions on a finite disjoint union of copies of K . Its compact dual is a finite direct sum of $\mathcal{O}[[K]]$. This also proves admissibility and exact coefficient reduction. Compatibility of the transformation characters on their intersections is part of the definition; the central character is totally even, since both χ and ε are odd.

The identification of locally algebraic vectors is the definite double-coset description of algebraic automorphic forms. Jacquet–Langlands transfers the regular noncharacter packets to Hilbert cusp forms [28, Theorems 14.4 and 16.1]; their Galois representations, weights and de Rhamness are given by [4, Theorem 2.1.1]. At the dyadic places we use the potentially semistable comparison, including ramified types: [3, Corollary 1.2]. For the principal types used here, the two distinct inertial characters force monodromy to be zero. Thus the local representation is potentially crystalline and crystabelian, with the weights specified by the algebraic factor.

We spell out the density assertion with fixed type. Let $K_0 = \prod_{v|2} \text{GL}_2(\mathcal{O}_{F_v})$. Matrix coefficients of algebraic representations of central degree zero form a subalgebra of the continuous functions on $K_0/Z(K_0)$. They contain constants and separate points: the adjoint representation is faithful on the projective group. Nonarchimedean polynomial approximation, equivalently Stone–Weierstrass for this compact totally disconnected space, makes this algebra dense.

Choose a nonzero smooth type τ and an algebraic representation V with the needed central degree. The matrix coefficients $a_1, \dots, a_N$ of $\tau \otimes V$ have no common zero, since every representation matrix is invertible. For a continuous section f with the prescribed central transformation, choose a finite clopen partition subordinate to sets on which some a_i is bounded away from zero. The functions f/a_i on these clopens, extended by zero, are continuous of central degree zero. Approximate them by the algebra just described and multiply by a_i . Every resulting product is a matrix coefficient of $\tau \otimes V \otimes A$, where A is algebraic of central degree zero. Complete reducibility in characteristic zero decomposes $V \otimes A$ into algebraic irreducibles of exactly the same central degree. The smooth factor τ is unchanged. Averaging over the finitely many projective stabilizers gives density in the double-coset function space; its fixed denominators cause no problem over E .

At every dyadic factor we can choose the two inertial characters to be distinct with the required product, using characters of sufficiently large 2-power order. The associated compact principal-series type forces the smooth local representation to be principal series of that type. Here we use Henniart’s compact types [14, Appendix A, Sections A.1.1–A.1.4 and A.2.3–A.2.4], whose scope includes residue cardinality two. In particular it excludes norm characters. The algebraic central degree is common at all embeddings, as required for classical weights. This proves the assertion about a dense span of regular cuspidal packets.

The integral Hecke algebra and its residual factors. Form the Hecke algebra as the closure of the integral Hecke operators in the endomorphisms of the finite Iwasawa module. Reduction modulo its maximal augmentation ideal gives a finite-dimensional commuting algebra. Its topologically nilpotent kernel permits lifting idempotents, giving finitely many residual factors.

Interpolation as an identity of integral operators. We next construct a continuous Galois trace in this integral Hecke algebra. To pass from classical traces to actual operators, we first specify finite tests for the operator topology. Work with the full compact module $M \simeq \Lambda_K^r$, where $\Lambda_K = \mathcal{O}[[K]]$ and $\mathfrak{a} = (\varpi, \ker(\Lambda_K \rightarrow \mathcal{O}))$. The topology on $\text{End}_{\Lambda_K}(M)$ is its $\mathfrak{a}$ -adic matrix topology. An operator A is zero at depth j precisely when $A(M) \subseteq \mathfrak{a}^j M$, or equivalently when its dual kills the finite $\mathcal{O}$ -module

$$Q_j = \text{Hom}_{\mathcal{O}}(M/\mathfrak{a}^j M, E/\mathcal{O}).$$

Choose finitely many generators of Q_j . Exact coefficient reduction in the function model represents them by $\varpi^{-e_i} f_i$ modulo integral functions, with each f_i integral. Thus the finitely many tests $A^\vee f_i \in \varpi^{e_i} \tilde{S}_\psi(U^2, \mathcal{O})$ detect depth j .

Approximate these f_i , to the required precisions, by finite sums of classical eigenvectors. Clear the denominators of this finite collection once. Continuity of the finitely many classical Galois traces then makes the corresponding Hecke operators agree to those precisions on the approximations whenever their Frobenius conjugacy classes are sufficiently close. Integral Hecke operators and their differences have norm at most one, so the approximation errors are controlled uniformly. Hence a Chebotarev sequence approaching any Galois conjugacy class is Cauchy at every depth j . Completeness gives a unique integral limit, independent of the sequence; the same finite tests prove continuity of the resulting trace function. Restriction to the integral Hecke-idempotent summands gives the assertion for each residual factor, with its inherited operator topology.

The pseudodeformation quotient and its support. The determinant is the fixed character χ . The polynomial identities for a two-dimensional determinant hold on all classical packets and hence on their dense span. Boundedness and torsion-freeness make these identities integral. The universal property gives a continuous map from $R_F^{\text{ps}, \chi}$ onto the completed trace algebra: the Frobenius traces generate its dense subalgebra, and the source is compact.

An operator vanishing at every classical eigenpacket vanishes on the dense classical span, hence is zero. Classical Hecke actions are semisimple. A nilpotent operator therefore vanishes there, proving reducedness; the action on integral functions proves $\mathcal{O}$ -torsion-freeness. The same test proves the asserted closure of the classical coefficient points. $\square$

4.2 Blocks, integral finiteness, and Banach fibers

Fix a residual factor $\mathbb{T}$. Extend k so that the residual local parameters and the simple objects of the corresponding local blocks are absolutely split. For each $v \mid 2$, choose one representative from each simple isomorphism class in the smooth block with central character ψ_v . Let P_v be the projective envelope of the direct sum of their duals, taken in the compact dual category, and put $E_v = \text{End}(P_v)$. This is the standard block generator of [48, Section 4.1]. The block parameter is the semisimplification of $\bar{r}_v \otimes \bar{\varepsilon}$. Denote the corresponding local pseudodeformation ring by R_v^{ps} ; its determinant is $\psi_v \varepsilon = \chi_v \varepsilon^2$. Set

$$R_{\text{loc}}^{\text{ps}} = \widehat{\bigotimes_{v \mid 2, \mathcal{O}} R_v^{\text{ps}}}, \quad E_{\text{loc}} = \widehat{\bigotimes_{v \mid 2, \mathcal{O}} E_v}, \quad P = \widehat{\bigotimes_{v \mid 2, \mathcal{O}} P_v}.$$

Untwisting by ε identifies this local base with the fixed- χ_v local parameters of the global problem.

Proposition 4.2. *The residual Hecke summand belongs to the product block just specified. The exact projective-generator functor gives*

$$L = \text{Hom}_{\text{cts}, G_2}(P, M),$$

taken in the compact dual category of the specified product block, with the right E_{loc} -action by precomposition. It is faithful over $\mathbb{T}$, finite over $R_{\text{loc}}^{\text{ps}}$, and finite over $\mathbb{T}$. The local pseudodeformation action on L is the restriction of the global Galois action, with the stated cyclotomic twist. In particular, $\mathbb{T}$ is finite over $R_{\text{loc}}^{\text{ps}}$.

At a coefficient point $x : \mathbb{T} \rightarrow E'$, after finite extension of E' , its Banach Hecke eigenspace contains

$$\widehat{\bigotimes_{v|2}} \Pi_v,$$

where Π_v is an irreducible unitary Banach representation with local parameter $(r_x|_{G_{F_v}} \otimes \varepsilon)^{\text{ss}}$, of determinant $\chi_v \varepsilon^2$, in the block normalization of Section 2. If every local parameter is absolutely irreducible, de Rham, and regular, the eigenspace has a nonzero locally algebraic vector. If the untwisted parameter at v is $\gamma_{1,v} \oplus \gamma_{2,v}$, with ratio outside $\{1, \varepsilon, \varepsilon^{-1}\}$, the possibilities are the continuous unnormalized upper-Borel inductions

$$\text{Ind}_B^{\text{GL}_2(\mathbb{Q}_2)}(\gamma_{i,v\varepsilon} \otimes \gamma_{j,v}), \quad \{i, j\} = \{1, 2\}.$$

Proof. The one-factor input. We first specify the one-factor results. The integral center map and finiteness theorem of Paškūnas–Tung apply to every block, including the scalar residual block at 2 [48, Theorems 1.1–1.4]. In particular E_v is finite over R_v^{ps} ; an integral-surjectivity claim about the center is unnecessary. More precisely, integral finiteness is Theorem 1.3; the center comparison in Theorem 1.4 can have integral cokernel killed by 2. Their compact projective-generator equivalence is exact. Their generic reducible and absolutely irreducible characteristic-zero fibers are described in [48, Corollary 6.10 and Proposition 6.11]. The all-prime correspondence and locally algebraic comparison are supplied by [18, Theorems 1.1 and 1.3] and [17, Theorem 0.20]. Thus the old residual-scalar restriction at 2 does not enter the present application.

The product category. These statements extend to the required product category by the argument of [26, Lemmas B.7–B.8], adapted from the projective group to a fixed central character using the all-prime inputs just stated. Here are the details. Given a smooth admissible vector, fix compact opens in all but one factor. Its cyclic span under the remaining factor is finitely generated and admissible, so is finite length by [21, Lemma 2.3.6 and Theorem 2.3.8]. The commuting actions give a surjection from the tensor product of the one-factor cyclic spans onto the full cyclic span. Absolutely split simple factors tensor to simple product objects; filtering proves finite length. Hence the smooth product representation is locally finite.

Apply the one-factor compact equivalences successively. On dual finite-length objects this can be checked on simple tensors and extended by exactness. Taking inverse limits gives the equivalence for the compact locally finite category, with endomorphism ring E_{loc} . Admissibility gives finite multiplicities in the cosocle: a homomorphism from a simple is detected by a nonzero compact-fixed generating vector. There are only finitely many simples in the block. Compact Nakayama therefore makes $\text{Hom}(P, M)$ finite over E_{loc} .

Identifying the actual local and global actions. It remains to identify the block and its central action with the global Galois restriction. Test one factor at a time on the dense classical principal types of Proposition 4.1. Local-global compatibility gives a regular crystabelian Galois parameter with distinct inertial characters. For an irreducible parameter, the universal unitary completion of its algebraic-smooth principal series is the corresponding Banach representation by [8, Theorem 4.3.1 and Corollaries 5.3.1–5.3.4] and the locally algebraic comparison in [18]. Distinct inertia excludes both equal-root and special degeneracies.

For an ordinary parameter with cyclotomic exponents $a > b$, the algebraic factor is

$$\text{Sym}^{a-b-1} E^2 \otimes \det^{b+1}.$$

The ordinary completion theorem [13, Proposition 2.2.1] identifies the relevant locally algebraic induction with a dense subspace of the continuous unitary induction in the order

$$(\eta_l \varepsilon, \eta_h),$$

where η_h, η_l have exponents a, b . The following character calculation fixes the inducing order on the dense ordinary test types; that order will matter again in Section 8. For the precise normalization, put $n = a - b - 1 \geq 0$ and $\delta = (\eta_l \varepsilon) \eta_h^{-1}$. Twisting by the full unitary character $\eta_h^{-1} \circ \det$ reduces this induction to $\text{Ind}_B^{\text{GL}_2(\mathbb{Q}_2)}(\delta \otimes 1)^{C^0}$. With x the identity character and $\varepsilon = x|\cdot|$, the source characters in the cited proposition, with $k = n + 2$, are

$$\chi_1 = |\cdot| x^{-n}, \quad \chi_2 = \delta |\cdot|^{-1} x^n.$$

Since δ is unitary of algebraic exponent $-n$, both χ_2 and $(\chi_2/\chi_1)x^{-n}$ are smooth, and

$$v_2(\chi_1(2)) = -n - 1, \quad v_2(\chi_2(2)) = n + 1.$$

The two characters in its completion formula are $\chi_1|\cdot|^{-1}x^n = 1$ and $\chi_2|\cdot| x^{-n} = \delta$, in the reversed inducing order $(\delta, 1)$. Twisting back by $\eta_h \circ \det$ gives the displayed order and changes the algebraic factor $\det^{-n} \text{Sym}^n$ into $\det^{b+1} \text{Sym}^{a-b-1}$. On each lower-coordinate disk its vectors are precisely bounded-degree polynomial sections. The theorem concerns this locally algebraic subspace, not the whole locally analytic induction. When $n = 0$, the excluded equality $\chi_1 = \chi_2|\cdot|^2$ is exactly $\delta = 1$, which is ruled out by the distinct finite inertial characters. The ordinary and nonordinary statements together supply every test type used in the density argument; see also [48, Propositions 5.3–5.6].

A map from such a classical constituent into completed forms extends continuously from its universal unitary completion. After scaling lattices, it is a bounded map in the compact block category. Naturality of the center then identifies the central trace action on its image with the Galois parameter given by classical local-global compatibility. The normalizations give $\rho \otimes \varepsilon$, as stipulated above. The dense classical span proves equality of the two actions on the whole Banach space. Torsion-freeness of the integral lattice gives integral equality and shows that every other block projection vanishes.

Integral finiteness and coefficient fibers. It follows from integral Paškūnas–Tung finiteness that L is finite over $R_{\text{loc}}^{\text{ps}}$. Exactness and the equivalence imply that the action of $\mathbb{T}$ on L is faithful. The same finite generators generate L over $\mathbb{T}$. Moreover,

$$\mathbb{T} \hookrightarrow \text{End}_{R_{\text{loc}}^{\text{ps}}}(L);$$

the right side is finite over the noetherian local base. Thus $\mathbb{T}$ is finite over that base.

At x , the quotient of $L[1/2]$ by the Hecke maximal ideal is nonzero and finite-dimensional. Select a simple quotient after a finite coefficient extension. The commuting images of the factor endomorphism rings generate a finite-dimensional algebra. Their radicals kill the simple quotient; their central idempotents select semisimple factors; after splitting those factors, the quotient is a tensor product of simple modules for matrix algebras. The original integral actions preserve a lattice. All full finite lattices in this quotient are commensurable. Under the inverse compact equivalence, the tensor lattice corresponds to the completed tensor product of the individual Banach lattices. Dualizing the quotient therefore embeds the asserted Banach tensor product in the eigenspace.

For an absolutely irreducible regular de Rham parameter, the local factor has a nonzero locally algebraic vector, so their tensor does as well. For a generic reducible parameter, substituting $\gamma_i \varepsilon$ into the fiber formula of [48, Corollary 6.10] gives exactly the two displayed inducing orders. $\square$

4.3 A lower bound on every Hecke component

We next obtain a dimension bound that will supply the initial support size for patching. For a finitely generated compact $k[[K]]$ -module, its growth degree is the degree of growth of its length modulo powers of the augmentation ideal.

Lemma 4.3. *Let k be the characteristic-two residue coefficient field of Section 2. The compact dual of an absolutely irreducible admissible smooth k -representation of $\mathrm{GL}_2(\mathbb{Q}_2)$ with central character has growth degree at most one on a sufficiently deep determinant-one compact subgroup. Finite-length objects for d factors have growth degree at most d .*

Proof. Write $K_n = 1 + 2^n M_2(\mathbb{Z}_2)$, $n \geq 2$. Characters have bounded invariant dimension. For a principal series, the double cosets indexed by $\mathbb{P}^1(\mathbb{Z}/2^n\mathbb{Z})$ give $O(2^n)$ invariants. The Steinberg case follows from the constants–principal-series–Steinberg exact sequence: its possible $H^1(K_n, k)$ correction is bounded, since K_n has four generators.

For a supersingular π , let

$$I_1 = \begin{pmatrix} 1 + 2\mathbb{Z}_2 & \mathbb{Z}_2 \\ 2\mathbb{Z}_2 & 1 + 2\mathbb{Z}_2 \end{pmatrix}, \quad \Pi = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}.$$

Thus I_1 is the standard pro-2 Iwahori subgroup. Let M_π be the span of the translates of π^{I_1} under $\begin{pmatrix} 2^n & b \\ 0 & 1 \end{pmatrix}$, where $n \geq 0$ and $b \in \mathbb{Z}_2$; this space is Iwahori-stable [47, Definition 4.5 and Lemma 4.6]. Use the exact sequence of Iwahori representations

$$0 \longrightarrow \pi^{I_1} \longrightarrow M_\pi \oplus \Pi M_\pi \longrightarrow \pi \longrightarrow 0$$

from [47, Theorem 1.2, proved as Theorem 6.3]. That statement explicitly includes $p = 2$. By [47, Proposition 4.7 and Lemma 4.10], M_π is the sum of two injective envelopes for the upper unipotent group $\mathbb{Z}_2$. Each has 2^n -dimensional invariants under $2^n\mathbb{Z}_2$. Conjugation by Π shifts the congruence level by at most one. Since $\dim \pi^{I_1} = 2$, the invariants sequence gives

$$\dim_k \pi^{K_n} \leq 2 \cdot 2^n + 2 \cdot 2^{n+1} + 8.$$

The last term bounds $H^1(K_n, \pi^{I_1})$. The classification of Barthel–Livné and Breuil [5, 6], in the form of [12, Corollary 1.2], shows that the cases above exhaust the irreducible admissible smooth representations with central character.

Let I be the augmentation ideal of a fixed deep uniform compact group. In characteristic two, the operators $g^{2^n} - 1 = (g - 1)^{2^n}$ belong to I^{2^n} . Thus the compact quotient modulo I^{2^n} is bounded in dimension by the dual K_{n+c} -invariants, for a fixed shift c . Choosing 2^n comparable to an arbitrary exponent gives linear augmentation growth. The central character is trivial on a sufficiently deep scalar subgroup, and the product of that subgroup with a deep determinant-one subgroup has finite index. The same upper bound therefore holds on the latter. Finite extensions preserve upper growth bounds, and tensor products add them. This proves the product assertion. $\square$

Proposition 4.4. *For every irreducible component C of $\mathrm{Spec} \mathbb{T}$,*

$$\dim C \geq 1 + 2d.$$

Proof. Reducedness and prime avoidance give an element of $\mathbb{T}$ that vanishes on every component except C and is nonzero on C . Its image on M is a nonzero submodule M' supported on C . The full M is finite free over a sufficiently deep Iwasawa algebra $A = \mathcal{O}[[K]]$. The residual summand and M' embed in this free module.

The maximal augmentation filtration, including ϖ , has polynomial associated graded ring of dimension $1 + 3d$, after taking K sufficiently deep. A nonzero submodule of a finite free A -module has that full growth degree: it contains a cyclic copy of A , and Artin–Rees compares its induced filtration with its intrinsic filtration. Consequently M' has growth degree $1 + 3d$.

Let $r = \dim C$, and choose r parameters of the supporting Hecke quotient. Write J for their ideal. Proposition 4.2 and exactness of the block functor show that M'/JM' is dual finite length. Lemma 4.3 bounds its growth degree by d . Each Hecke parameter is topologically nilpotent; on the finite quotient $M'/\mathfrak{m}_A M'$, some fixed power is zero. Hence there is $c > 0$ such that

$$J^{cn} M' \subseteq \mathfrak{m}_A^n M' \quad (n \geq 1).$$

Filter $M'/\mathfrak{m}_A^n M'$ by J . The degree- j layer is a quotient of at most $\binom{j+r-1}{r-1}$ copies of

$$(M'/JM')/\mathfrak{m}_A^n (M'/JM').$$

Only cn layers occur. Their total length is therefore $O(n^{r+d})$. Comparing with the degree $1 + 3d$ gives $1 + 3d \leq r + d$, which is the claim. $\square$

4.4 Auxiliary level and solvable base change

The dimension bound supplies the initial support size. Patching also requires that removing auxiliary level recover the original operators, not just modules of the same rank. The following exact statements connect completed forms to the finite stages of localized patching. An auxiliary prime has residue cardinality $q_v \equiv 1 \pmod{2^n}$. Inside its Iwahori group, take the kernel of the diagonal-ratio map to a cyclic quotient of residue units of order 2^n . This subgroup contains the scalars. For h auxiliary primes, the deck group is $\Delta = (\mathbb{Z}/2^n)^h$.

Lemma 4.5. *Assume that χ is unramified at the auxiliary primes. Localize at a characteristic-two point of the completed Hecke spectrum at which their Frobenii have distinct eigenvalues. The point may be the generic point of an enhanced curve, rather than the original finite-field residual point. The following assertions hold for the localized completed-forms block modules, and continue to hold after the coefficient extensions, completions, and smooth framings used in Section 5.*

1. *The auxiliary-level module is flat over $\mathcal{O}[\Delta]$. Diamond augmentation is the Iwahori module; after localization at the indicated point it is the direct sum of 2^h copies of the old hyperspecial module, with its Galois-ring action factoring through the old unramified quotient.*
2. *If U_j is a diamond generator and τ_j is the corresponding tame inertia element, then*

$$U_j^2 - T(\tau_j)U_j + 1 = 0.$$

Congruent tame transformation characters give isomorphic residue modules with the same deformation actions.

3. *A quadratic character trivial above 2 and at infinity, unramified away from the auxiliary primes, and factoring through the auxiliary character quotient, acts by twisting forms. It gives an actual semilinear isomorphism of the block module with its Galois twist, preserving the central character, the level, and the characteristic-two completion point.*

Proof. The finite disjoint-union function model used in Proposition 4.1 is free under the deck action, since the ratio quotient kills scalars and the level was chosen to remove projective stabilizers. Its compact dual is therefore flat over $\mathcal{O}[\Delta]$, with augmentation equal to the Iwahori-level module.

Apply the exact projective functor to the finite-presentation ideal tests for flatness. This proves flatness of the block module and commutation of reduction with that functor.

At one auxiliary prime, let $\alpha : M_{\text{old}}^{\oplus 2} \rightarrow M_{\text{Iw}}$ be the two degeneracy maps and $\beta : M_{\text{Iw}} \rightarrow M_{\text{old}}^{\oplus 2}$ their trace maps. Their composite $C = \beta\alpha$ has diagonal entries $q_v + 1$ and off-diagonal entries given by the unramified Hecke operator, up to central units. Write $D = \det C$. On the Iwahori module, D denotes the same scalar expression in the interpolated Frobenius trace and central operators. In particular this is one compatible scalar operator on both levels, not a determinant defined only on the old module. The standard degeneracy calculation expresses it, up to a central unit, as

$$(q_v + 1)^2 - q_v \chi(\phi_v)^{-1} T(\phi_v)^2,$$

where ϕ_v is a Frobenius lift. At the chosen characteristic-two point, its value is a unit times the nonzero square of the Frobenius trace. Thus D becomes invertible in this localization.

Before inverting D , we have the integral identity

$$\alpha \operatorname{adj}(C)\beta = D \operatorname{id}_{M_{\text{Iw}}}.$$

The local alternatives here are the Iwahori component of Henniart's description [14, Appendix A, Sections A.1.3 and A.2.3]. Indeed an Iwahori-spherical classical cuspidal local representation is unramified principal series or special. In the former case its two Iwahori vectors are spanned by the degeneracies, and the identity follows from the matrix adjugate identity. In the latter case the old space is zero and the Frobenius eigenvalue ratio is $q_v^{\pm 1}$, so the displayed scalar expression for D is zero as well. Classical density and torsion-freeness prove the identity on integral completed forms and their compact duals. Consequently $D^{-1} \operatorname{adj}(C)\beta$ is both a left and a right inverse to α after the selected localization. Iterating gives 2^h old copies.

The maps are Hecke equivariant. Interpolation identifies their Galois trace actions; at the absolutely irreducible enhanced curve, the framed representation action is consequently the actual old unramified action. Thus the augmentation factors through the old deformation quotient, with its original action, rather than only having the same multiplicity. The flatness and Iwahori augmentation established before localization remain valid at the selected point on both sides. The exact block functor and the finite-ideal tests used in Section 5 preserve them under the indicated flat coefficient extensions, localizations, completions, and smooth framings.

On a classical point with a diamond eigencharacter, local Langlands gives semisimple inertia characters equal to that character and its inverse, because χ is unramified. Their sum is $T(\tau_j)$. The displayed monic quadratic identity follows at every such point, and hence on the integral module by density. Fixed tame character transformations congruent modulo ϖ identify their residue function spaces. The associated trace comparisons show that this identification also preserves the residue deformation actions.

For the last assertion, multiply a form by the quadratic character composed with reduced norm. The central character is unchanged because its value on a scalar norm is a square. On the ratio-defined auxiliary subgroup, the two residue diagonal entries have the same image in the character quotient; their product is therefore killed by a quadratic character. Thus the level is preserved. Triviality at 2 and infinity preserves the local actions there, and the Galois parameter twists by the same character. Its residual reduction is one, so it fixes the characteristic-two point and its residual frames. This gives the asserted semilinear intertwiner.

Finally, adding a coefficient variable preserves flatness by the complete-module local criterion. Localization, completion of finite noetherian modules, and smooth framing extension preserve the stated flatness and module identities. The intertwiners are local and preserve maximal-ideal filtrations. Hence every assertion survives the operations used at the finite patching stages. $\square$

Lemma 4.6. *Let F'/F be a solvable totally real Galois extension split completely at 2. The restriction to F' of a pro-modular trace locus over F is pro-modular over F' , at one common tame level for the whole restricted locus. If all its bad inertial types are killed, this level can be taken Iwahori at the bad non-dyadic primes and hyperspecial elsewhere away from 2. For a general solvable totally real extension split completely at 2, the restricted locus is potentially pro-modular.*

Proof. First suppose that F'/F is Galois. Transfer the dense classical points by solvable base change, iterated through prime-degree cyclic extensions [2, Chapter III, Theorems 4.2 and 5.1]; for the original GL_2 construction see [38, Chapter 11, Lemma 11.3 and Proposition 11.4]. Apply the unitary formulation after a norm twist when necessary, then undo the twist. Use Jacquet–Langlands [28, Theorems 14.4 and 16.1]. A regular cuspidal packet stays cuspidal along each prime-degree cyclic step: a loss of cuspidality would make it induced from a character of a totally real quadratic field, whose parallel algebraic exponents contradict regularity. Local-global compatibility and the uniform inertia control of Lemma 3.9 bound the conductors at a fixed finite set of places. The wild kernel is common and the rank is two, so the possible tame monodromy contribution is uniformly bounded. Thus a single tame level contains every transferred packet.

Their trace closure contains the restricted locus by density and continuity. Thus this locus is pro-modular over the specified field F' , at the common tame level just chosen. After the inertial types are killed, only unramified principal or special local factors remain at a bad non-dyadic prime, both with Iwahori invariants. The stated level follows.

For a general solvable extension, pass to its solvable Galois closure. It is still totally real and completely split at 2. The Galois case then proves potential pro-modularity of the restricted locus. We use only this forward base-change assertion here; classical descent is applied later to an actual cuspidal packet. $\square$

5 Propagation from a characteristic-two curve

The input to this section is a pro-modular curve whose generic representation is not virtually solvable. We prove that completed forms have full support on the deformation germ at that curve. The residue field of the germ is a Laurent-series field; accordingly, the patching argument below uses finite jets, rather than compactness of a finite residue field.

The argument follows the auxiliary-level framework of Taylor–Wiles [57], its formulation relative to local rings by Kisin [36, Section 3.3], and the characteristic-two determinant-twist construction of Khare–Wintenberger [32, Section 9.1.3]. Pan’s localized propagation argument is the global predecessor [45]. Here the curve residue field requires explicit control of finite jets and fixed old-level quotients. We first construct finite auxiliary stages, then preserve their filtration in the limit, and finally use twists, sign intertwiners and diamond augmentation to recover support with the original deformation action.

Theorem 5.1 (Localized propagation). *Let F be a totally real solvable field of even degree, completely split at 2, and let S contain its real and dyadic places and the allowed finite ramification. Fix the determinant $\chi = \det r = \chi_{\mathrm{fin}} \varepsilon^w$ of a target r as in Section 2, a semisimple residual representation with determinant $\bar{\chi}$, and let $R_F^{\mathrm{ps}, \chi}$ be the corresponding complete local global pseudodeformation ring. Let C be a characteristic-two curve in $\mathrm{Spec} R_F^{\mathrm{ps}, \chi}$. Suppose that C is potentially pro-modular, that its generic representation is not virtually solvable, and that its local images at the non-dyadic finite places in S are finite. Every irreducible closed locus of $\mathrm{Spec} R_F^{\mathrm{ps}, \chi}$ containing C is potentially pro-modular, with ramification allowed only at the specified set S before base change.*

We will prove the theorem at the end of the section. First make a totally real solvable Galois base change, split completely above 2, on which the curve is pro-modular. Lemma 3.9, applied both to the source deformation rings and to the Hecke algebras, permits a further such base change with the following properties: the curve representation is trivial at every bad finite place; the determinant χ is unramified there; the source has unipotent inertia there; and each bad residue cardinality is 1 modulo 4. By Lemma 4.6, the curve is then in the support of forms with Iwahori level at the bad places and hyperspecial level elsewhere away from 2. The assertion about the source is set-theoretic, which is sufficient for the theorem.

Retain the notation F, S after these base changes, and put

$$d = [F : \mathbb{Q}], \quad u = \#\{v \in S : v \nmid 2\infty\}, \quad s = 2d + u.$$

Use Lemma 3.4 to enhance the curve by z and complete. Its coefficient ring and residue field are

$$\mathcal{L} = \widehat{\mathcal{O}[[z]]}_{(\varpi)}, \quad K = k((t)), \quad z \mapsto t.$$

Finite extensions of these coefficients are allowed throughout. Write $\rho_* : G_{F,S} \rightarrow \mathrm{GL}_2(K)$ for the curve representation. It is absolutely irreducible, and remains so under the base changes just made. All local and global deformation rings in this section are completed at this same enhanced curve and have residue field K .

5.1 Auxiliary primes and characters

Character twists will enlarge the support obtained from completed forms. Since twisting multiplies the determinant by the square of the character, we need a variable-determinant problem as well as the fixed-determinant one. In the former, the determinant is fixed at the dyadic and real places and on bad inertia, but is allowed to vary globally. Enlarge $\mathcal{O}$ so that $i^2 = -1$ belongs to it. At a bad place, choose a tame inertia generator τ and an arithmetic Frobenius ϕ with $\phi\tau\phi^{-1} = \tau^{q_v}$. For $\eta \in \{1, i\}$ impose

$$\det \rho(\tau) = 1, \quad \mathrm{tr} \rho(\tau) = \eta + \eta^{-1}.$$

The determinant of $\rho(\phi)$ is unrestricted in the variable problem. The local curve is trivial, so wild inertia and prime-to-two tame inertia disappear in these completed problems. Denote the completed product of the framed local rings at S by $R_{\mathrm{loc}, \eta}$. The $\eta = 1$ condition contains the source with unipotent bad inertia. The $\eta = i$ condition provides an irreducible comparison model: Proposition 3.10 identifies its special fiber with that of the 1-model and specifies the horizontal components through each generic special-fiber point. We will use this comparison to transfer support to every component of the 1-model.

For a set Q_n of auxiliary primes, let $R_{\eta, n}$ be the S -framed global ring with these local conditions and unrestricted additional ramification at Q_n . A superscript f denotes the quotient imposing the global determinant χ . Write R_η and R_η^f for the corresponding rings without Q_n . Thus forgetting the additional ramification gives quotient maps

$$R_{\eta, n} \twoheadrightarrow R_\eta, \quad R_{\eta, n}^f \twoheadrightarrow R_\eta^f.$$

The deformation interpretation over $\mathcal{L}$ and noetherianity follow from Lemma 3.4 and the relative tangent and obstruction calculation of Proposition 3.6.

Lemma 5.2 (Auxiliary-prime bound). *There is a sufficiently large integer h , independent of n , and sets Q_n of h primes outside S such that:*

1. every prime in Q_n splits in $F(\zeta_{2^n}, 2^{1/2^n})$ and has residue cardinality 1 modulo 2^n ;
2. its ρ_* -Frobenius is regular semisimple, with both eigenvalues in one fixed finite extension of K ;
3. there are surjections

$$B_\eta := R_{\text{loc}, \eta}[[X_1, \dots, X_g]] \twoheadrightarrow R_{\eta, n}, \quad g = 2h + 1. \quad (13)$$

The two sets of maps in (13), for $\eta = 1, i$, can be chosen to agree modulo ϖ . For each η , their composites with $R_{\eta, n} \twoheadrightarrow R_\eta$ can also be chosen independently of n .

Proof. Put $M = \text{ad } \rho_*$. The pairing $(A, B) \mapsto \text{tr}(AB)$ on $M_2(K)$ is perfect also in characteristic two. Thus the dual coefficient module is again M : the mod-two cyclotomic twist is trivial. The relative Selmer condition is zero at S and full at Q_n . The dual condition therefore consists of classes unramified outside S whose localizations at Q_n vanish.

Set

$$F_\infty = F(\zeta_{2^\infty}, 2^{1/2^\infty}), \quad H = \ker(G_{F, S} \longrightarrow \text{Gal}(F_\infty/F)).$$

The tower group is solvable. Nonvirtual solvability of ρ_* therefore implies that the geometric reduced Zariski closure of $\rho_*(H)$ contains SL_2 ; the same is true for every open subgroup of H . Indeed, take the closure over an algebraic closure of K . Its smooth connected derived group lies in SL_2 . If proper, it has dimension at most two and is solvable, forcing the identity component itself to be solvable, a contradiction [43, Corollaries 6.19 and 17.27]. Consequently regular semisimple elements are dense in every open coset of H .

We give the detection argument explicitly. Let $c \in Z^1(G_{F, S}, M)$ have zero diagonal evaluation at every regular semisimple $\sigma \in H$. Then the first-order representation

$$\rho_c(\sigma) = (1 + \epsilon c(\sigma))\rho_*(\sigma), \quad \epsilon^2 = 0,$$

has the same trace as ρ_* on those elements. Indeed, the quotient

$$M/(\text{Ad}(\rho_*(\sigma)) - 1)M$$

is the diagonal subspace in an eigenbasis, and zero evaluation kills both diagonal entries. Continuity and the density just observed give equality of traces on all of H .

Choose $h_1, \dots, h_4 \in H$ such that $A_j = \rho_*(h_j)$ form a basis of $M_2(K)$. The lifts $\rho_c(h_j)$ form a basis over $K[\epsilon]$. The Gram matrix $\text{tr}(A_j A_k)$ is invertible, and equality of traces on products of two and three group elements shows that this Gram matrix and all multiplication structure constants are unchanged after lifting. The linear map taking A_j to $\rho_c(h_j)$ is therefore an algebra automorphism of $M_2(K[\epsilon])$, reducing to the identity. Such an automorphism is conjugation by a matrix congruent to the identity: this follows, for example, by carrying a system of matrix units to its image and choosing the corresponding basis. Pairing with the four basis matrices then shows that this conjugation carries $\rho_*|_H$ to $\rho_c|_H$ on every element of H . Hence $[c]|_H = 0$.

The invariants M^H are the scalars. Inflation–restriction bounds the kernel of all these evaluations by

$$H^1(\text{Gal}(F_\infty/F), K).$$

This space has dimension at most two. To see the last assertion, the corresponding tower over $\mathbb{Q}$ is pro-two and unramified away from 2. Its quadratic subfields are $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, and $\mathbb{Q}(\sqrt{-2})$. Its elementary two-character rank is therefore two. The field F is disjoint from that tower: a nontrivial intersection would have a quadratic subfield, whereas none of these three fields has 2 split completely. Thus the tower group over F has the same two-character rank.

The ambient cohomology space is finite dimensional. A finite list of regular elements consequently detects its quotient by a subspace of dimension at most two. Extend K once to split the eigenvalues of this list. Both regular split eigenvalues and the nonvanishing minors of the evaluation matrix persist on sufficiently small neighborhoods. For each n , Chebotarev approximation supplies Frobenius elements in these neighborhoods which are trivial at the required finite stage of the tower [42, V, Theorem 3.23]. All conditions concern finite quotients after the neighborhoods have been chosen. Extra such primes can be added, so the cardinality can be any fixed sufficiently large h . The dual Selmer dimension is then at most two.

The auxiliary evaluations have reduced the undetected dual Selmer space to at most two scalar tower directions. The next count turns this into the uniform presentation bound $g = 2h + 1$. For completeness, write $h_v^j = \dim_K H^j(G_{F_v}, M)$. Local duality and Euler characteristics give

$$h_v^1 = \begin{cases} 2h_v^0 + 4[F_v : \mathbb{Q}_2], & v \mid 2, \\ 2h_v^0 - 4, & v \text{ real}, \\ 2h_v^0, & v \nmid 2\infty. \end{cases}$$

For trivial complex conjugation, the dimensions are $(h_v^0, h_v^1) = (4, 4)$. For nontrivial unipotent complex conjugation they are $(2, 0)$. Thus the dyadic contribution $4d$ cancels the real contribution $-4d$. The localization image in the direct sum of the local H^1 spaces is its own annihilator under Poitou–Tate. For a subspace of local conditions L , elementary linear algebra therefore gives the Selmer difference as $\dim L - \frac{1}{2} \sum_v h_v^1$. Here L is zero at S and full at Q_n . At each auxiliary prime regularity gives $h_v^0 = 2$ and $h_v^1 = 4$. Consequently the identity is

$$h_{\text{Sel}}^1(M) - h_{\text{Sel}^\perp}^1(M) = - \sum_{v \in S} h_v^0 + 2h.$$

The finite-coefficient formulas are [40, I, Corollary 2.3, Theorems 2.8, 2.13, 4.10, and 5.1]; their passage to K is the lattice-and-limit argument of Lemma 3.5. With independent S -frames the relative tangent dimension is

$$h_{\text{Sel}}^1(M) + \sum_{v \in S} h_v^0 - h^0(G_{F,S}, M) = h_{\text{Sel}^\perp}^1(M) + 2h - 1 \leq 2h + 1.$$

Generators with a fixed old composite. To obtain the last assertion as well as the variable bound, write $A_\eta = R_{\text{loc}, \eta}$ and $I_{\eta, n} = \ker(R_{\eta, n} \rightarrow R_\eta)$. The relative cotangent spaces are

$$Q_{\eta, n} = \mathfrak{m}_{\eta, n} / (\mathfrak{m}_{\eta, n}^2 + \mathfrak{m}_{A_\eta} R_{\eta, n}), \quad Q_\eta = \mathfrak{m}_\eta / (\mathfrak{m}_\eta^2 + \mathfrak{m}_{A_\eta} R_\eta).$$

The first maps onto the second. Its kernel is the image of the actual ideal $I_{\eta, n}$: if a representative maps into $\mathfrak{m}_\eta^2 + \mathfrak{m}_{A_\eta} R_\eta$, lift a decomposition of its image into $\mathfrak{m}_{\eta, n}^2 + \mathfrak{m}_{A_\eta} R_{\eta, n}$ and subtract it. This leaves a representative in $I_{\eta, n}$ of the same relative cotangent class.

Choose a basis of the old space Q_η with fixed representatives $r_{\eta, 1}, \dots, r_{\eta, q}$ in R_η . Choose their reductions first in the common special-fiber diagram and then lift to both values of η . Their relative cotangent spaces agree modulo ϖ , so the number q is the same. For each n , lift these fixed old representatives to $R_{\eta, n}$ and complete their independent cotangent classes to a basis using representatives in $I_{\eta, n}$. The tangent bound just proved shows that at most $g = 2h + 1$ elements are needed; pad the list with zeros to length g .

These choices can have identical reductions for $\eta = 1, i$. First choose the lifts and the kernel-complement elements in the common special-fiber diagram. Lift an element arbitrarily to $R_{\eta, n}$. If its image in R_η differs from the required fixed old value by ϖa , subtract ϖ times any preimage of a

in $R_{\eta,n}$. This preserves the chosen special-fiber reduction and makes the old image exact. For a kernel-complement element the required old image is zero. No flatness of the old global ring over the coefficient DVR is used.

Complete relative Nakayama now gives (13). Every composite to R_η sends the first q variables to the fixed $r_{\eta,j}$ and all remaining variables to zero, independently of n . The two presentations still agree modulo ϖ . The variable count, and hence every later ambient dimension, is unchanged. $\square$

Lemma 5.3 (Rank of auxiliary character twists). *Increase h so that $b = h - 2d + 2 > 0$. There are quotients*

$$A_n = (\mathbb{Z}/2^{a_n}\mathbb{Z})^b, \quad a_n \longrightarrow \infty,$$

of the ray class groups defining characters trivial at every dyadic and real place and unramified outside Q_n . At Q_n they use only the residue-unit quotients of order 2^n .

Proof. Let S_2 be the dyadic places. Take the ordinary ray class group of modulus $\prod_{q \in Q_n} q$, and quotient by the dyadic ideal classes. Omitting real places from the modulus requires them to split; at the dyadic places, both the units and uniformizers are trivial in this quotient. The ray class exact sequence is

$$\mathcal{O}_{F,S_2}^\times \longrightarrow \prod_{q \in Q_n} k(q)^\times \longrightarrow \Gamma_n \longrightarrow \text{Cl}_{S_2}(F) \longrightarrow 0.$$

This is the usual ideal-theoretic ray sequence, with the primes in S_2 inverted [42, V, Theorem 1.7]. The group of dyadic S -units has free rank

$$(d-1) + d = 2d-1.$$

Here we use the S -unit theorem [41, Theorem 5.11]. The direction generated by 2 is primitive in its free part: its valuation at each dyadic prime is one, since 2 splits completely in F . Since F is totally real, generators for these units may therefore be chosen as $-1, 2$ and $2d-2$ further units. For $n \geq 2$, put $a_n = n-1$ and take the cyclic residue-unit quotient of order 2^{a_n} at each auxiliary prime. Both -1 and 2 vanish in these quotients: the first is a 2^{a_n} th power because $\mu_{2^{a_n}+1}$ is present, and the second is such a power by the imposed Kummer splitting. The elementary divisor theorem shows that quotienting $(\mathbb{Z}/2^{a_n}\mathbb{Z})^h$ by the images of the remaining $2d-2$ generators leaves a quotient $(\mathbb{Z}/2^{a_n}\mathbb{Z})^{h-2d+2}$. The image of the local units in Γ_n thus has such a quotient. A finite abelian group containing a subgroup with this quotient also has this quotient: in invariant-factor form, the subgroup forces at least this many factors of order at least 2^{a_n} in the containing group. Apply this to Γ_n . Class field theory turns its quotient into the required characters [42, V, Theorems 5.3 and 5.5]. Discarding the first finitely many indices is harmless. $\square$

The resulting $b = h - 2d + 2$ character parameters will supply the support dimensions missing from the fixed-determinant estimate. Their vanishing at the old local conditions is what permits twisting without changing those conditions. Let H_n denote the formal character group of A_n , completed at the trivial character and ϖ . Its coordinate ring is

$$\mathcal{L}[[l_1, \dots, l_b]] / ((1 + l_j)^{2^{a_n}} - 1 : 1 \leq j \leq b).$$

Its tautological Galois character will be denoted θ_n . Twisting a fixed-determinant representation by θ_n preserves the prescribed determinants at the dyadic and real places and on bad inertia. It therefore defines a map from the variable global problem to the fixed global problem with these character parameters.

The local product $R_{\text{loc},\eta}$ in (13) is obtained from the enhanced real-dyadic base A^{rd} of Proposition 3.10 by adjoining the u raw tame factors of (6), with $\det \phi_j$ unrestricted. Adjoining the $g = 2h + 1$ presentation variables gives the rings B_η of that proposition. They therefore have dimension

$$e = 1 + 8d + 4u + g = 8d + 4u + 2h + 2. \quad (14)$$

Their special fibers are identified. The underlying spectrum of B_i is irreducible and nonvertical. All components of B_1 are horizontal of dimension e , and correspond bijectively to the top-dimensional components of its special fiber, of dimension $e - 1$. The generic point of each such special-fiber component lies on just its corresponding component of B_1 . These precise component assertions, and not only the dimension in (14), will be used below.

5.2 The modules and their finite-level actions

Let $N_{\eta,n}$ be the block module of completed forms at the auxiliary ratio level $U_1(Q_n)$, with tame type (η, η^{-1}) at the bad places. Add z , make the enhanced completion, and pull back to the independent S -frame space. Define N_η in the same way at old level. The modules are finite over $R_{\eta,n}^f$ and R_η^f , respectively, by Propositions 4.1 and 4.2. We also regard them as modules for the variable-determinant rings through their fixed-determinant quotients.

Choose the quotient of the diamond group $\Delta_n = (\mathbb{Z}/2^n\mathbb{Z})^h$, with generators $U_1, \dots, U_h$, and write $V_j = U_j - 1$. Lemma 4.5 gives

$$N_{\eta,n} \text{ is flat over } \mathcal{L}[\Delta_n], \quad (15)$$

$$N_{\eta,n}/(V_1, \dots, V_h)N_{\eta,n} \simeq N_\eta^{\oplus 2^h}, \quad (16)$$

$$\overline{N} := N_1/\varpi N_1 \simeq N_i/\varpi N_i \neq 0, \quad \dim \text{Supp } \overline{N} \geq 2d + 4s - 1. \quad (17)$$

The isomorphism (16) is an isomorphism of $R_{\eta,n}$ -modules through $R_{\eta,n} \twoheadrightarrow R_\eta$. In particular, it identifies the actual Galois-ring actions. The n -level modules and their ring and diamond actions for the two values of η likewise agree modulo ϖ .

We recall why the operations just performed preserve the assertions. Flatness over $\mathcal{O}[\Delta_n]$ holds in the definite function-space model and passes through the exact block functor. The z -adic flatness criterion applies after adjoining z , and base change from $\mathcal{O}[[z]]$ to $\mathcal{L}$ preserves flatness. The subsequent localizations, noetherian completions, and smooth frame extensions are flat maps of the commuting deformation rings. Tensoring a finite-ideal flatness test over $\mathcal{L}[\Delta_n]$ with these flat extensions preserves injectivity, proving (15) after all the stated operations. The degeneracy maps giving (16) are Hecke equivariant. At the regular Frobenius curve their composite is invertible, and the special contribution is absent. The augmented Galois action is consequently unramified at Q_n and factors through the old quotient; the classical comparison identifies it with the old action. Finally, Proposition 4.4 gives dimension at least $1 + 2d$ on each Hecke component. Enhanced completion preserves this dimension, independent S -frames add $4s - 1$, and reduction by ϖ loses at most one. This proves the bound in (17). Nonvanishing follows from the pro-modular curve and the faithfulness of the old block module.

For chosen tame inertia elements $\tau_{j,n}$ at the auxiliary primes, the commuting diamond actions also satisfy

$$U_j^2 - T(\tau_{j,n})U_j + 1 = 0. \quad (18)$$

Here T denotes the trace of the Galois deformation acting on the module. This is an identity of operators, as supplied by Lemma 4.5.

We will need actual twist isomorphisms, rather than only a numerical invariance of support. For every sign tuple $\sigma \in \{1, -1\}^b$, specialize θ_n to the quadratic character $\theta_{\sigma,n}$. Multiplication of forms by $\theta_{\sigma,n} \circ \text{Nrd}$ gives a semilinear isomorphism of $N_{\eta,n}$ for the corresponding automorphism of $R_{\eta,n}^f$. The central character is unchanged because this character is quadratic. It is trivial above 2 and at infinity and unramified on bad inertia. It preserves the ratio level: on that level the two diagonal character values agree, so their product is one. Its reduction is one, hence the enhanced curve and its residual frames are fixed. These observations prove that the isomorphisms act on precisely the completed modules under consideration.

5.3 Exact patching with finite jets

Fix a nonprincipal ultrafilter $\mathfrak{U}$ on the positive integers. For a fixed complete noetherian local ring $(B, \mathfrak{m})$, define

$$B^* = \varprojlim_j \prod_{\mathfrak{U}} B/\mathfrak{m}^j.$$

Its residue field is the ultrapower K^* of the residue field K . This deliberate auxiliary scalar extension is separate from the finite coefficient extensions used to define the original curves and representations. For a fixed finite B -module M , use the same definition with $M/\mathfrak{m}^j M$. For varying quotients of B and varying finite modules, a subscript ∞ will denote this construction using the maximal-ideal filtration induced by B .

Lemma 5.4 (Finite jets and uniform exactness). *The following assertions hold.*

1. *The ring B^* is complete noetherian local, and $B \rightarrow B^*$ is faithfully flat. For fixed finite modules the construction is exact and equals extension of scalars. It preserves their support dimensions. If B is a domain, so is B^* .*
2. *For varying quotients of B and modules generated by a bounded number of elements, the patched objects are finite over B^* . Surjections and cokernels of a fixed finite number of compatible operators commute with patching.*
3. *Let $(D, \mathfrak{a})$ be a fixed noetherian local ring and C a fixed finite D -module. For arbitrary finite-module extensions*

$$0 \longrightarrow P_n \longrightarrow H_n \longrightarrow C \longrightarrow 0$$

there is a constant c , independent of the extension, such that

$$P_n \cap \mathfrak{a}^j H_n \subseteq \mathfrak{a}^{j-c} P_n \quad (j \geq c). \tag{19}$$

Thus a filtration of fixed length with fixed successive quotients patches exactly whenever its ambient jet filtration is uniformly cofinal with the $\mathfrak{a}$ -adic filtrations on its pieces.

Proof. At each depth, taking an ultrapower extends the scalars in each finite-dimensional associated graded piece from K to K^* . Thus the graded ring of B^* is generated by the images of any fixed finite set of generators of $\mathfrak{m}$. Successive expansion in these generators, followed by completeness, identifies the jet filtration with powers of the ideal they generate. The complete filtered-ring argument then proves noetherianity: initial ideals are finitely generated in the noetherian associated graded ring, and successive approximation lifts those generators.

For an exact sequence of fixed finite modules, Artin–Rees makes the induced and intrinsic filtrations on its submodule cofinal. Use the induced filtration to obtain exact sequences of jets; cofinality identifies their inverse limits with the construction using intrinsic filtrations. The resulting

limit is exact. A fixed finite presentation identifies M^* with $B^* \otimes_B M$. This gives flatness by the finite-ideal criterion, and the nonzero residue extension gives faithful flatness. Lengths of the graded pieces are unchanged after residue-field extension, so their Hilbert growth, and hence the support dimension, is unchanged.

Suppose next that B is a domain and is not a field. Choose a discrete valuation v of its fraction field centered at $\mathfrak{m}$, for example one supplied by the normalized blowup of $\mathfrak{m}$. It is positive on $\mathfrak{m}$. The ideals

$$I_a = \{x \in B : v(x) \geq a\}$$

have zero intersection. Chevalley's lemma for a complete local ring implies that for every q there is an $a(q)$ with $I_{a(q)} \subseteq \mathfrak{m}^q$. Thus being nonzero modulo a fixed $\mathfrak{m}^q$ bounds $v(x)$ from above, uniformly in x . If $c_0 > 0$ is the minimum valuation of fixed generators of $\mathfrak{m}$, then $v(\mathfrak{m}^r) \geq rc_0$. For two fixed nonzero jets, these inequalities give a depth at which the product of any representatives of the two jets is nonzero. Applied on a set belonging to $\mathfrak{U}$, they show that the product of two nonzero elements of B^* cannot vanish. The assertion is immediate when B is a field.

For the varying construction, choose a uniform number of generators at each level. Each jet ultraproduct is then a quotient of a fixed finite free module over the corresponding jet of B^* and has finite length. Images of kernels at any fixed depth form a descending chain in a finite-length module, hence stabilize. The Mittag-Leffler argument gives surjectivity on inverse limits and a finite generating set over B^* . The same argument applied to a right-exact sequence proves the cokernel assertion. This does not assert exactness for arbitrary varying submodules.

For the uniform assertion, choose one finite free presentation $F \rightarrow C$, with kernel L , and lift it to $f_n : F \rightarrow H_n$. Then $H_n = P_n + f_n(F)$. If $x \in P_n \cap \mathfrak{a}^j H_n$, write

$$x = p + f_n(y), \quad p \in \mathfrak{a}^j P_n, \quad y \in \mathfrak{a}^j F.$$

Since x maps to zero in C , we have $y \in L$. Artin-Rees for the one fixed inclusion $L \subset F$ gives $y \in \mathfrak{a}^{j-c} L$, with c independent of n . As $f_n(L) \subset P_n$, this proves (19). Applying this successively to the fixed quotients of a finite filtration gives uniform strictness for each inclusion. Uniform cofinality transfers these bounds to the ambient jets, proving the final statement. $\square$

We now use the jet lemma in three specific ways: scalar extension of fixed modules, bounded-generation patching of the auxiliary modules, and uniform strictness of filtrations with fixed old quotients. It is the fixed quotient in the third assertion, not arbitrary varying-module exactness, that preserves the regular sequence below.

The domain assertion in Lemma 5.4 can be applied to each quotient by a minimal prime. Exactness and nilpotence of the nilradical then show that a fixed finite list of irreducible components is preserved by starring. The same applies to the special fiber and to inclusions between its components and the ambient components. In particular all the geometric assertions following (14) remain valid for B_η^* . This observation does not require B_i itself to be reduced.

Apply the construction to the surjections $B_\eta \twoheadrightarrow R_{\eta,n}$ and their fixed-determinant quotients. The module generator numbers are uniformly bounded. Indeed modulo the maximal ideal of $R_{\eta,n}$, (18) gives $V_j^2 = 0$. The quotient by all the V_j is the fixed old module in (16). The monomials $V_1^{e_1} \cdots V_h^{e_h}$ with $0 \leq e_j \leq 1$ therefore give a uniform bound, and Nakayama supplies the required generators. We obtain finite $N_{\eta,\infty}$ and surjections

$$B_\eta^* \twoheadrightarrow R_{\eta,\infty} \twoheadrightarrow R_{\eta,\infty}^f \twoheadrightarrow (R_\eta^f)^*.$$

The commuting U_j patch as well. They satisfy the patched versions of (18); the joint action algebra is consequently module finite over $R_{\eta,\infty}$. Modulo its original maximal ideal all the V_j are nilpotent, so adjoining their actions does not change the complete separated module topology.

Proposition 5.5 (The patched regular sequence). *On $N_{\eta,\infty}$ the sequence $\varpi, V_1, \dots, V_h$ is regular. Moreover*

$$\dim \operatorname{Supp}_{R_{\eta,\infty}} N_{\eta,\infty} \geq 2d + 4s + h, \quad (20)$$

$$N_{\eta,\infty}/(V_1, \dots, V_h)N_{\eta,\infty} \simeq (N_{\eta}^{\oplus 2^h})^*. \quad (21)$$

Proof. Polynomial layers with a fixed old action. Fix η , and write $\mathfrak{m}$ for the maximal ideal of B_η . The presentations of Lemma 5.2 give surjections $B_\eta \rightarrow R_{\eta,n}$ whose composites to R_η are independent of n . Regard

$$C = \overline{N}^{\oplus 2^h}$$

as a fixed finite B_η -module through this common composite. Let $J = (\varpi, V_1, \dots, V_h)$ denote the ideal of commuting operators on the auxiliary modules and on their patched limit.

Fix a truncation length r . For all sufficiently large n , the relations $(1 + V_j)^{2^n} - 1$ have J -order greater than r . Flatness in (15), together with the actual old-action identification (16), gives multiplication isomorphisms

$$\operatorname{gr}_J^{<r} N_{\eta,n} \simeq C \otimes_K K[T_0, T_1, \dots, T_h]_{<r}, \quad (22)$$

where T_0, T_j represent ϖ, V_j . These are isomorphisms of B_η -modules. Indeed the Galois action commutes with the J -operators, so its action on each layer is obtained from its action on $N_{\eta,n}/JN_{\eta,n} = C$. The fixed old composite therefore makes every degree a fixed finite direct sum of C . Only the extensions joining these layers vary with n .

Exactness at each fixed truncation length. Put $H_n = N_{\eta,n}/J^r N_{\eta,n}$ and $F_j = J^j H_n$ for $0 \leq j \leq r$. All these modules are finite over B_η . Their successive quotients are the fixed modules in (22), so their numbers of generators are bounded independently of n for this fixed r . If $\mathfrak{q}_n$ is the maximal ideal of $R_{\eta,n}$, the surjection from B_η gives

$$\mathfrak{m}R_{\eta,n} = \mathfrak{q}_n.$$

Thus the $\mathfrak{m}$ -adic filtrations are precisely the maximal-ideal filtrations used for patching these modules.

Apply Lemma 5.4(3) over the one fixed ring B_η to

$$0 \longrightarrow F_{j+1} \longrightarrow F_j \longrightarrow F_j/F_{j+1} \longrightarrow 0.$$

Its fixed-quotient Artin–Rees bound makes each inclusion uniformly strict. Composing the finitely many bounds compares the induced filtration from H_n with the intrinsic filtration on every F_j . The constants may depend on r , but are independent of n . Consequently this finite filtration patches exactly. No exactness for arbitrary varying submodules is required.

The ideal J^r is generated by a fixed finite number of monomials in its commuting generators. The cokernel assertion of Lemma 5.4(2) therefore identifies the patched H_n with $N_{\eta,\infty}/J^r N_{\eta,\infty}$; the same argument identifies its quotients by J^j . These identifications respect the multiplication maps and are compatible as r varies. We obtain

$$\operatorname{gr}_J N_{\eta,\infty} \simeq C^* \otimes_{K^*} K^*[T_0, T_1, \dots, T_h]. \quad (23)$$

Here C^* is the scalar extension of the fixed B_η -module C from Lemma 5.4(1).

Regularity, dimension, and augmentation. The finite joint action algebra generated by the U_j is complete local: modulo the deformation-ring maximal ideal, every V_j is nilpotent. Its module

topology agrees with the one used for patching. Since J lies in its maximal ideal, the finite module $N_{\eta,\infty}$ is J -adically separated and complete. A nonzero element killed by ϖ would therefore have a first nonzero J -initial form, contradicting injectivity of T_0 in (23). The same initial-form argument gives

$$\varpi N_{\eta,\infty} \cap J^m N_{\eta,\infty} = \varpi J^{m-1} N_{\eta,\infty} \quad (m \geq 1).$$

Hence the associated graded module after quotienting by ϖ is obtained by setting $T_0 = 0$. Repeating the argument for $V_1, \dots, V_h$ proves that the stated sequence is regular.

The final quotient by this regular sequence is C^* . Its dimension is at least $2d + 4s - 1$ by (17) and Lemma 5.4(1). Each member of the regular sequence raises support dimension by one on passing back to the original module. Dimensions over the finite joint action algebra and over $R_{\eta,\infty}$ agree, giving (20). Finally, apply cokernel patching to the diamond augmentation (16), without reducing by ϖ . The old B_η -action is fixed by the same common composite, so this gives (21) with that actual old deformation action. $\square$

5.4 Twisting and full support

Proposition 5.6 (Full support after patching). *The untwisted patched module satisfies*

$$\text{Supp}_{R_{1,\infty}} N_{1,\infty} \supseteq \text{Spec } R_{1,\infty}^f.$$

Consequently the old module has full support: $\text{Supp}_{R_1^f} N_1 = \text{Spec } R_1^f$.

Proof. Finite twisting and its dimension contribution. The finite characters of Lemma 5.3 give a local ring map after patching,

$$\Theta_\eta : R_{\eta,\infty} \longrightarrow R_{\eta,\infty}^f[[l_1, \dots, l_b]]. \quad (24)$$

Here one takes bounded total jets in the ring and character variables. At each such depth the equations $(1 + l_j)^{2^{a_n}} - 1$ disappear for large n . Coefficientwise patching of the finitely many monomials at each depth identifies the target with the displayed power-series ring. This explains both existence of the map and its target.

The map (24) is finite. Since A_n is a character-group quotient, choose, for each n and each j , a Galois element detecting exactly its j th coordinate. Its determinant divided by χ maps under twisting to $(1 + l_j)^2$. Although these Galois elements may vary with n , their determinant functions determine elements in every patched jet, and hence in $R_{\eta,\infty}$. The resulting elements have residue one. On the closed fiber of (24) they impose $l_j^2 = 0$. At $l_1 = \dots = l_b = 0$ the map is the fixed-determinant surjection. Its closed fiber is therefore an Artinian local algebra, of length at most 2^b , with the same residue field. Complete Nakayama proves finiteness.

Let T_η denote $N_{\eta,\infty}[[l_1, \dots, l_b]]$, restricted along (24) and then regarded as a finite B_η^* -module. Its support is the closed image of the power-series-extended support. Finiteness of the map preserves its dimension, and (20) gives

$$\dim \text{Supp}_{B_\eta^*} T_\eta \geq 2d + 4s + h + b = 8d + 4u + 2h + 2 = e.$$

Furthermore ϖ is regular on T_η , since it is regular before power-series extension and restriction of scalars. No minimal support prime of T_η is vertical.

Full support on the i -model and transfer to the 1-model. The dimension estimate now reaches the entire ambient dimension after twisting. The spectrum of B_i^* is irreducible of dimension e . It follows that T_i has full support. On reducing modulo ϖ , the finite-stage modules, actions, and twisting

maps for $\eta = 1, i$ are identical. Reduction is a cokernel, and power-series extension commutes with it and with the jets. Thus $T_1/\varpi T_1$ and $T_i/\varpi T_i$ are the same module over the identified special fibers. For any finite module T , local Nakayama gives

$$\mathrm{Supp}(T/\varpi T) = \mathrm{Supp}(T) \cap V(\varpi).$$

It follows that $\mathrm{Supp} T_1$ contains the generic point of every top-dimensional special-fiber component of B_1^* .

We give the component argument explicitly. Let $\mathfrak{q}$ be one of these generic points, and let $\mathfrak{p}$ be the unique minimal prime of B_1^* below it. The local geometry gives $\dim B_1^*/\mathfrak{p} = e$ and $\dim B_1^*/\mathfrak{q} = e - 1$; equivalently $\mathfrak{q}/\mathfrak{p}$ has height one. The latter also follows from the principal ideal theorem, since it is a minimal prime of the divisor ϖ on this component. Choose a minimal support prime $\mathfrak{a} \subseteq \mathfrak{q}$ of T_1 . The uniqueness assertion forces $\mathfrak{p} \subseteq \mathfrak{a}$. The prime $\mathfrak{a}$ is horizontal, so it is not $\mathfrak{q}$; there is no prime strictly between $\mathfrak{p}$ and $\mathfrak{q}$. Thus $\mathfrak{a} = \mathfrak{p}$. Every ambient component is obtained this way, and T_1 has full support.

Removing twists by actual sign intertwiners. It remains to remove the formal twists. For each fixed sign tuple $\sigma \in \{1, -1\}^b$, the finite-level quadratic twists described above give local ring automorphisms and semilinear module isomorphisms. They preserve maximal-ideal filtrations, and therefore patch, together with their inverses. In particular the support of $N_{1,\infty}$ is stable under the resulting sign automorphisms of $R_{1,\infty}^f$. Evaluation of (24) at $l_j = \sigma_j - 1$ is the fixed-determinant quotient followed by this patched sign twist, by its finite-level construction. The substitution is well-defined on formal series, since its values are 0 or -2 .

Let $x \in \mathrm{Spec} R_{1,\infty}^f$. Full support of T_1 supplies a point y in the support of $N_{1,\infty}[[l_1, \dots, l_b]]$ mapping to x . The determinant calculation, since x has fixed determinant, gives in the residue field $\kappa(y)$

$$(1 + l_j)^2 = 1, \quad \text{or equivalently} \quad l_j(l_j + 2) = 0.$$

For every j , primality therefore chooses $l_j = 0$ or -2 . Choose the corresponding sign tuple. Quotienting by these sign equations leaves a nonzero localized module, by Nakayama, and identifies the quotient with $N_{1,\infty}$ at a point x' of its support. Its image x is the sign transform of x' . Sign stability puts x in the untwisted support. In characteristic two the two signs coincide, with the same conclusion. This proves the first assertion.

Diamond augmentation and the old action. Finally consider the old quotient $(R_1^f)^*$. Let $\mathfrak{p}$ be any of its primes, viewed in $\mathrm{Spec} R_{1,\infty}$. It belongs to the support just proved. In the finite joint action algebra generated by the U_j , there is a support prime $\mathfrak{P}$ above $\mathfrak{p}$. On the old quotient each $\tau_{j,n}$ is trivial, so (18) becomes

$$U_j^2 - 2U_j + 1 = V_j^2 = 0.$$

Hence $V_j \in \mathfrak{P}$ for all j . The nonzero finite module localized at $\mathfrak{P}$ remains nonzero after quotienting by all the V_j , by Nakayama. Equation (21) shows that $(N_1^{\oplus 2h})^*$ has support at every prime of $(R_1^f)^*$. Faithful flatness in Lemma 5.4 now gives $\mathrm{Supp}_{R_1^f} N_1 = \mathrm{Spec} R_1^f$. $\square$

Proof of Theorem 5.1. We have proved full support on the enhanced completed global deformation germ with independent S -frames, global determinant χ , and unipotent bad inertia. After the initial base changes, the source locus has exactly these local conditions. At its absolutely irreducible curve, the representation and pseudo-deformation problems agree after a smooth choice of frames. Adding z , passing to a finite curve-field extension, and taking this completed localization are faithfully flat on the indicated germ, by Lemma 3.4. Consequently the closed Hecke support contains that source germ.

To pass from the germ to the asserted irreducible closed locus, work on its reduced irreducible coordinate ring. The image of any equation of the Hecke support is nilpotent in the completed source germ, by the set-theoretic containment just proved. Faithful flatness descends this nilpotence to the localization at the enhanced curve. That localization is reduced, so the image is zero there. It therefore vanishes on the entire irreducible locus: localization of a domain is injective. Thus the locus is pro-modular over the field reached by the allowed solvable base changes. This is precisely potential pro-modularity and proves the theorem. $\square$

6 Fields with controlled Selmer groups

The connectedness argument needs large totally real fields, very few allowed non-dyadic primes, and a strict Selmer bound at a small proportion of the dyadic places. We construct these fields here. The original allowed set, including the primes needed for the modular seed, is fixed before this construction.

Let Σ be a finite set of odd rational primes, and let $\bar{r}$ be a semisimple representation of $G_{\mathbb{Q}}$ over a sufficiently large finite field k of characteristic two, unramified outside $\Sigma \cup \{2\}$. If $\bar{r}$ is reducible, write its two characters as $\bar{\chi}_1, \bar{\chi}_2$ and put $\alpha = \bar{\chi}_1/\bar{\chi}_2$. Thus α has odd order, is even, and is unramified at 2. The last assertion follows from $\mathbb{Z}_2^\times$ being a pro-2 group. Write α_2 for its restriction to $G_{\mathbb{Q}_2}$. For a number field E and a finite set S_E containing the dyadic and real places, all Selmer groups below are subspaces of $H^1(G_{E,S_E}, k(\beta))$; at places outside S_E their local condition is unramified.

Proposition 6.1 (Field preparation). *There is a constant U , depending only on the fixed residual data and Σ , with the following property. One can construct abelian, totally real 2-extensions*

$$\mathbb{Q} \subset A \subset F$$

in which 2 splits completely, together with allowed sets S_A, S_F , such that the following assertions hold.

1. *The allowed sets contain the real and dyadic places and all places above Σ . Their numbers u_A, u_F of non-dyadic finite places are at most U . The degrees $d_A = [A : \mathbb{Q}]$ and $[F : A]$ can be made arbitrarily large, with $d = [F : \mathbb{Q}] = [F : A]d_A$.*
2. *If $\bar{r}$ is absolutely irreducible, its restriction to G_F is absolutely irreducible. In this case there are no marked places.*
3. *In the reducible case, the characters remain distinct on G_F if they were distinct on $G_{\mathbb{Q}}$. If $\alpha_2 \neq 1$, there are again no marked places. If $\alpha_2 = 1$, there is a dyadic place v_0 of A and the marked set M consists of all its extensions to F . Set $m = |M|$; thus $m = [F : A]$ and $m/d = 1/d_A$ in the marked case, whereas $m = 0$ otherwise.*
4. *In the reducible case, for $\beta = \alpha, \alpha^{-1}$ the following group is zero:*

$$\left\{ c \in H^1(G_{F,S_F}, k(\beta)) : \text{loc}_v(c) = 0 \text{ for every } v \in S_F \setminus M \right\} = 0. \quad (25)$$

Here the zero conditions include all real places. The condition at a marked place is unrestricted.

5. *In the reducible case, restriction preserves residual extension classes: for either ratio β , the map*

$$H^1(G_{A,S_A}, k(\beta)) \longrightarrow H^1(G_{F,S_F}, k(\beta))$$

is injective.

In the reducible case let $H_{\text{str}}^1(F, k(\beta))$ denote the Selmer group with zero condition at M , full condition at $S_F \setminus M$ (including the real places), and unramified condition elsewhere. Then

$$\dim_k H_{\text{str}}^1(F, k(\beta)) \leq \begin{cases} d + u_F, & \alpha_2 \neq 1, \\ 2d - 3m + u_F, & \alpha_2 = 1. \end{cases} \quad (26)$$

The field construction has three cases. For absolutely irreducible residue, a cyclic extension disjoint from the residual splitting field will suffice. For scalar residue, a narrow-class-group argument will give the required vanishing. For distinct residual characters, we will choose conductor primes successively so that changing the local Selmer conditions removes classes without introducing new ones. In each case we must bound the number of allowed primes independently of the degrees of the fields; this is what permits the strict dimension inequalities used in Section 7.

We use class field theory with real places in the modulus [42, Chapter V, Theorems 1.7, 3.5, and 3.6], the Kummer identification, and inflation–restriction. Local and global duality give the following Selmer dimension formula for a finite coefficient module V and linear local conditions $\mathcal{L}$:

$$\begin{aligned} \dim H_{\mathcal{L}}^1(V) - \dim H_{\mathcal{L}^\perp}^1(V^\vee(1)) &= h^0(V) - h^0(V^\vee(1)) \\ &+ \sum_v (\dim \mathcal{L}_v - h_v^0(V)). \end{aligned} \quad (27)$$

This is obtained by taking dimensions in Poitou–Tate and using the global Euler formula [40, Chapter I, Theorems 4.10 and 5.1, Remark 5.2(a)]. The real places are retained; their h_v^0 in this formula is ordinary invariant dimension. Our coefficient characters are trivial at real places, so local H^1 there has dimension one.

6.1 Cyclic fields and a scalar genus argument

Fix integers $n > a \geq 1$. If an odd prime r satisfies

$$r \equiv 1 \pmod{2^{n+1}}, \quad 2 \in (\mathbb{F}_r^\times)^{2^n}, \quad (28)$$

the degree- 2^n cyclic subfield $L(r, n)$ of $\mathbb{Q}(\zeta_r)$ is real and has 2 split completely. It is totally ramified at r and unramified at other finite primes. For an odd prime $\ell \neq r$, the Frobenius of ℓ generates its Galois group precisely when ℓ is a nonsquare modulo r . Indeed, both assertions are read in the cyclic quotient of $\mathbb{F}_r^\times$ of order 2^n ; an element generates that quotient if and only if its exponent in $\mathbb{F}_r^\times$ is odd.

The power conditions in (28) say that r splits in the Galois 2-extension

$$E_n = \mathbb{Q}(\zeta_{2^{n+1}}, 2^{1/2^n}).$$

This extension is unramified at odd primes. Consequently, independent square-root conditions at specified odd primes are compatible with splitting in E_n : any nontrivial product of their squareclasses is ramified at an odd prime, whereas every quadratic subfield of E_n is unramified there. Chebotarev [42, Chapter V, Theorem 3.23] therefore supplies r satisfying (28) with every $\ell \in \Sigma$ nonsquare modulo r . One can avoid any further fixed finite set of primes.

This already proves the field assertion for absolutely irreducible $\bar{r}$: take $F = L(r, n)$ and its degree- 2^a subfield for A . Choose r unramified in the residual splitting field. Every nontrivial subextension of $F/\mathbb{Q}$ is ramified at r , so F is disjoint from that splitting field. The residual image is therefore unchanged. Each $\ell \in \Sigma$ has one prime above it at every cyclic layer. The prime r need

not be allowed, since restricting a representation does not introduce ramification at a prime where it was unramified.

This completes the construction for absolutely irreducible residue: the residual image is unchanged and the number of allowed primes is bounded independently of the cyclic degree. For scalar residue we also need the Selmer vanishing in (25). We obtain it from a different fixed base field and the following elementary form of the one-prime genus argument.

Lemma 6.2. *Suppose that K/E is a quadratic extension of totally real fields and that exactly one finite prime of E ramifies. If the narrow class number of E is odd, the narrow class number of K is odd. For every totally real field with odd narrow class number, each totally positive unit is a square.*

Proof. If the narrow class number of K were even, the nonzero $\mathbb{F}_2$ -space $\text{Cl}^+(K)/2$ would have a nonzero invariant linear functional for $\text{Gal}(K/E)$. Here we use the elementary fact that a nonzero module of a finite 2-group in characteristic two has nonzero invariants, applied to the dual module. Class field theory would give a quadratic extension H/K , unramified at every finite prime, which is Galois over E . The group $G = \text{Gal}(H/E)$ has order four. Above the unique finite prime ramified in K/E , its inertia group I has order two and maps isomorphically to $\text{Gal}(K/E)$. It thus has trivial intersection with $\text{Gal}(H/K)$. This excludes $G \simeq \mathbb{Z}/4\mathbb{Z}$, whose only subgroup of order two is $\text{Gal}(H/K)$. Hence G is the Klein four group. All inertia groups above that prime equal I , since G is abelian, and all other finite inertia groups are trivial. The quadratic extension H^I/E is consequently unramified at all finite primes, contrary to oddness of the narrow class number of E . Ramification at infinity is allowed by the narrow class field and causes no exception to this argument.

For the last assertion, the signature cokernel of the units is a 2-group embedded in the narrow class group, so it is zero. For a totally real field of degree e , Dirichlet's unit theorem [41, Theorem 5.1] gives $|\mathcal{O}_E^\times/(\mathcal{O}_E^\times)^2| = 2^e$. The signature map from this group onto $\{\pm 1\}^e$ is therefore an isomorphism. Its kernel is the group of totally positive units modulo squares. $\square$

Now suppose $\alpha = 1$. Put

$$B = \mathbb{Q}(\sqrt{17}), \quad \pi = \frac{5 + \sqrt{17}}{2}.$$

The Minkowski bound [41, Theorem 4.3], here $\sqrt{17}/2 < 3$, says that every ideal class of B has a representative of norm one or two. Both primes above two are principal, generated by π and its conjugate, so the ordinary class number is one. The units -1 and $4 + \sqrt{17}$ realize every signature; the latter has norm -1 . Thus the narrow class number is also one.

Choose r as above, additionally inert in B , and put

$$F = BL(r, n), \quad A = BL(r, n)_a,$$

where the subscript denotes the degree- 2^a layer. Inertness in B is the extra condition that 17 be a nonsquare modulo r . It is compatible with all the preceding conditions, including when $17 \in \Sigma$, in which case it repeats an existing condition. Each quadratic step of $BL(r, n)/B$ has exactly one finite ramified prime, the prime over r . Lemma 6.2 implies that F has odd narrow class number and that its totally positive units are squares. Take S_A, S_F to contain only the prescribed places above Σ , two, and infinity; in particular r is not allowed. There are at most two primes above any $\ell \in \Sigma$.

Let w_1 be the dyadic prime of B generated by π , and w_2 the other dyadic prime. Both embeddings of π into $\mathbb{R}$ are positive. At w_2 it is a unit congruent to seven modulo eight: the odd root of $X^2 - 5X + 2$ is seven modulo eight. Thus π is not a square in $B_{w_2} = \mathbb{Q}_2$.

Choose any v_0 of A above w_1 . We prove a stronger version of (25), allowing arbitrary localization at every place of F above w_1 . It suffices to work with $\mathbb{F}_2$ coefficients. By Kummer theory a class in

this enlarged group is represented by a totally positive $x \in F^\times$ modulo squares. Its valuations are even outside the set T of dyadic places above w_1 , and it is locally a square at all places above w_2 . The valuation-parity map

$$[x] \mapsto (v(x) \bmod 2)_{v \in T} \quad \text{into } \mathbb{F}_2^T \quad (29)$$

is injective even on the larger space of totally positive squareclasses with even valuations outside T . Indeed, zero parity gives $(x) = \mathfrak{a}^2$; odd ordinary class number makes $\mathfrak{a}$ principal, and division by a square leaves a totally positive unit, which is a square by Lemma 6.2.

Our Selmer subspace is stable under the 2-group $\text{Gal}(F/B)$. If nonzero, it has a nonzero invariant vector. The action on T is transitive, so injectivity of (29) forces the parity vector of that invariant to be the all-ones vector. The unique totally positive squareclass with this parity is $[\pi]$: it has valuation one at every place above w_1 , and the injectivity just proved gives uniqueness. But π is nonsquare at every place above w_2 , since these completions are still $\mathbb{Q}_2$. This contradiction proves the required vanishing, also after extending coefficients from $\mathbb{F}_2$ to k .

Finally, the kernel of restriction of scalar H^1 from A to F consists of characters factoring through the cyclic group $\text{Gal}(F/A)$. It is generated over k by the unique nonzero $\mathbb{F}_2$ -valued quadratic character, and every nonzero multiple is ramified above r . As $r \notin S_A$, these characters are excluded. This proves the restriction assertion in the scalar case.

6.2 Switching local conditions

The scalar case is now complete. With distinct residual characters, we will replace the class-group argument by successive changes of local conditions. The next lemma explains why a switch removes classes when the opposite Selmer group supplies enough local evaluations.

For the rest of the construction assume $\alpha \neq 1$. Put $k_0 = \mathbb{F}_2(\alpha)$ and $q = |k_0| \geq 4$. Both α and α^{-1} are defined over k_0 ; the desired conclusions over k follow by extension of scalars, since the continuous cochain complexes and all the specified local conditions commute with this finite scalar extension. Write K_α for the cyclic, totally real, odd-degree field cut out by α .

This construction uses the viewpoint of Ramakrishna's auxiliary-prime switching strategy [49]; see also the exposition in Tang [56, Section 3A2, proof of Theorem 3.4]. The rank-one characteristic-two variant needed here is proved below, rather than obtained by applying either reference's lifting theorem.

We first record the exact local switch used below. At an odd place where the coefficient character is trivial and the residue cardinality is one modulo four, local H^1 is two-dimensional. Its unramified line L_{ur} is self-orthogonal. Any line generated by a ramified quadratic character is transverse to it and is also self-orthogonal, for the pairing with the opposite coefficient. Indeed, the mod-two Hilbert pairing is alternating when -1 is a local square, and a line in a two-dimensional nondegenerate alternating space is its own annihilator.

Lemma 6.3 (Local switching). *Let X be a Selmer group over a number field E with one-dimensional coefficient $k_0(\beta)$, and let Y be the Selmer group with coefficient $k_0(\beta^{-1})$ and orthogonal local conditions. Let T be a finite set of odd places of E where the coefficient character is trivial and the residue cardinality is one modulo four. Suppose that X has the unramified local condition at every place in T , and that evaluation of Y on unramified Frobenius gives a surjection*

$$Y \longrightarrow k_0^T.$$

Replacing the unramified lines at T by arbitrary transverse ramified lines changes X to

$$\ker(X \longrightarrow k_0^T).$$

In particular the switch introduces no new classes, and it strictly decreases X if its own evaluation is nonzero.

Proof. Let X^{rel} have full condition at T and the old conditions elsewhere. The relevant part of the Poitou–Tate comparison sequence is

$$0 \longrightarrow X \longrightarrow X^{\text{rel}} \longrightarrow \bigoplus_{v \in T} H_v^1/L_{\text{ur},v} \longrightarrow Y^\vee.$$

The last arrow is the transpose of localization on Y under the perfect pairings $H_v^1/L_{\text{ur},v} \times L_{\text{ur},v} \rightarrow k_0$. It is injective by the hypothesis, so $X^{\text{rel}} = X$. The new Selmer group is a subspace of this unchanged relaxed group. Its localizations must belong both to the old unramified line and the new transverse line, hence must be zero. This gives the asserted kernel exactly. $\square$

6.3 Frobenius evaluations with splitting conditions

Take $B = \mathbb{Q}$ if $\alpha_2 \neq 1$. If $\alpha_2 = 1$, choose a prime $q_0 \notin \Sigma \cup \{2\}$ with

$$q_0 \equiv 1 \pmod{8}, \quad \alpha(\text{Frob}_{q_0}) \neq 1,$$

and take $B = \mathbb{Q}(\sqrt{q_0})$. Such a prime exists by Chebotarev, since the odd-degree extension K_α is disjoint from $\mathbb{Q}(\zeta_8)$. In this second case let w_1, w_2 be the two dyadic places of B and let σ be its nontrivial automorphism.

All later conductor primes will satisfy (28), split in K_α , and avoid the original primes and q_0 . The first conductor r_1 also splits in B , with every $\ell \in \Sigma$ nonsquare modulo r_1 . Compatibility follows from the independence of the odd squareclasses just used and from the coprimality of $[K_\alpha : \mathbb{Q}]$ with every auxiliary 2-degree. For each subsequent conductor r , require every $\ell \in \Sigma$ and every earlier conductor to be a square modulo r . These conditions say that r splits in a finite Galois 2-extension E_2 obtained by adjoining E_n and those square roots. In the quadratic- B case, E_2 is unramified at q_0 . Consequently $B \cap E_2 = \mathbb{Q}$; the same is true of $B \cap K_\alpha E_2$. We can therefore impose either splitting or inertness in B at a new conductor. In particular, B is not included among the fields required to split at the final inert step.

Lemma 6.4 (Evaluation freedom). *Fix finitely many cohomology classes over B with coefficients $k_0(\beta)$, where $\beta = \alpha$ or α^{-1} , and impose the preceding splitting conditions in $K_\alpha E_2$. Evaluations use chosen places above the rational prime and chosen Frobenius representatives in a common finite quotient through which the coefficient actions and the finitely many cocycles factor. Numerical prescriptions refer to these choices.*

1. *At primes split also in B , a Frobenius evaluation on any k_0 -linearly independent collection of classes can be prescribed arbitrarily. If B is quadratic, the pair of evaluations on a space W at the two places above such a prime is given by one functional restricted to W and σW . It is uniformly distributed as two independent functionals on W whenever $W \cap \sigma W = 0$.*
2. *In the quadratic case, for primes inert in B the evaluations on W are uniformly distributed in $W^\vee$ whenever $1 + \sigma$ is injective on W . The evaluations on c and σc at the single place above such a prime are equal.*
3. *Collections for different Frobenius orbits of coefficient characters can be prescribed independently. If $\alpha^{-1} = \alpha^{2^j}$, coefficient Frobenius identifies the two collections semilinearly, and preserves all zero and rank conditions used here.*

Every compatible prescription is realized, for suitable choices of these places and representatives, at infinitely many rational primes. Uniform distribution refers to the finite evaluation quotients used for this choice; no analytic density estimate is required. Chebotarev determines simultaneous conjugacy classes. The applications below use only rank and nonvanishing conditions, unchanged by the resulting coefficient scalings and interchange of the two split places.

Proof. Set $N = G_{BK_\alpha E_2}$. The finite quotient

$$\text{Gal}(BK_\alpha E_2/B) = C \times P$$

has $C = \text{Gal}(K_\alpha/\mathbb{Q})$ of odd order and P a 2-group. On $k_0(\beta)$ the group C acts nontrivially. Averaging over C and using $k_0(\beta)^C = 0$ gives $H^i(C \times P, k_0(\beta)) = 0$ for all $i \geq 0$. Inflation–restriction thus makes restriction to N injective. On N the coefficients are trivial, so cocycles are additive homomorphisms and coboundaries are zero.

For independent classes $c_1, \dots, c_t$, their joint evaluation map $N \rightarrow k_0^t$ has additive image stable under multiplication by $\beta(C)$. These scalars generate k_0 over $\mathbb{F}_2$, so its image is a k_0 -subspace. If it were proper, a nonzero linear functional annihilating it would give a nontrivial linear combination of the c_i restricting to zero on N , a contradiction. Its image is therefore all of k_0^t . The paired assertion follows by applying this to $W + \sigma W$; trivial intersection makes its two summands independent. The conjugation formula for cocycles identifies these with the evaluations at the two conjugate places.

For the inert assertion choose $g \in G_{\mathbb{Q}}$ acting nontrivially on B and trivially on $K_\alpha E_2$. Then $g^2 \in N$, and g represents σ on cohomology. Since $\beta(g) = 1$, for $h \in N$ the cocycle identity gives

$$c((gh)^2) = c(g^2) + \text{ev}_h((1 + \sigma)c). \quad (30)$$

The first term is a fixed affine offset. The proven evaluation surjectivity on $(1 + \sigma)W$ shows that this expression is uniform in $W^\vee$ if $1 + \sigma$ is injective on W . Also $c(g^2) = (\sigma c)(g^2)$ and $(1 + \sigma)\sigma c = (1 + \sigma)c$ in cohomology. Evaluations on N do not see coboundaries, so (30) gives equality for c and σc , including the affine offset.

For different Frobenius orbits, regard these evaluation images as $\mathbb{F}_2[C]$ -modules. They have different simple isotypes; semisimplicity of $\mathbb{F}_2[C]$ makes their joint image the product of the individual images. In the remaining case the automorphism $x \mapsto x^{2^j}$ of k_0 carries cocycles with coefficient α to those with coefficient α^{-1} . It commutes with restriction, conjugation, and evaluation. Every ramified line below is generated by an $\mathbb{F}_2$ -valued quadratic character, so these lines, as well as the zero, full, and unramified conditions, are preserved. All evaluation matrices are entrywise Frobenius transforms and have the same ranks.

The finite collection of cocycles factors through a finite quotient after adjoining their coefficient fields. Chebotarev realizes the conjugacy class of the chosen element of this quotient. Choosing a place in the corresponding finite Galois extension above the rational prime then chooses a Frobenius representative; one may choose the prescribed representative in that class. Conjugacy scales coefficient evaluations and may interchange the two split places, which preserves every rank and nonvanishing condition in use. Removing finitely many primes leaves infinitely many choices. $\square$

6.4 The distinct-character construction

Given the chosen conductors $r_1, \dots, r_t$, take a diagonal cyclic quotient of the product of the degree- 2^n fields $L(r_j, n)$: if $\gamma_j : G_{\mathbb{Q}} \rightarrow \mathbb{Z}/2^n\mathbb{Z}$ defines its j th factor, use $\gamma = \sum_j c_j \gamma_j$ with every c_j odd. Let L be the field of γ , let L_a be its degree- 2^a layer, and write

$$F = BL, \quad A = BL_a, \quad \delta = \gamma \bmod 2.$$

When B is quadratic, choose any v_0 of A above w_1 and mark all its extensions to F . The character δ is the quadratic sum of the conductor characters. Every r_j is totally ramified in L , and 2 is split. At each original $\ell \in \Sigma$, the first summand of $\gamma(\text{Frob}_\ell)$ is odd and all later summands are even, so its Frobenius has order 2^n . At a previous conductor, adjoining a new conductor does not change the local quadratic line $k_0\delta$: the new quadratic character is unramified and trivial there. This follows from the imposed square conditions and quadratic reciprocity, since all conductors are one modulo four.

The allowed sets now include primes above all the r_j , in addition to the original allowed places. They do not need to include q_0 . For $\beta = \alpha, \alpha^{-1}$ define X_B^β as follows:

- its real conditions are zero;
- if $B = \mathbb{Q}$, its dyadic condition is zero; if B is quadratic, its dyadic conditions are full at w_1 and zero at w_2 ;
- its other conditions are unramified, except above the r_j , where they are the lines $k_0\delta$.

At a conductor the coefficient is trivial, since r_j splits in K_α . These definitions therefore make sense, including at a possible inert conductor.

The remaining objective is to kill both spaces X_B^β while keeping the number of conductors bounded. First we explain why this vanishing gives the desired Selmer condition over the large field F ; we then construct the switches that achieve it.

Killing both spaces X_B^β suffices for (25). To see this, in the quadratic case first enlarge the possible marked set to all places above w_1 . The resulting Selmer space over F is stable under the cyclic 2-group $\text{Gal}(F/B)$. If it were nonzero it would have a nonzero invariant class. Since $k_0(\beta)^{G_F} = 0$, inflation–restriction identifies these invariant classes with classes over B . At two and infinity the local conditions descend unchanged, because these places split completely. Away from the conductors, F/B is unramified, so a class whose restriction is locally unramified has unramified localization downstairs. This applies both outside the allowed set and where localization was required to vanish. At a conductor the local extension is cyclic and totally ramified of degree 2^n , and the kernel of local restriction is precisely the line of its quadratic character, namely $k_0\delta$. Thus the descended class belongs to X_B^β , giving the asserted sufficiency. For $B = \mathbb{Q}$ the same argument has no enlarged marked set.

The case $B = \mathbb{Q}$. Suppose first that $B = \mathbb{Q}$, so $\alpha_2 \neq 1$. For X_B^β the Selmer dimension formula gives

$$\dim X_B^\beta - \dim(X_B^\beta)^{\text{dual}} = -1. \quad (31)$$

Here the dual group has coefficient β^{-1} and orthogonal conditions. The real zero condition contributes -1 ; the dyadic zero condition contributes zero, since its local invariants vanish; all other local conditions contribute zero. Both global invariant terms vanish. Each opposite dual is therefore nonzero.

Append a conductor split in $K_\alpha E_2$ at which every opposite dual has nonzero evaluation, and every nonzero X_B^β also has nonzero evaluation. Such a choice exists by Lemma 6.4: per coefficient isotype there are at most two required nonvanishings, and the failure probability is at most $2/q < 1$. Different isotypes can be prescribed separately; Frobenius-conjugate inverse characters impose identical conditions. Lemma 6.3 says that neither space gains classes and that each nonzero space decreases. Repeating kills both spaces.

Preparing the quadratic case. It remains to carry out this argument when B is quadratic, so $\alpha_2 = 1$. We first impose an auxiliary vanishing over $\mathbb{Q}$. Let X_0^β have zero dyadic condition, full real condition, and the same non-dyadic conditions as X_B^β (unramified away from the conductors, and $k_0\delta$ at

them). Its dimension minus that of its opposite dual is again -1 : the dyadic contribution is -1 , the real contribution is zero, and all other contributions are zero. Restriction $H^1(G_{\mathbb{Q}}, k_0(\beta)) \rightarrow H^1(G_B, k_0(\beta))$ is injective, since $H^0(G_B, k_0(\beta)) = 0$. Rational classes and their opposite duals can therefore be included in the evaluation spaces over B in Lemma 6.4. Use the preceding one-coordinate switching argument to kill both X_0^β , requiring all the added conductors to split also in B . Lemma 6.4 permits this extra splitting. Keep all the resulting local switches.

Let Y_B^β be the group with coefficient β whose conditions are dual to those of $X_B^{\beta^{-1}}$. Thus its real conditions are full, and its dyadic conditions are zero at w_1 and full at w_2 . Its other conditions are the same as those of X_B^β . The dimension formula and conjugation give

$$\dim Y_B^\beta = \dim X_B^{\beta^{-1}} + 1, \quad \sigma X_B^\beta \subset Y_B^\beta. \quad (32)$$

Indeed, for X_B the two dyadic contributions are 2 and -1 , the two real contributions are -1 each, and the remaining contributions vanish. The inclusion follows from the local conditions: conjugation exchanges the dyadic conditions and the real zero conditions are contained in the full ones.

The following table summarizes the four groups used in the quadratic case. Every row has coefficient $k_0(\beta)$, and “full” means the full local H^1 . The two entries for B are ordered as (w_1, w_2) ; the single entry for $\mathbb{Q}$ is the condition at 2.

Group	Base field	Real places	Dyadic places
X_B^β	B	0	(full, 0)
Y_B^β	B	full	(0, full)
X_0^β	$\mathbb{Q}$	full	0
$(X_0^{\beta^{-1}})^{\text{dual}}$	$\mathbb{Q}$	0	full

At every selected conductor all four groups use the line $k_0\delta$, and at every other non-dyadic finite place they use the unramified condition. The rational opposite dual in the last row will keep $X_0 = 0$ while the paired conditions over B are switched.

Lemma 6.5. *As long as all conductors used so far split in B and both X_0^β are zero, one has*

$$W \cap \sigma W = 0 \quad \text{for } W = X_B^\beta \text{ and } W = Y_B^\beta.$$

Proof. The intersection is stable under σ . If nonzero it has a nonzero invariant class, which descends to $\mathbb{Q}$ because $k_0(\beta)^{G_B} = 0$. Its dyadic localization is zero: intersecting either pair of conjugate dyadic conditions imposes zero at both w_1 and w_2 . At infinity it satisfies at least the full condition defining X_0^β ; this observation also covers $W = Y_B^\beta$. At every conductor, splitting in B makes descent of the rational line $k_0\delta$ immediate. At other primes unramified in $B/\mathbb{Q}$, the unramified condition descends by restriction to inertia. There is no condition to check at q_0 : the coefficient is unramified with nontrivial Frobenius there, so local H^0 and H^2 vanish and the local Euler formula gives $H^1 = 0$. Thus the nonzero descended class lies in X_0^β , a contradiction. $\square$

We have arranged $X_0^\beta = 0$ for both coefficient characters. The next split switches must preserve these vanishings while decreasing the nonzero spaces X_B^β . The preceding lemma, together with Lemma 6.4, supplies the freedom to choose paired evaluations needed for each new split conductor.

Split switches. Now suppose both X_B spaces are nonzero. Include the restrictions of the rational opposite-dual classes in the evaluation spaces just described. Choose a new split conductor with the following three properties, for each coefficient isotype:

1. the paired evaluation on Y_B^β has rank two;

2. the paired evaluation on X_B^β is nonzero;
3. the evaluation on the opposite dual of the zero space $X_0^{\beta^{-1}}$ is nonzero.

The third dual has dimension one, by its dimension difference. The first space has dimension at least two by (32). Lemmas 6.4 and 6.5 make the paired evaluations on each of the first two spaces independent uniform functionals. For $t = \dim Y_B^\beta \geq 2$, the rank failure probability is

$$1 - (1 - q^{-t})(1 - q^{1-t}) \leq q^{-1} + q^{-2}.$$

The second failure probability is at most q^{-2} , and the third is q^{-1} . Their sum satisfies

$$(q^{-1} + q^{-2}) + q^{-2} + q^{-1} < 1. \quad (33)$$

No independence between these three failure events is needed for this union bound. Different character isotypes are independent; conjugate inverse characters give identical requirements under coefficient Frobenius. Hence an admissible conductor exists.

Switching at its two places makes both X_B spaces smaller and introduces no new classes. The third requirement has a separate purpose: Lemma 6.3, now over $\mathbb{Q}$, preserves $X_0^\beta = 0$. Thus Lemma 6.5 remains applicable at the next split step. Continue until both X_B spaces vanish or exactly one remains.

The final inert switch. In the latter situation write $X_B^\beta \neq 0$ and $X_B^{\beta^{-1}} = 0$. Formula (32) implies

$$\dim X_B^\beta = \dim Y_B^\beta = 1, \quad Y_B^\beta = \sigma X_B^\beta. \quad (34)$$

If the inverse characters were Frobenius conjugate, their X_B spaces would have equal dimension, so this case could not arise.

Choose one final conductor inert in B , still split in $K_\alpha E_2$. Immediately before this step Lemma 6.5 holds, so $1 + \sigma$ is injective on each relevant X_B and Y_B : its kernel would be a fixed vector in $W \cap \sigma W$. At this conductor require nonzero evaluation on both opposite duals $Y_B^\alpha, Y_B^{\alpha^{-1}}$ and on the remaining X_B^β . There is no third independent demand in the surviving isotype. By (34) and the exact equality in Lemma 6.4, the evaluations on Y_B^β and X_B^β coincide. The affine offset in (30) does not change this equality. Uniform evaluation and the union bound give the conservative failure bound $2/q < 1$ per isotype. The switch at the single place over this inert prime therefore kills the last X_B space, while the opposite zero space stays zero. This is the final step: preservation of $X_0 = 0$ after it is neither required nor asserted. Its pre-step vanishing was used precisely to justify injectivity of $1 + \sigma$.

6.5 Uniformity, restriction, and the numerical bounds

The constructions now give the required dual Selmer vanishing. To finish Proposition 6.1, we bound the number of conductors, check preservation of residual data, and convert that vanishing into the strict Selmer bounds. The conductor count is the point that allows both cyclic exponents to grow.

The number of conductors in the preceding argument is bounded independently of n and a . Here is an explicit accounting which also avoids making any uniformity assumption about their sizes. Let $b = [B : \mathbb{Q}] \leq 2$, and let

$$C_E = \sum_{\beta \in \{\alpha, \alpha^{-1}\}} \dim_{k_0} H^1(G_{E, T_E}, k_0(\beta)), \quad E = \mathbb{Q}, B,$$

where T_E is the fixed set above Σ , two, and infinity. In the quadratic case one may include q_0 in T_B without change to the relevant local cohomology. Count both characters in this sum even if they

are Frobenius conjugate; the resulting bound is only larger. Allowing one new rational prime freely adds at most $2b$ to each global H^1 dimension over B , by restriction to inertia or by the localization sequence. Thus it adds at most $4b$ to their sum. This deliberately generous bound applies uniformly to every conductor and every quadratic line chosen there.

After the first conductor, the sum of the two dimensions over $\mathbb{Q}$ is at most $C_{\mathbb{Q}} + 4$. The first loop killing the auxiliary X_0 spaces therefore has length

$$N_0 \leq C_{\mathbb{Q}} + 4.$$

During it, the sum of the B -dimensions is at most $C_B + 4b(1 + N_0)$. Every later split switch decreases that sum by at least one and never increases either space. There is at most one final inert switch. Hence in the quadratic case the total number t of conductors satisfies, for example,

$$t \leq 2 + C_B + 4b + (1 + 4b)(C_{\mathbb{Q}} + 4). \quad (35)$$

For $B = \mathbb{Q}$ the direct loop gives the simpler bound $t \leq 1 + C_{\mathbb{Q}} + 4$. These constants depend only on the fixed original data and the fixed choice of B .

Every original $\ell \in \Sigma$ has full Frobenius order in L and its degree- 2^a layer, so has at most b primes in either A or F . Every conductor is totally ramified in both cyclic layers, so likewise has at most b primes. Consequently

$$u_A, u_F \leq b(|\Sigma| + t).$$

The analogous scalar bound is $2|\Sigma|$, and the absolutely irreducible bound is $|\Sigma|$. This proves the required common constant U . The degrees are $d_A = b2^a$, $d = b2^n$, and $m = 2^{n-a}$ in the marked cases. Taking a and $n - a$ large makes d_A and $[F : A]$ large independently. In the marked cases this makes m large and gives $m/d = 1/d_A$.

Since $K_{\alpha}/\mathbb{Q}$ has odd degree, α remains nontrivial on every field constructed in the distinct-character case. Inflation–restriction, with $H^0(G_F, k(\beta)) = 0$, proves injectivity of restriction from A to F . Together with the scalar and absolutely irreducible cases proved above, this establishes all the field and restriction assertions of Proposition 6.1.

It remains to derive (26). Since the mod-two cyclotomic character is trivial, the Tate dual of $k(\beta)$ is $k(\beta^{-1})$. Its Selmer conditions dual to the strict ones are exactly those in (25). The dual Selmer group is zero. The global invariant terms in the Selmer dimension formula cancel, whether $\beta = 1$ or $\beta \neq 1$. At a real place, the full condition contributes $\dim H_v^1 - h_v^0 = 1 - 1 = 0$. At an unmarked dyadic place, local Euler characteristic gives contribution one if $\alpha_2 \neq 1$ and two if $\alpha_2 = 1$. At a marked place the zero condition contributes $-h_v^0 = -1$. At a non-dyadic allowed place the full condition contributes $h_v^2 \leq 1$, and an unramified condition elsewhere contributes zero. Adding these terms gives exactly the two bounds in (26).

For later use, the degree choices can in particular ensure the strict inequalities

$$d + u_F + 3 < 2d - 1 \quad \text{in the unmarked reducible case,} \quad (36)$$

$$2d - 2m + u_F + 3 < 2d - m - 1 \quad \text{in the marked case,} \quad (37)$$

$$2d - 2m - 4 - u_F > d. \quad (38)$$

For example, it is enough in the marked case to have $m > U + 4$ and $m(d_A - 2) > U + 4$. The same freedom makes d_A larger than any prescribed fixed cost in the initial trace cuts. These choices are legitimate because (35) and the bounds for u_A, u_F are independent of both cyclic exponents.

7 Connectedness and potential pro-modularity

We now connect a modular seed to the target. The field construction of Section 6 makes the reducible locus small after recording local lines at selected dyadic primes. The unrestricted propagation theorem of Section 5 can then cross successive component intersections.

Theorem 7.1. *Let r satisfy the hypotheses of Theorem 1.1 and be not virtually solvable. Then r is pro-modular over some solvable totally real Galois extension of $\mathbb{Q}$, of even degree and split completely at 2.*

More generally, fix a complete integral framed deformation family over $\mathbb{Q}$, with determinant $\chi = \det r$, containing the coefficient point defined by r , and with ramification confined to a finite set. The closed trace image of this family is pro-modular after restriction to one common extension of the stated kind, at one fixed tame level.

To prove the family assertion, we will compare a curve in a Hecke component with a curve in an unrestricted deformation component containing the target family. The first curve requires a cuspidal lift of the target's semisimple residue. We choose that lift before preparing the fields, so that its tame primes belong to the fixed allowed set used in the field estimates.

7.1 A modular residual seed

Lemma 7.2. *Every semisimple two-dimensional residual representation arising from the target has a classical cuspidal lift over $\mathbb{Q}$, after finite coefficient extension and allowing a finite enlargement of the tame level.*

Proof. For absolutely irreducible residue this is Serre's modularity theorem, including characteristic two [33, Theorem 0.2]; the results are due to Khare–Wintenberger and Kisin [31, 32, 35]. The characteristic-two parity condition is automatic.

For reducible residue, the existence of a cuspidal lift, including in characteristic two, is already proved by Billerey–Menares [9, Theorem 2.1]. The geometric argument below uses the same Eisenstein difference as their [9, Theorem 2.3]; we retain the details of integrality, cuspidality, and lifting needed here.

Write the reducible residue as $\bar{\chi}_1 \oplus \bar{\chi}_2$ and put $\alpha = \bar{\chi}_1/\bar{\chi}_2$. After a finite-order twist it suffices to lift the eigensystem of $1 \oplus \alpha$. The character α has odd order, is unramified at 2, and has an even Dirichlet lift of odd conductor. Choose an even weight $k' \geq 4$ and a normalized Eisenstein series f with this character pair and $a_1(f) = 1$. Take an odd level $N \geq 5$ divisible by the conductor. Let ℓ be a prime congruent to one modulo $N2^a$, where a will be sufficiently large. Put

$$g(\tau) = f(\tau) - f(\ell\tau).$$

Regard this as a form of level $\Gamma_1(N\ell)$. Its positive Fourier coefficients at infinity are 2-integral, and its constant term there is zero. The tame modular curve has good reduction, with geometrically integral special fiber [30, Corollary 10.9.5(2) and Table 10.9.6]. The q -expansion principle [29, Theorem 1.6.1 and Corollary 1.6.2] therefore extends g integrally: a vertical pole would be detected at that component's cusp expansion.

We verify cuspidality modulo ϖ at every cusp. For a cusp matrix $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, the primitive-column decomposition is

$$\begin{pmatrix} \ell & 0 \\ 0 & 1 \end{pmatrix} \gamma = \gamma' \begin{pmatrix} e & t \\ 0 & \ell/e \end{pmatrix}, \quad e \in \{1, \ell\}.$$

Since $\ell \equiv 1 \pmod{N}$, one can choose $\gamma' \equiv \gamma \pmod{N}$. The two constant terms in $g|_{k'}\gamma$ therefore differ by the factor

$$1 - (\ell/e)^{-k'}.$$

There are only finitely many constant terms of f at level N . Choose a larger than their denominator valuations, with an additional power sufficient for reduction modulo ϖ . The displayed factor makes every constant term of g vanish in the residue field. Its coefficient of q is one, so the reduction is nonzero. Away from $N\ell$, it is an eigenvector with the Eisenstein residual eigensystem $1 \oplus \alpha$.

This mod- ϖ cusp form lifts to the characteristic-zero cuspidal space in the same weight. Indeed on the tame modular curve, Kodaira–Spencer gives $\Omega^1 \simeq \omega^2(-\text{cusps})$. For this formula and positivity of ω , use [29, Section 1.5 and Appendix A1.5.4]: after adjoining the required roots of unity, a symplectic full-level component maps to the Γ_1 -curve by a finite cover of odd degree. The logarithmic Kodaira–Spencer isomorphism descends along this tame cover, and positivity descends along the finite map. On the special fiber, Serre duality [58, Lemma 53.4.1 and Remark 53.4.3, Tags 0BS2 and 0BS4] identifies the obstruction $H^1(\omega^{k'}(-\text{cusps}))$ with the dual of $H^0(\omega^{2-k'})$, which is zero for $k' \geq 4$. The corresponding integral H^1 -module is finite by proper coherent cohomology [58, Lemma 30.19.2, Tag 02O6]. The multiplication-by- ϖ exact sequence and the special-fiber vanishing make multiplication by ϖ surjective on this module. Nakayama’s lemma makes it zero, so reduction of cuspidal sections is surjective. The finite integral cuspidal Hecke module has a characteristic-zero eigenpacket specializing to this residual eigensystem: its Hecke algebra is finite and torsion-free over $\mathcal{O}$, and the residual maximal ideal lies above a horizontal minimal prime. Finally undo the finite-order twist. The construction also covers $\alpha = 1$. $\square$

7.2 Field choices

Fix the target’s semisimple residue $\bar{r}$ and determinant χ . Choose a cuspidal lift of this residue by Lemma 7.2, and fix a finite allowed set over $\mathbb{Q}$ containing the bad primes of the target, the prescribed family when present, and this lift. This choice precedes the field construction below; it ensures that the bound for the number of allowed primes also covers the modular seed. In the reducible case write $\bar{r}^{\text{ss}} = \bar{\chi}_1 \oplus \bar{\chi}_2$ and $\alpha = \bar{\chi}_1/\bar{\chi}_2$. Use the fields

$$\mathbb{Q} \subset A \subset F$$

of Proposition 6.1, with $d_A = [A : \mathbb{Q}]$, $d = [F : \mathbb{Q}]$. There are no marks when $\bar{r}$ is absolutely irreducible or $\alpha|_{G_{\mathbb{Q}_2}} \neq 1$. Otherwise choose the distinguished place v_0 of A and mark its $m = [F : A]$ lifts to F . The numbers u_A, u_F of allowed non-dyadic primes are uniformly bounded as these degrees grow.

Choose the degrees sufficiently large that

$$d > u_F + 4, \quad m > u_F + 4 \quad \text{if there are marks,} \quad 2d - 2m - 4 - u_F > d. \quad (39)$$

In the unmarked case set $m = 0$. Since $m/d = 1/d_A$ in the marked case and $m/d = 0$ otherwise, the further fixed lower bounds on d_A below can be imposed simultaneously.

7.3 Charts with transverse local lines

Over the characteristic-two fixed-determinant pseudo-base, take the finite-type framed representation scheme and adjoin a local invariant projective line at every mark. Call the resulting scheme Y . In its closed residual fiber, call a point *transverse* if none of its marked lines is globally invariant. The condition is geometric: it must hold after extending its residue field to an algebraic closure. There

is no additional condition when there are no marks. The transverse locus is open in the residual fiber, since the incidence of a global invariant line equal to a specified marked line is proper.

At a transverse point with marks, the local residual representation at each mark is scalar after untwisting. Indeed there is a global invariant line, and the different marked invariant line splits its restriction. The two local semisimple characters coincide. This simple observation is responsible for the strict Selmer condition in the next lemma.

Lemma 7.3. *Let F , the marked set, and the fixed-determinant representation scheme Y be as above, with the degree inequalities (39). Let y be a transverse closed residual point of Y , with finite residue field, and let R'_y be the completion of Y at y . Every component of $\text{Spec } R'_y$ has dimension at least $2d - m$. Its components can be joined by a chain whose successive intersections have dimension at least $2d - m - 1$. The globally reducible locus has dimension strictly less than $2d - m - 1$.*

Proof. Retain the real local problems and the m marked dyadic line problems as the local base. Reducing this base changes no spectrum. By Lemma 3.12, it is a complete local domain of dimension

$$2d + 5m.$$

The relative framed presentation of Proposition 3.6 has generators minus relations at least

$$3 - 6m - h^0((\text{ad}^0)^\vee(1)) \geq -6m.$$

The last inequality uses the three-dimensionality of $(\text{ad}^0)^\vee(1)$. A quotient of a complete local domain with this presentation has all components of dimension at least $2d - m$. Grothendieck's connectedness theorem gives connectedness in dimension at least $2d - m - 1$: in the form used here, cutting a complete local domain of dimension N , after adding a formal variables, by b equations gives connectedness in dimension at least $N + a - b - 1$. This is the component-chain formulation of the connectedness theorem [27, Exposé XIII, Théorème 2.1].

We bound the reducible locus separately. It is empty for absolutely irreducible residue. Otherwise add a global invariant line by proper incidence. At a closed point of that incidence, choose an adapted basis and first hold this line fixed. Its movement costs one parameter. The diagonal global characters with fixed product vary in dimension at most one, by abelian Leopoldt over F ; see Lemma 3.8. Freeze those character parameters.

At a marked place the closed local matrices are scalar and the marked line is transverse to the chosen global line. An infinitesimal upper extension, with diagonal characters frozen, must therefore vanish locally: a nonzero upper nilpotent matrix cannot preserve a second transverse line. Thus its cohomology class is strict at the marks. Moving a marked line costs at most one further tangent parameter, giving m in total. There is at most one upper coboundary parameter. The strict Selmer bounds in (26) therefore give the following dimensions:

$$\begin{cases} d + u_F + 3, & \alpha|_{G_{\mathbb{Q}_2}} \neq 1, \\ 2d - 2m + u_F + 3, & \alpha|_{G_{\mathbb{Q}_2}} = 1. \end{cases}$$

Here the three remaining parameters account for the diagonal base, the global line, and the possible coboundary. The relative tangent estimate bounds the number of generators over the diagonal base, so it is an upper bound for the whole formal reducible chart, not only for its tangent space at a fixed diagonal character. Properness of the global-line incidence transfers the bound to its image. Finite residue extensions do not affect these dimensions.

The first two inequalities in (39) make both bounds smaller than $2d - m - 1$. This proves the lemma. $\square$

Lemma 7.4. *At a transverse closed residual point, either every component of Y through that point has potentially pro-modular pseudo-image, or none does. It suffices to begin with one component whose pseudo-image contains a potentially pro-modular, non-virtually-solvable curve having finite local images at the allowed non-dyadic places.*

Proof. Work first in the completed chart. By Lemma 7.3, two successive components in a connecting chain intersect in dimension at least $2d - m - 1$. Choose an irreducible component of maximal dimension in that intersection. Its generic global representation is absolutely irreducible, because the reducible locus has smaller dimension. Pass first to the completed trace-and-character image, retaining the characters on the marked lines. Forgetting the global frame and the line choices loses at most $3 + m$ dimensions, by Lemma 3.2. Proposition 3.1 makes this image finite over its trace image, so forgetting the recorded characters causes no further dimension loss. Thus the pseudo-image of the chosen component has dimension at least

$$2d - 2m - 4.$$

At each allowed non-dyadic prime impose that the trace of a chosen Frobenius lift equal its residual value. These u_F equations leave a closed locus of dimension at least $2d - 2m - 4 - u_F > d$. The bad pseudo-locus bounds of Lemma 3.8 and curve avoidance therefore produce a non-virtually-solvable curve in it. Lemma 3.9 gives finite local images at the non-dyadic allowed places.

If the preceding component is potentially pro-modular, this curve is potentially pro-modular. Theorem 5.1 makes the next component potentially pro-modular. A starting curve as in the statement first makes its containing component potentially pro-modular by that same theorem. Iterating crosses the entire chain. Only finitely many extensions and levels occur. Take the compositum of their Galois closures and a common deeper level, using Lemma 4.6.

Completion is faithfully flat. Every component of Y through the closed point is the contraction of a minimal prime of the completed chart. The component pseudo-images are the corresponding trace contractions. The conclusion therefore descends from the chart to Y . $\square$

The local chart statement must next be joined across different residual extension classes. The transverse residual condition was chosen so that this joining does not require changing the semisimple residue.

Lemma 7.5. *If nonempty, the transverse closed residual fiber of Y is geometrically connected. Consequently, if one transverse chart has potentially pro-modular component images, every transverse chart does.*

Proof. Work over an algebraic closure of the residue field. For scalar residue with marks, a transverse residual representation is a nonzero additive extension after a scalar twist. It has a unique global invariant line. Its cocycle is zero at the marks. With the global line fixed, these cocycles form a vector space with the origin removed. The marked complementary lines form products of affine line charts. Conjugating the basis gives a connected parameter space whose image is the entire transverse fiber.

For distinct residual characters, use both upper triangular orientations. In either orientation, with its global line fixed, the extension cocycles form a vector space. In the marked case they vanish locally at the marks. The allowed choices of marked lines form a nonempty open subset of the product with projective lines, hence an irreducible parameter space. Varying the basis preserves connectedness. The two orientations meet at the split representation; when there are marks, choose lines avoiding both global invariant lines. Without marks there is no transversality condition, and the same two families meet at that split representation. For absolutely irreducible residue, the

residual fiber is a conjugacy orbit. These parameter spaces are finite-dimensional algebraic spaces of cocycles: this follows either from the residual representation scheme or from the finite-dimensional residual H^1 . This proves geometric connectedness.

There are finitely many components of the noetherian scheme Y . Their nonempty intersections with the transverse residual fiber form a finite closed cover. Make a graph with these components as vertices, joining two vertices when they meet in that fiber. The graph is connected because the fiber is connected. Choose a spanning tree and, for each edge, a closed point of the corresponding intersection in the transverse locus, after a finite coefficient extension. At that point Lemma 7.4 gives the same potential pro-modularity status to the two components.

Only finitely many such points and completed-chart component chains are needed. Their field extensions have a common Galois compositum, and forward base change gives one common deeper tame level. Thus the conclusion holds simultaneously for every component meeting the transverse fiber, and hence for every transverse chart. Every component through a transverse point is accounted for by completion, as in the preceding lemma. The finite coefficient extensions are harmless because the integral forms and their Hecke support commute with them. $\square$

7.4 From the two curves to the target family

The proof first constructs large seed and target loci over A , then chooses a characteristic-two curve in each. Appropriate lattices put both curves in transverse charts over F . Connectedness transfers potential pro-modularity from the seed curve to the target curve; unrestricted propagation then reaches the entire containing target locus.

Proof of Theorem 7.1. The seed and target loci over A . For the seed and fields already chosen, solvable base change and Jacquet–Langlands give a cuspidal residual packet over A . It remains cuspidal by regularity, as in Lemma 4.6.

Its central character need agree with $\psi = \chi\varepsilon$ only modulo ϖ . At a sufficiently deep tame level their transformations coincide on reduction. The exact integral function model of Proposition 4.1 therefore gives residual occurrence also for the specified central character ψ . Proposition 4.4 supplies a pro-modular component C_{seed} in the fixed-determinant pseudo-base over A with

$$\dim C_{\text{seed}} \geq 1 + 2d_A.$$

Independently choose an unrestricted global fixed-determinant framed component through $r|_{G_A}$, or containing the restricted framed support of the prescribed family. Such a component exists by taking a minimal prime below that support. Its generic representation is absolutely irreducible because it contains the absolutely irreducible target point. The relative presentation over the real local rings gives dimension at least

$$1 + 2d_A + 3 - h^0((\text{ad}^0)^\vee(1))$$

before forgetting the frame. The loss under trace contraction is at most three. Its pseudo-image C_{target} therefore has dimension at least $2d_A - 2$. This large unrestricted component contains the trace image of the initially prescribed family; no lower dimension bound on that smaller family is required.

Choosing curves in the two loci. In each of C_{seed} and C_{target} , pass to the special fiber and prescribe residual Frobenius trace at every bad non-dyadic place. If there are marks, prescribe in addition that the entire local pseudorepresentation at v_0 is its constant residual one. This last condition costs at most the number of topological generators of one fixed local pseudo-ring for $\mathbb{Q}_2$, a constant

independent of d_A . All the other costs are bounded by $1 + u_A$, with a fixed additional constant for the target dimension estimate. Taking d_A sufficiently large leaves dimension greater than d_A . By Lemma 3.8, each locus contains a non-virtually-solvable curve. The Frobenius constraints make its bad local images finite.

Entering the transverse fiber over F . We explain why both curves enter transverse charts over F . Without marks, any stable lattice suffices. With marks, the local pseudo at v_0 is constant and scalar after untwisting, so the curve-field representation has an invariant line there after a finite coefficient extension. Choose a nonzero vector v on this line and let

$$\mathcal{V} = \sum_{g \in G_A} k'[[t]] \rho(g)v.$$

Compactness bounds this module inside a fixed lattice. Global irreducibility makes it full rank, and it is stable by construction. The vector v is primitive in $\mathcal{V}$: if it belonged to $t\mathcal{V}$, all its translates would, and Nakayama would force $\mathcal{V} = 0$. Its reduction lies in no global invariant line, since its orbit spans $\mathcal{V}/t\mathcal{V}$.

Restrict the lattice to G_F . No new global invariant line appears in the residue. For distinct residual characters, a nonsplit extension remains nonsplit by the restriction injectivity in Proposition 6.1; a split representation retains its two distinct character lines. For scalar semisimplification, the spanning property above rules out a split scalar residue, and its nonzero additive extension remains nonzero by the same injectivity. These descriptions hold after extending the residue field. Since F/A is Galois, G_A permutes the globally invariant lines of the restricted residue. Transporting the chosen local line to every prime over v_0 therefore preserves its avoidance of all these lines. Hence every transported marked line is transverse. Both curves pass through transverse points of the same residual scheme Y .

Returning from the target curve to its containing locus. The seed curve remains potentially pro-modular after restriction to F . Its finite bad local images allow Theorem 5.1 to start the propagation in its transverse chart. Lemma 7.5 then gives potentially pro-modular images at every transverse chart. In particular the restricted target curve is potentially pro-modular. By definition, the original target curve over A is therefore potentially pro-modular as well. Apply Theorem 5.1 over A to this original curve and its containing locus C_{target} . The whole locus, and therefore the target and the prescribed family trace image, becomes pro-modular after a common allowed extension.

The finite component graph in Lemma 7.5 and the finite chains inside its selected charts require only finitely many propagation steps. Replace their fields by one Galois compositum and their levels by one deeper tame level, using forward base change. Total reality, complete splitting at 2, and solvability over $\mathbb{Q}$ are preserved. This proves both assertions. $\square$

The distinction between the two spaces in this proof is important. Marks and transverse lines produce connectedness in the global special fiber. The propagation theorem subsequently applies to unrestricted dyadic deformation loci. No ordinary or potentially crystalline restriction has been imposed on the target. The next section extracts its classical vectors.

8 Classicality and descent

Theorem 7.1 places the target representation in the closed Hecke support after a solvable totally real base change. We now produce a locally algebraic vector with its eigensystem. The additional work is needed when the local representation at 2 is reducible: the two possible orders in a unitary principal series need not have the same locally algebraic vectors. Throughout this section the target

r is not virtually solvable, as arranged in the preliminary reductions. In particular, its restriction to every finite extension is absolutely irreducible.

We retain the conventions of Section 4: $\chi = \det r$, the central character of completed forms is $\psi = \chi\varepsilon$, and the local block parameter is $r \otimes \varepsilon$. Cyclotomic exponents are counted with $\exp(\varepsilon) = 1$. Thus a de Rham character of exponent a has Hodge–Tate weight $-a$ in the convention used by the ordinary modularity theorem below. The field F will always be totally real, solvable over $\mathbb{Q}$, of even degree, and split completely at 2. Until the final subsection, assume that $r|_{G_{\mathbb{Q}_2}}$ is reducible. We will retain its higher-exponent line in a global family, show that the joint trace-and-character support is finite over weight space, and use dense points that become modular after normalization and dualization. Those points determine the inducing order to retain at the target. Specialization will then give an actual vector, whose local algebraicity is proved directly. The locally absolutely irreducible case will use Proposition 4.2 in the final subsection.

8.1 The local line and its global family

We first record the rank-one consequence of Bloch–Kato that fixes the orientation. For a de Rham character θ of $G_{\mathbb{Q}_2}$, write $H_g^1(\mathbb{Q}_2, \theta)$ for the subspace of de Rham extension classes. The local Euler characteristic, duality, and the Bloch–Kato formulas give

$$\dim H^1(\mathbb{Q}_2, \theta) = 1 + \dim H^0(\mathbb{Q}_2, \theta) + \dim H^0(\mathbb{Q}_2, \theta^{-1}\varepsilon), \quad (40)$$

$$\dim H_g^1(\mathbb{Q}_2, \theta) = \dim D_{\text{dR}}(\theta)/\text{Fil}^0 + \dim H^0(\mathbb{Q}_2, \theta) + \dim D_{\text{cris}}(\theta^{-1}\varepsilon)^{\varphi=1}. \quad (41)$$

The second formula follows from the exact annihilators of [10, Proposition 3.8], the H_f^1 and H_e^1 dimension formulas of [10, Corollary 3.8.4], and local duality; it is the rank-one consequence used here. These formulas have no restriction excluding 2.

Lemma 8.1. *If $r|_{G_{\mathbb{Q}_2}}$ is reducible and has distinct cyclotomic exponents, it has an invariant line whose exponent is greater than that of the quotient.*

Proof. Characters occurring as subquotients of a de Rham representation are de Rham. Suppose a chosen subcharacter has smaller exponent than the quotient, and let θ be their ratio. Its exponent is strictly negative, so $D_{\text{dR}}(\theta)/\text{Fil}^0 = 0$ and $H^0(\mathbb{Q}_2, \theta) = 0$. The character $\theta^{-1}\varepsilon$ has strictly positive exponent. If its finite inertial part is nontrivial, its crystalline module is zero; otherwise its crystalline Frobenius eigenvalue has nonzero valuation and cannot be 1. Equation (41) therefore gives $H_g^1 = 0$. The extension splits, and its other summand is the required line. $\square$

Lemma 8.2. *Suppose $r|_{G_{\mathbb{Q}_2}}$ is reducible, and record the line of Lemma 8.1. Fix $\chi = \det r$. After enlarging coefficients, there is a complete local integral $\mathcal{O}$ -algebra $\mathcal{D}$, together with its fixed global determinant χ , its global trace, and a recorded local subcharacter η_1 , with the following properties.*

1. $\mathcal{D}$ is horizontal, contains the target point, and $\dim \mathcal{D} \geq 2$. Put $\eta_2 = \chi\eta_1^{-1}$. On its globally absolutely irreducible locus the local representation has subcharacter η_1 and quotient η_2 .
2. For one field F as above, the restricted trace lies in the Hecke support of completed forms at a fixed tame level.
3. $\mathcal{D}$ is finite over the closed image of the global pseudodeformation ring for F . It is also finite over the image of the product of the dyadic pseudodeformation rings.

Proof. Use a framed global deformation chart with fixed determinant and an invariant line at 2, in an integral basis for the target and its line. Consider its completed local ring at the characteristic-zero

target point. Corollary 3.7 gives dimension at least five for the local line chart and dimension two for the odd real factor, over the characteristic-zero coefficient field. The relative presentation of Proposition 3.6 has generators minus relations at least -3 : its dual invariant term vanishes because the global representation is not virtually solvable. The global completed local ring therefore has dimension at least $5 + 2 - 3 = 4$. The characteristic-zero dimension comparison in Lemma 3.4 then gives a horizontal integral component through the target in the original chart of dimension at least five.

Contract that component to its global trace and local subcharacter values. On the absolutely irreducible locus the only positive dimensional fibers are conjugacy fibers, of dimension three. The local characters are distinct on a dense open set, so recording the character does not leave an extra moving-line fiber there. The dimension comparison of Lemma 3.2 gives a domain $\mathcal{D}$ of dimension at least two. Properness of the invariant-line incidence preserves the recorded subcharacter condition on the irreducible locus. Theorem 7.1, applied to an unrestricted framed component containing this family, gives one common F and one tame level with the asserted trace support.

We give the finiteness argument, including the recorded character. Write $T_{\mathcal{D}}$ for the trace of the family. Let $A \subset \mathcal{D}$ be the closed restricted trace image, and let $q = [G_{\mathbb{Q}} : G_F]$. For every $g \in G_{\mathbb{Q}}$, $g^q \in G_F$ after replacing q by the order of the finite quotient if necessary. The identity

$$T_{\mathcal{D}}(g^q) = P_q(T_{\mathcal{D}}(g), \chi(g))$$

has $P_q(X, Y)$ monic of degree q in X . Hence every downstairs trace is integral over A . A local character value satisfies

$$\eta_1(g)^2 - T_{\mathcal{D}}(g)\eta_1(g) + \chi(g) = 0, \quad g \in G_{\mathbb{Q}_2},$$

and is integral as well. The global determinant algebra has finitely many topological trace generators, and a continuous local character is determined topologically by its values on the uniformizer, -1 , and 5 . Adjoining these finitely many integral elements to A gives a finite complete algebra, hence a closed subalgebra of $\mathcal{D}$; it contains its topological generators and is all of $\mathcal{D}$. Finally, Proposition 4.2 makes the Hecke algebra finite over the product of the local central pseudodeformation images. Transitivity of finiteness proves the last assertion. $\square$

We may discard the proper globally reducible loci, including reducibility after restriction to F . We may also require

$$\eta_1\eta_2^{-1} \notin \{1, \varepsilon, \varepsilon^{-1}\}. \quad (42)$$

Indeed, each exceptional equality and the fixed determinant imply $\eta_1^2 = \chi\varepsilon^a$ for $a \in \{0, 1, -1\}$. The local trace image then has dimension at most one: its character values are integral over the fixed coefficient ring. Because 2 splits in F and the family comes from $\mathbb{Q}$, the same holds at all dyadic factors. Finiteness in Lemma 8.2 shows that none of these loci contains $\text{Spec } \mathcal{D}$. Characteristic-zero points with finite coefficient fields satisfying these conditions are consequently dense.

8.2 The ordinary torsion module

We now have a family carrying both global traces and the chosen local character. To control its character parameter, we construct a torsion space on which the traces and torus act jointly. Its finiteness over torus weight space will constrain the dimension of $\mathcal{D}$.

Fix the Hecke summand of the completed forms over F carrying the restricted family, and write $\mathcal{H}$ for its torsion function space with coefficients $E/\mathcal{O}$. Put

$$G_2 = \prod_{v|2} \text{GL}_2(\mathbb{Q}_2), \quad N_0^- = \prod_{v|2} n^-(\mathbb{Z}_2), \quad T_1 = \prod_{v|2} (1 + 4\mathbb{Z}_2)^2,$$

where

$$n^-(a) = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}, \quad n^+(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \quad \mathbf{p} = (\text{diag}(1, 2))_{v|2}.$$

All matrix formulas involving several places are interpreted coordinatewise. On $\mathcal{H}^{N_0^-}$ define

$$U = \sum_{a \in (\mathbb{Z}/2\mathbb{Z})^{[F:\mathbb{Q}]}} n^-(a) \mathbf{p}. \quad (43)$$

The actions are the right translation actions with the representation group law. Let

$$W = \{f \in \mathcal{H}^{N_0^-} : U^N f = f \text{ for some } N > 0\}, \quad M = W^\vee, \quad \Omega = \mathcal{O}[[T_1]].$$

The dual is the compact Pontryagin dual. We transpose the commuting operator actions without inversion, so their scalar eigenvalues retain the conventions above.

Proposition 8.3 (Ordinary weight finiteness). *The space W is an $\mathcal{O}$ -submodule stable under T_1 , the tame Hecke operators, and $U^{\pm 1}$, and these actions commute. Its dual M is finite over Ω . The closed commutative algebra generated by these actions on M is finite over Ω .*

Proof. In one factor the identity

$$n^-(a + 2b) \text{diag}(1, 2) = n^-(a) \text{diag}(1, 2) n^-(b) \quad (44)$$

proves independence of representatives and preservation of lower invariance. Conjugation by $t = \text{diag}(u_1, u_2) \in T_1$ replaces a by $(u_2/u_1)a$, which permutes the residue classes. Thus $tU = Ut$ on the lower-invariant space. The tame Hecke actions commute with these local actions. A common multiple of two periods is a period of the sum of the corresponding vectors. Finally U is bijective on W : on a vector of period N , its inverse is U^{N-1} , independently of the chosen period.

We show that one compact open subgroup fixes $W[\varpi]^{T_1}$. For $j \geq 2$, $x \in 2^j \mathbb{Z}_2$, and $a \in \mathbb{Z}_2$, direct multiplication gives

$$\begin{aligned} n^+(x) n^-(a) \text{diag}(1, 2) &= n^-\left(\frac{a}{1+xa}\right) \text{diag}(1, 2) \\ &\quad \cdot \text{diag}(1+xa, (1+xa)^{-1}) n^+\left(\frac{2x}{1+xa}\right). \end{aligned} \quad (45)$$

If f is T_1 -fixed and upper- 2^{j+1} -invariant, the final two factors fix f . The map $a \mapsto a/(1+xa)$ permutes the lower residue classes, and (44) absorbs the changes of lift. Thus Uf is upper- 2^j -invariant. Once upper-4 invariance is reached, it is preserved by the same identity.

Each smooth vector of $W[\varpi]^{T_1}$ starts at some finite upper level. Iterate the preceding improvement far enough, taking an iteration number divisible by its period. The resulting vector is the original vector, so it is fixed by the single open compact subgroup generated by N_0^- , T_1 , and $\prod_{v|2} n^+(4\mathbb{Z}_2)$. Proposition 4.1 gives smooth admissibility; therefore $W[\varpi]^{T_1}$ is finite-dimensional over k . Compact Nakayama now gives finite generation of M over Ω .

Since Ω is noetherian and M is finite, $\text{End}_\Omega(M)$ is finite over Ω . The closed algebra in the statement is an Ω -submodule of this finite module, and hence is finite too. $\square$

Let R denote the fixed-determinant global pseudodeformation ring over F . The trace and torus actions define

$$Q = R \hat{\otimes}_{\mathcal{O}} \Omega \longrightarrow \text{End}_\Omega(M).$$

Write S for its image. It is finite over Ω and is closed: the source is compact and the target is Hausdorff. The larger action algebra in Proposition 8.3, denoted by $\mathcal{A}$, is finite over S and contains both U and U^{-1} .

8.3 Finite joint images and the weight map

Set

$$\Lambda = \mathcal{O}[[1 + 4\mathbb{Z}_2]] \simeq \mathcal{O}[[T]], \quad [5] = 1 + T.$$

The map $\Lambda \rightarrow \mathcal{D}$ sends $[5]$ to $\eta_1(5)$, using local reciprocity. At each place of F above 2, there are two possible inducing assignments

$$(\eta_1\varepsilon, \eta_2), \quad (\eta_2\varepsilon, \eta_1). \quad (46)$$

For each choice s of an assignment at every dyadic place, the fixed determinant and these formulas define a homomorphism $s : \Omega \rightarrow \Lambda$. Only characters on $1 + 4\mathbb{Z}_2$ enter this map; the unramified character values remain in the trace algebra.

Lemma 8.4 (Finite joint-image argument). *Let $\mathcal{D}$ be the domain of Lemma 8.2, let $M = W^\vee$ be the ordinary module above, and use the weight map $\Lambda \rightarrow \mathcal{D}$ defined by $[5] \mapsto \eta_1(5)$. Suppose a dense set of finite-coefficient characteristic-zero points of $\mathcal{D}$ occurs in the joint trace-and-weight support of M , with one of the finitely many assignments s in (46) at each point. Then $\mathcal{D}$ is finite and dominant over Λ ; in particular, $\dim \mathcal{D} = 2$.*

Proof. For each s , form $S_s = S \hat{\otimes}_{\Omega, s} \Lambda$, and let T_s be the image of $R \hat{\otimes}_{\mathcal{O}} \Lambda$ in S_s . Both are finite Λ -algebras, and their images are closed. Let I_s be the kernel of the latter map. Each point in the dense set kills some I_s after mapping to $\mathcal{D}$. Hence

$$\mathrm{Spec} \mathcal{D} = \bigcup_s V(I_s \mathcal{D}).$$

The spectrum is irreducible, so one of these finitely many closed sets is the whole spectrum. Equivalently, the prime kernel of $R \hat{\otimes} \Lambda \rightarrow \mathcal{D}$ contains one I_s . The closed trace-plus-weight image B in $\mathcal{D}$ is therefore a quotient of T_s and is finite over Λ .

By Lemma 8.2, $\mathcal{D}$ is finite over the restricted trace image A , and $A \subset B$. The same finite set of A -module generators generates $\mathcal{D}$ as a B -module. Thus $\mathcal{D}$ is finite over Λ . Its dimension is at least two, whereas $\dim \Lambda = 2$; a nonzero prime kernel in the domain Λ would lower that dimension. The map is therefore injective, and the dimension is exactly two. $\square$

The hypotheses of the lemma hold here. At each finite-coefficient characteristic-zero point outside the excluded loci, Proposition 4.2 supplies a Banach subrepresentation of completed forms whose local factors are unitary continuous inductions in one of the orders (46); the two possibilities are precisely those of [48, Corollary 6.10(3)] for the normalized parameter $r \otimes \varepsilon$. In $\mathrm{Ind}_B^{\mathrm{GL}_2(\mathbb{Q}_2)}(\delta_1 \otimes \delta_2)$, with B the upper Borel and unnormalized induction, take the function supported on $Bn^-(\mathbb{Z}_2)$ and equal to 1 on $n^-(\mathbb{Z}_2)$. It is lower-invariant and has torus character (δ_1, δ_2) . Subdivision into the two lower disks shows that its U -eigenvalue is $\delta_2(2)$, a unit. The product vector has the analogous properties at all dyadic places.

Its nonzero image in completed forms can be scaled into the integral lattice. Modulo every power of the coefficient uniformizer its unit eigenvalue has finite multiplicative order, so the reduction belongs to W . Duality consequently places this trace-and-weight point in $\mathrm{Spec} S$. Finite coefficient extension is harmless, either by flat extension of the torsion space or by taking scalar coordinates. Lemma 8.4 applies.

Thus the family is finite and dominant over the one-variable weight space. We can now choose a dense set of character values there and lift them to the family. At these points the local extension will satisfy the ordinary modularity theorem after the normalization and dualization described next.

8.4 A dense set of ordinary modular points

Let $w = \exp(\chi)$. After a finite extension of E , choose c with $c^2 = 5^{w+1}$. Consider weights

$$\eta_1(5) = c\zeta, \quad \zeta \in \mu_{2^\infty}. \quad (47)$$

These are continuous characters of $1 + 4\mathbb{Z}_2$: this group is procyclic with generator 5, and $c\zeta$ lies in the open unit neighborhood of 1 in its finite coefficient field. We are specifying a character value, not asserting that $(w+1)/2$ is an integral Hodge–Tate weight.

The weights in (47) are Zariski dense in $\text{Spec } \Lambda$. Indeed, after dividing by its maximal scalar power, a nonzero element of $\mathcal{O}[[T]]$ is a unit times a distinguished polynomial by Weierstrass preparation and has only finitely many zeros in the open unit disk. Finite dominance then gives a dense set of coefficient points of $\mathcal{D}$ above these weights, even after removing any prescribed proper closed subset. To see the last assertion directly, a nonzero ideal of the finite integral domain $\mathcal{D}$ meets Λ nontrivially: an integral equation of minimal degree for a nonzero element of the ideal has nonzero constant term. Thus every proper closed subset has proper closed image in $\text{Spec } \Lambda$.

On inertia, the ratio at these points has the form

$$\eta_1 \eta_2^{-1} = \varepsilon \xi, \quad (48)$$

where ξ has finite order. This follows from η_1^2/χ on $1 + 4\mathbb{Z}_2$, and the remaining unit quotient is finite. The finite part of χ is fixed; at most finitely many choices of ζ make ξ trivial. We discard them, as well as the globally reducible loci before and after restriction to F .

There is a single continuous normalizing character over the family. The character η_2 on inertia factors through the local unit group $\mathbb{Z}_2^\times$ by local class field theory. Denote this unit character by $\eta_{2,u}$, and put

$$\lambda(g) = \eta_{2,u}(\varepsilon(g))^{-1}, \quad g \in G_{\mathbb{Q}}. \quad (49)$$

This character is continuous and unramified outside 2. At a point chosen above, write $r' = r_x \otimes \lambda_x$ and $\eta'_j = \eta_{j,x} \lambda_x$. Then

$$\eta'_1 = u_h \xi \varepsilon, \quad \eta'_2 = u_l, \quad (50)$$

with u_h, u_l unramified and ξ of nontrivial finite inertia.

Lemma 8.5. *Every representation r' just constructed is odd, absolutely irreducible, finitely ramified, ordinary, and potentially crystalline with cyclotomic exponents $\{1, 0\}$. It is modular.*

Proof. The two diagonal characters in (50) are de Rham. Their ratio $\theta = (u_h/u_l)\xi\varepsilon$ has $H^0(\theta) = H^0(\theta^{-1}\varepsilon) = 0$, the latter because the finite inertial character is nontrivial. Thus (40) gives $\dim H^1(\mathbb{Q}_2, \theta) = 1$. In (41) the de Rham tangent term has dimension one, while the other two terms vanish. Hence every extension class belongs to H_g^1 and r' is de Rham.

The p -adic monodromy theorem makes it potentially semistable [7, Theorems 0.6 and 5.19]. Its two finite inertial characters are distinct. Monodromy commutes with inertia, so it preserves their two eigenspaces; it vanishes on each by nilpotence. Thus it is potentially crystalline.

Twisting preserves absolute irreducibility and finite ramification. For a complex conjugation c_∞ , $\lambda(c_\infty)^2 = 1$, so the determinant remains odd. Finally, in the convention $\text{HT}(\varepsilon) = -1$, the dual has weights $\{0, 1\}$ and ordinary shape

$$(r')^\vee|_{G_{\mathbb{Q}_2}} \sim \begin{pmatrix} u_l^{-1} & * \\ 0 & \varepsilon^{-1} u_h^{-1} \xi^{-1} \end{pmatrix}.$$

All the hypotheses of [59, Theorem D] are now satisfied. That theorem makes the dual modular, and hence makes r' modular. $\square$

The modular representations in this lemma have a common bound for their tame conductors. Indeed, the original noetherian Galois family has, at each non-dyadic bad prime, a common open inertia subgroup on which its action is unipotent, as in Lemma 3.9. The finite inertial quotient is fixed, and the rank-two monodromy contribution is bounded. This bounds the Artin conductor after the fixed base change to F . The normalizing characters add no non-dyadic inertia. Consequently, after replacing the tame compact subgroup by one smaller subgroup, all these modular forms occur at a single tame level. Solvable base change, iterated through prime-degree cyclic extensions, and Jacquet–Langlands put them in the definite forms over F ; see [38, Chapter 11, Lemma 11.3 and Proposition 11.4], [2, Chapter III, Theorems 4.2 and 5.1], and [28, Theorem 16.1].

8.5 The required inducing order

We next track the order at the dense modular points, rather than selecting it from a semisimple Banach fiber. In the geometric reciprocity convention $\varepsilon = x|x|$, one has $\text{WD}(\varepsilon) = |\cdot|$. For the normalized characters (50), the two characters of $\text{WD}(r' \otimes \varepsilon)$ are

$$\theta_h = u_h \xi |\cdot|^2, \quad \theta_l = u_l |\cdot|.$$

Here the same notation for a finite or unramified character denotes its character under local reciprocity. A regular algebraic Hilbert form on GL_2 is essentially self-dual via its central character. Local–global compatibility identifies the smooth principal series: the underlying semisimple Weil representation has these distinct inertial characters, and no monodromy can join them. This also follows from the compatibility up to the monodromy order of [3, Corollary 1.2], since that order preserves the underlying semisimple Weil representation and monodromy is zero in this case.

In the conventions of Proposition 4.2, put the smooth inducing characters in the order

$$(\theta_l, \theta_h |\cdot|^{-1}) = (u_l |\cdot|, u_h \xi |\cdot|).$$

The algebraic factor at cyclotomic exponents $\{1, 0\}$ is $\det$. Multiplication by that factor changes the pair to

$$(u_l \varepsilon, u_h \xi \varepsilon) = (\eta'_2 \varepsilon, \eta'_1).$$

For unnormalized smooth induction, the intertwiner changes an inducing pair (A, B) to $(B|\cdot|, A|\cdot|^{-1})$. The two orders here are therefore intertwined with the appropriate modulus factors, with no exceptional ratio because their inertial characters differ. Equivalently, this is the weight-two instance of the ordinary universal completion formula [13, Proposition 2.2.1]; that proposition applies at 2 and concerns the locally algebraic induction. In particular, the lower-disk vector used above already belongs to this locally algebraic representation.

Undo the twist by multiplying definite functions by λ_x^{-1} evaluated on the reduced norm, via global reciprocity. Rational reduced norms are totally positive, so this is well-defined on the definite quotient. It changes the central character back to $\chi \varepsilon$ and changes the inducing assignment to

$$(\eta_2 \varepsilon, \eta_1) \quad \text{at every } v \mid 2. \tag{51}$$

Lower invariance is preserved, and the U -eigenvalue is multiplied by a unit. All dyadic places have the same order because they are split over $\mathbb{Q}$ and the family and normalizing character descend from $\mathbb{Q}$.

At this common tame level, let M and S again denote the ordinary dual and its joint trace-and-weight image algebra. Write $s_{\text{fav}} : \Omega \rightarrow \Lambda$ for the single assignment (51), and set

$$S_{\text{fav}} = S \hat{\otimes}_{\Omega, s_{\text{fav}}} \Lambda, \quad I_{\text{fav}} = \ker(R \hat{\otimes}_{\mathcal{O}} \Lambda \longrightarrow S_{\text{fav}}).$$

Let T_{fav} be the image of this map. The untwisted vectors just constructed show that I_{fav} vanishes at the dense family points used in Lemma 8.5. Its image in the domain $\mathcal{D}$ is therefore zero. In particular, the original target defines a coefficient point of T_{fav} , including when its local character ratio satisfies one of the exceptional equalities excluded from the dense testing set.

The algebra S_{fav} is finite over T_{fav} , so lying over lifts this point to S_{fav} after a finite coefficient extension. Its restriction to S has the target trace and the torus character prescribed by s_{fav} . Since M is finite over S and the S -action is faithful, $\text{Supp}_S M = \text{Spec } S$. Thus this point belongs to the joint trace-and-weight support needed below.

The favorable character order has now reached the target. To use it in a classicality argument, we still need a nonzero vector carrying that order. The next step obtains one from a scalar quotient of the finite action module, without assuming that ordinary vectors or locally algebraic vectors commute with specialization.

8.6 Specialization to a genuine vector

Lemma 8.6 (Eigenvector specialization). *Let S be the joint trace-and-weight image algebra acting on $M = W^\vee$ at the fixed tame level above, and let $\mathcal{A}$ be its finite faithful action algebra containing $U^{\pm 1}$, as in Proposition 8.3. Let a finite-coefficient characteristic-zero point of $\text{Spec } S$ belong to the support of M . After a finite coefficient extension it gives a nonzero vector in completed forms with the specified trace and torus eigencharacters, invariant under N_0^- , and with a unit U -eigenvalue.*

Proof. The algebra $\mathcal{A}$ containing $U^{\pm 1}$ is finite over S . Since it is the faithful action algebra of M , and M is finite over it, its spectrum is the support of M . Lying over the given point and passing to a finite coefficient field E' therefore gives a joint scalar quotient of $M[1/2]$. Write $\mathcal{O}'$ for its ring of integers, with uniformizer ϖ' .

The corresponding nonzero map

$$M \otimes_{\mathcal{O}} \mathcal{O}' \longrightarrow E'$$

is continuous. Indeed, M is finite over weight space and its weight values lie in the formal open unit disk. Its image is a compact $\mathcal{O}'$ -submodule spanning E' , hence a lattice; rescale it to $\mathcal{O}'$. The scalar actions preserve this lattice. In particular the U -eigenvalue is a unit because U^{-1} preserves it as well.

Applying $\mathcal{O}'$ -linear Pontryagin duality, with values in $E'/\mathcal{O}'$, produces an equivariant injection

$$E'/\mathcal{O}' \hookrightarrow W_{\mathcal{O}'} \subset \mathcal{H}_{\mathcal{O}'}.$$

The image f_n of ϖ'^{-n} has exact order ϖ'^n and satisfies $\varpi' f_{n+1} = f_n$. Taking the Tate module gives a nonzero continuous integral completed-forms vector. Explicitly, a compatible sequence of continuous functions modulo ϖ'^n determines a continuous $\mathcal{O}'$ -valued function, because each of its reductions is continuous; conversely every such integral function gives that sequence. Exact orders ensure that the resulting vector is nonzero. All Hecke and torus eigenrelations, lower invariance, and the U -eigenrelation pass to this inverse limit. This construction finds a divisible copy inside W ; it does not require W itself to be divisible. $\square$

Apply the lemma to the target. Its torus eigencharacter, restricted to a sufficiently small open subgroup, is algebraic with exponents $e_{1,v}, e_{2,v}$ satisfying

$$e_{2,v} - e_{1,v} = \exp(\eta_1) - \exp(\eta_2) - 1 \geq 0. \quad (52)$$

The inequality follows from the choice of the higher line and the integrality and distinctness of the target weights. We have therefore obtained a genuine ordinary eigenvector whose torus exponents

have the order needed for an algebraic representation. It remains to identify the algebraic form of its orbit on a sufficiently small open subgroup.

8.7 A unit ordinary vector is locally algebraic

Lemma 8.7. *Let V be a continuous unitary Banach representation of G_2 over a finite extension $E'/\mathbb{Q}_2$. Suppose $0 \neq f \in V$ is fixed by N_0^- , has torus character with integral exponents $e_{1,v}, e_{2,v}$ on a sufficiently small open subgroup, and satisfies $Uf = \alpha f$ with $\alpha \in \mathcal{O}'^\times$. If $\delta_v = e_{2,v} - e_{1,v} \geq 0$ for all v , then f is locally algebraic, of algebraic type*

$$A = \bigotimes_{v|2}^{\epsilon_{1,v}} \det \operatorname{Sym}^{\delta_v}(E'^2).$$

Proof. Choose $m \geq 2$ so the stated torus character holds on $\prod_v (1+2^m \mathbb{Z}_2)^2$, and put $X = (2^m \mathbb{Z}_2)^{[F:\mathbb{Q}]}$. Lower invariance gives

$$U^j f = \sum_{b \bmod 2^j} n^-(b) \mathbf{p}^j f,$$

independently of the integral lifts of b . For $x \in X$ the map $a \mapsto a/(1-xa)$ permutes these residue classes. With $c = 1-xa$, direct matrix multiplication gives

$$n^+(x) n^-\left(\frac{a}{1-xa}\right) \mathbf{p}^j = n^-(a) \mathbf{p}^j \operatorname{diag}(c^{-1}, c) n^+(2^j xc). \quad (53)$$

Define the V -valued polynomial

$$P_j(x) = \alpha^{-j} \sum_{a \bmod 2^j} \prod_v (1 - x_v a_v)^{\delta_v} n^-(a) \mathbf{p}^j f.$$

Its multidegree is bounded by δ . Use a G_2 -invariant nonarchimedean norm on V . The torus eigenrelation, (53), and the ultrametric inequality give

$$\sup_{x \in X} \|n^+(x)f - P_j(x)\| \leq \sup_{y \in (2^{j+m} \mathbb{Z}_2)^{[F:\mathbb{Q}]}} \|n^+(y)f - f\| \longrightarrow 0. \quad (54)$$

There is no factor for the number of residue classes: the norm of a sum is bounded by the largest norm of a summand. The unit eigenvalue and unitarity also introduce no growing factor. Continuity at the identity proves the final convergence.

For each v , choose $\delta_v + 1$ distinct points in $2^m \mathbb{Z}_2$. Evaluation on their product grid and inversion of the resulting Vandermonde matrices recover all coefficients of a polynomial of multidegree at most δ , continuously in its values. This polynomial space is therefore closed in $C(X, V)$. Equation (54) implies

$$n^+(x)f = \sum_{0 \leq \nu \leq \delta} x^\nu v_\nu. \quad (55)$$

For completeness, we identify an algebraic representation containing the resulting orbit. Put $H = \prod_v (1 + 2^m M_2(\mathbb{Z}_2))$ and write $h_v = \begin{pmatrix} A_v & B_v \\ C_v & D_v \end{pmatrix}$. Its Gauss decomposition is

$$h_v = n^+(B_v/D_v) \operatorname{diag}(\det(h_v)/D_v, D_v) n^-(C_v/D_v).$$

Every factor lies in the neighborhoods already considered. Hence

$$hf = \sum_{0 \leq \nu \leq \delta} \prod_v \det(h_v)^{\epsilon_{1,v}} B_v^{\nu_v} D_v^{\delta_v - \nu_v} v_\nu. \quad (56)$$

These are regular algebraic functions on G_2 , also when some $e_{1,v}$ is negative, since powers of the determinant are invertible in its coordinate ring.

Let $z_0 = \bigotimes_v e_2^{\delta_v} \in A$, with the determinant factors understood. In the binomially scaled monomial basis E_ν of A , one has $n^+(x)z_0 = \sum_\nu x^\nu E_\nu$. Define $\Theta : A \rightarrow V$ by $\Theta(E_\nu) = v_\nu$. Formula (56) says $\Theta(hz_0) = hf$ for $h \in H$. The vectors kz_0 , $k \in H$, span A , by the same upper-unipotent interpolation. Thus

$$\Theta(hkz_0) = hkf = h\Theta(kz_0), \quad h, k \in H,$$

proves that Θ is H -equivariant. Since H is Zariski dense, its invariant subspace $\ker \Theta \subset A$ is an algebraic subrepresentation. The representation A is irreducible and $\Theta(z_0) = f \neq 0$, so Θ is injective. Its image contains f and has algebraic type A . This proves local algebraicity without any assertion that algebraic vectors commute with specialization. $\square$

The vector of Lemma 8.6 meets these hypotheses by (52). It is therefore a nonzero locally algebraic vector with the target eigensystem.

8.8 Completion of the proof over $\mathbb{Q}$

If $r|_{G_{\mathbb{Q}_2}}$ is absolutely irreducible, the preceding ordinary argument is unnecessary. Proposition 4.2 and the local correspondence give nonzero locally algebraic vectors in the target eigenmodule: for an absolutely irreducible parameter, regular de Rham is exactly the required condition [18, Theorems 1.1 and 1.3], [17, Theorem 0.20]. The all-prime correspondence covers the scalar residual case at 2. Since 2 splits completely in F , this applies at every dyadic factor. The parameter in this application is $r \otimes \varepsilon$. We translate the opposite Hodge–Tate sign convention in these local correspondence sources using cyclotomic exponents, obtaining the algebraic factor fixed in Proposition 4.2.

In either local case, the definite-forms comparison of Proposition 4.1 now gives a regular algebraic automorphic eigensystem on the definite quaternion algebra over F . It cannot be a reduced-norm character, since its global Galois representation is absolutely irreducible. Jacquet–Langlands therefore gives a regular algebraic Hilbert cusp form [28, Theorem 14.4]. Its Galois representation exists at 2 by [4, Theorem 2.1.1] and agrees with $r|_{G_F}$ by the unramified Hecke identities and Chebotarev.

Finally choose a tower of prime-degree cyclic extensions from $\mathbb{Q}$ to a solvable Galois closure of F , replacing F by that closure if needed. At each stage the Hilbert representation upstairs is invariant under the cyclic Galois group: its conjugates have the same unramified eigensystems because the Galois representation extends downstairs, so strong multiplicity one identifies them [38, Chapter 3, Lemma 3.1]. Cyclic descent supplies a cuspidal representation downstairs, unique up to characters of that cyclic extension [2, Chapter III, Theorems 4.2(d) and 5.1]; see also the original degree-two construction in [38, Chapter 11, Lemma 11.3 and Proposition 11.4]. When needed, apply the cited unitary formulation after twisting by a power of the idelic norm, and undo that twist after descent. Every real place splits in the totally real extension, so the corresponding local extension is $\mathbb{R}/\mathbb{R}$; archimedean base change therefore preserves the regular algebraic real-place type. The Galois representation of the descended packet and the given representation become isomorphic on restriction upstairs; absolute irreducibility of that restriction implies that they differ by a character of the cyclic quotient. Twist the descended automorphic representation by the inverse finite Hecke character to correct this difference, and continue down the tower. Over $\mathbb{Q}$ the resulting representation is attached to a classical cusp eigenform, with the Tate twist dictated by the original weights. This proves Theorem 1.1.

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